1. PURPOSE:

Purpose of this Global Design For Manufacturing Guideline is to define standardized concepts which provide the best practice for an engineering design that suits the manufacturing requirements in relation to handling and processing. This will enable the plants to deliver high quality harnesses at an optimized efficiency level.
In addition, the guideline should help to standardize the engineering design.

The concepts shown in this guideline are generated out of proposals collected from the manufacturing locations and the Customer Specific Business Unit (BU),the COSEE organization and the Quality department.
These proposals were analyzed by the Global Design For Manufacturing (DFM) Team and evaluated and confirmed by the production plants and COSEE.

If you have any new items or suggestions please contact the DFM group.
After investigation we will implement the proposals into the new release of the guideline.

In some cases, the DFM concept might deviate from the customer requirements or specifications.
In this case, the customer requirements are binding, but the design engineer should investigate to convince the customer to allow the YAZAKI preferred process. Furthermore, some items are only regionally valid.

Due to a technical requirement, it might be that the necessary design deviates from the guideline; in this case the design engineer has to contact the manufacturing plants in advance to get a feasibility approval.

In general, it can be said that with approval from manufacturing every deviation can be tolerated.

Please note that this collection is a guideline and not a specification.

The identical guideline is available in:
YEL with the number: EA-EN-WH-G-001 Revision 4 (Version11)
YNCA with the number: AA-EN-WH-G-001 Revision 9

2. SCOPE:

Chapter 4.1 of the Global DFM Guideline focuses on the items that may be needed before RFQ. The content of this chapter needs to be considered before the RFQ phase. During the architecture and general strategies phase, the general items from Chapter 4.1 have to be clarified. After a general decision on the items during design phase, the items from Chapter4. 1 do not have to be checked again. Chapters 4.2 through 4.7 of the Global DFM Guidelines focus on the items that may be needed after the RFQ has been established. The Design Engineer should concentrate on the DFM items, in order to meet the Quality Cost Delivery (QCD).

3. TERMS & DEFINITIONS:

CAN	Controller Area Network
CFA	Crimp Force Analysing
CFM	Crimp Force Monitoring
COSEE	Committee Of Standardisation Evaluation Equipment
C_{pK}	Process Capability Index
BU	Business Unit
DFM	Design For Manufacturing
GCT	Global Costing Tool
HV	High Voltage
J/C	Joint Connector
p/n	Part Number
QCD	Quality Cost Delivery
RFQ	Request For Quotation
TCSS	Terminal Crimping Standard Specification
TPO	Terminal Push Out
W/H	Wire Harness
WUT	Weld Under Test
YC	Yazaki Corporation
YEL	Yazaki Europe Limited
YNCA	Yazaki North & Central America

4. PROCESS DETAILS:

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4.1 Design Development Activities

Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

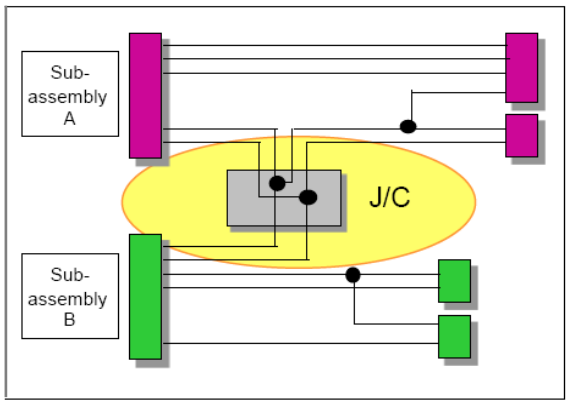
Design Develop Activities: 1
Item: 01

4.1.1 Use J/C when necessary to improve sub-assy in manufacturing

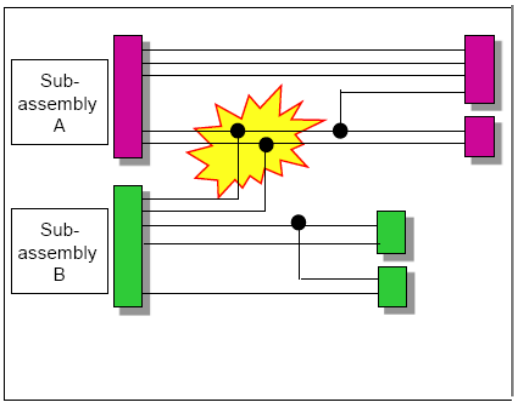
DFM Concept:

Use J/C (Joint Connector) to improve the manufacturing processes and increase the automation possibilities.
An appropriate location to package the J/C in the vehicle must be found. Recommended positions are inside or outside of protectors (if available) with tab provisions to fix it. Otherwise it should be taped back on the main trunk considering optimized wire lengths or full covered over the wiring harness and ensure that it might not cause any rattle in the vehicle.

OK



NG



Correct Method:
Use a J/C to allow smaller sub-assembly construction. Consider using J/C if one off the following condition is met:
a) The number of wires in a splice is higher than 6 wires.
b) When the sub-assembly has 2 or more splices interconnected and have at least one of the circuits with a length of more than 2.3 meters.
c) When the sub-assembly has 2 or more complicated processes interconnected as twisted pairs, shielded wires in splices.
d) If the design includes already J/C it shall not be removed without prior approval by Top Management

Incorrect Method:
Splices are used when J/C should be used.

Benefits/Effects:
Improve handling of sub-assemblies in manufacturing.
The J/C should be used, if after a detailed costs (labor, material, MUDA, quality cost, floor production space, investment, tooling, etc.) analysis, it is cheaper than the handling (tangling) of complex spiderwebs caused by complicated splices or interconnections between splices.
The usage of J/C will allow the automated manufacturing of sub-assemblies (DFA) and support the elimination of manual splicing operations.

Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

Design Develop Activities: 1
Item: 02

4.1.2 Use of ground blocks for bonder terminals

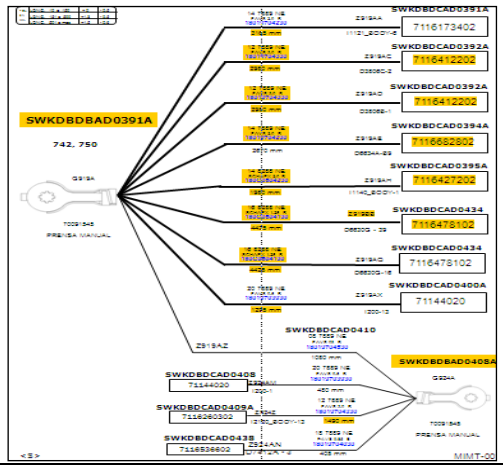
DFM Concept:

To avoid complex large bonder subassemblies that require coiling and handling consider using ground blocks

OK



NG



Correct Method:
Use of ground block which eliminates the complexity and handling of the subassembly.

Incorrect Method:
Bonder terminal which complicates the handling of the subassembly

Benefits/Effects:
In some cases, the total cost could be reduced by reducing the complexity and the handling of the subassembly and therefore, increasing the efficiency.

Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

Design Develop Activities: 1
Item: 03

4.1.3 Ultrasonic / bonder terminals for multi-crimp wires

DFM Concept:
Avoid using mechanical crimps for multi-crimp wires, use instead ultrasonic terminals.

Please see also DFM concept: 6.18.

(According to USCAR38, no collateral damage may result from terminal being welded if the wire length is equal or more than 500mm. If this can't be done Inspection must include components on the same wire within 500mm and welded prior to WUT. If OEM Customer and/or Supplier specify something different, you MUST follow what they are requiring).

OK



NG



Correct Method:
The preferred method is to use ultrasonic/bonder type welded eyelets.

The wiring harness design engineer specifies only circuits with a minimum length of 500 mm between an ultrasonic/resistance welded terminal and the opposite crimped terminal for the same circuit.

Incorrect Method:
The use of mechanical crimp terminals for multiple wire does not guarantee a good performance of C_{pK}.

Benefits/Effects:
The equipment for ultrasonic and bonder welding is higher cost than manual crimping presses, but from a quality assurance point of view multi-crimping is not an equivalent process.
Resistance/sonic weld eyelets were developed to meet the need for multi-wire eyelet terminals because the weld process bonds the conductor to the terminal rather than compacting them into the grip. Bonding is much more tolerant of variation in the position of the wires relative to each other in the nugget.

Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

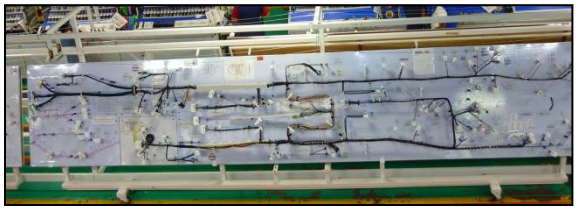
Design Develop Activities:	1
Item:	04

4.1.4 Avoid big complex W/H clarify max. possible wire lengths

DFM Concept:

Harnesses should be able to fit on a standard size jig board.
The standard sizes will vary from region to region.
To use especially full automatic machine (cutting, crimping and terminal insertion) please contact in advance the manufacturing plant to clarify the maximum possible wire lengths.

OK



NG





Correct Method:
Reduce W/H complexity, when possible.

Incorrect Method:
Huge W/H (heavy, too long to handle).

Benefits/Effects:
Increased floor space
Reduced capacity
Low efficiency
Wire harnesses (W/H) more than e.g., 4 meters will be too heavy and hard to manage.

Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

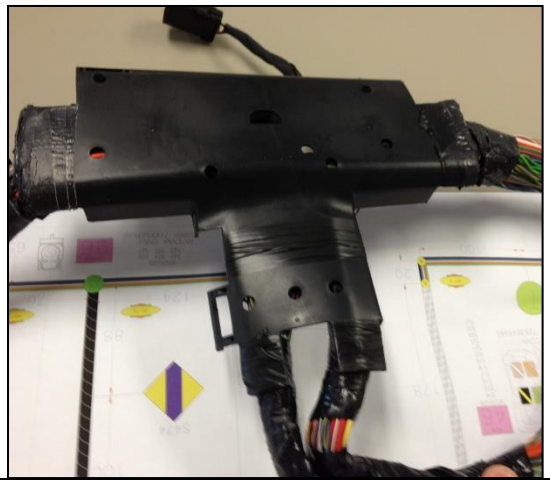
Design Develop Activities: 1
Item: 05

4.1.5 Use protectors with fixing tabs to fasten the wire bundle

DFM Concept:

All Protectors must have fixing tabs, ribs or tie strap fixing to fasten the wire bundle to improve dimensional accuracy during manufacturing.

OK



Correct Method:
Tape tabs help "anchor" protector to the harness.

NG



Incorrect Method:
No tape tabs on the protector.

Benefits/Effects:
This concept helps to ease the assembly process on the assy line. It is much easier for the operator to handle the harness within this area if there is adequate fixing for the bundle. This concept helps prevents the harness from slipping out of the protector and reduce dimensional accuracy after receipt of the product.

Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

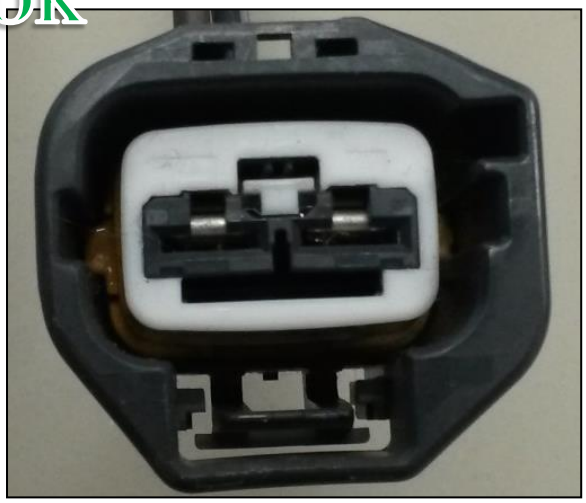
Design Develop Activities: 1
Item: 06

4.1.6 Use connectors that have considerations for circuit checkers

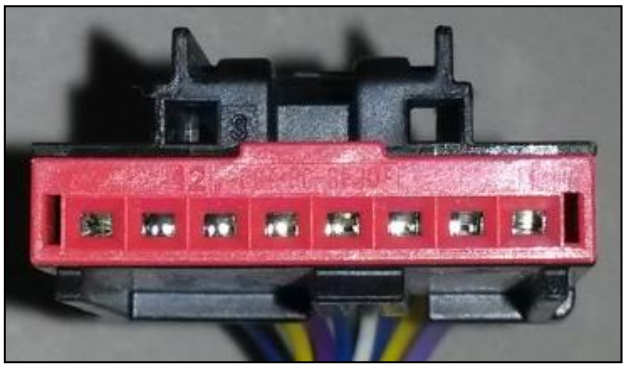
DFM Concept:

Avoid using connectors that do not have "windows" that allow for pogo-pin contact points for electrical checking.
The direct contact with female terminal spring causes spread terminals.

OK



NG



Correct Method:
The gap or "window" in the insulator will result in proper pogo pin/terminal contact.

Incorrect Method:
This insulator does not have a contact area, therefore the pogo pin needs to come in contact with the inside of the female terminal - Not preferred.

Benefits/Effects:
If you have connectors without these "windows" electrical checker fixture pogo pins may make direct contact to the terminal "spring". There is a risk for the terminal quality anytime contact is made on the inside portion of the terminal.

Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

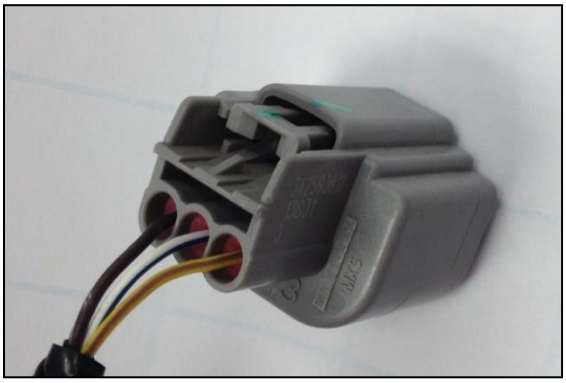
Design Develop Activities:	1
Item:	07

4.1.7 Avoid connector that prevents easy circuits insertion

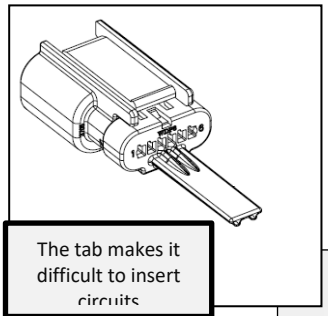
DFM Concept:

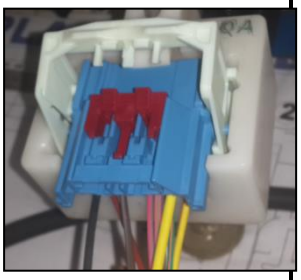
Don't use connector that prevents easy circuits insertion in order to avoid Terminal Push Out (TPO).

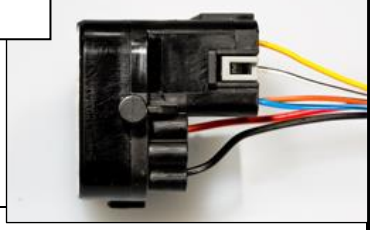
OK



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Correct Method:
Connector has a clear area to insert circuits.

Incorrect Method:
Non clear area in the back of the connector.

Benefits/Effects:
Difficulty in inserting circuits can result in terminals not seating fully and can cause potential Terminal Push Out (TPO).
Increased Tact Time due to extra processing (inspection).

Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

Design Develop Activities:	1
Item:	08

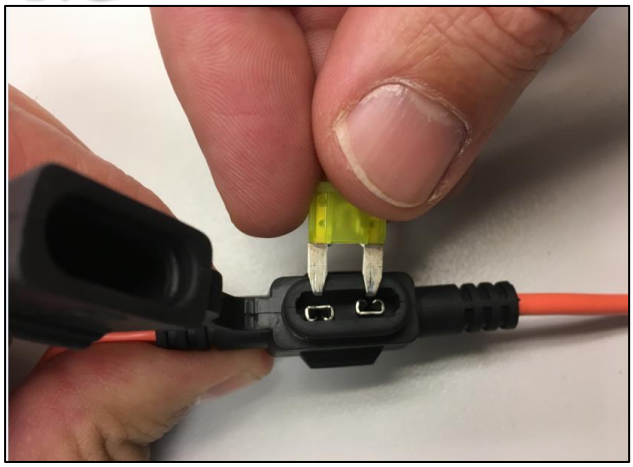
4.1.8 Avoid components that contain difficult assembly process

DFM Concept:

Avoid using connectors that requires additional steps such as fastening circuits using screws, bolts, insertion of fuses into mis-aligned terminals, etc..

This results in longer cycle time which typically isn´t captured in the quote.

NG









Correct Method:
Choose components that have minimal steps to assemble to minimize processing time.

Incorrect Method:
Components that have additional steps to process (bolts, screws and/or additional steps to assemble).

Benefits/Effects:
Cycle Time can exceed Takt Time.
Extra processing time in order to perform inspection during the assembly. Torque and terminal position are difficult to repeat for complex connectors that require a screwing operation. Special machinery/equipment may be required.

Assembly:
Cost:
Environment & Safety:
Quality:
High Voltage:

Design Develop Activities: 1
Item: 09

4.1.9 Molded-on grommets vs. applied grommets (pre-molded)

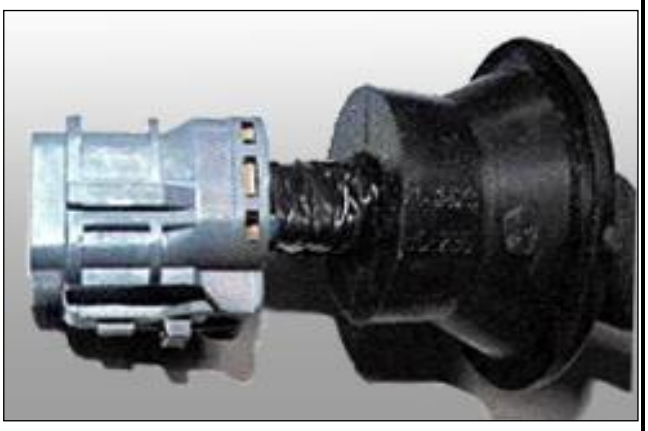
DFM Concept:

Unless required by the customer, avoid using grommets that have to be molded directly on the wiring harness. This applies to grommets being molded during W/H build or as a post process. If an overmolded grommet is required, the relationship between the bundle min/max diameter and tooling size must be understood.

OK



NG



Correct Method:
Pre-Molded grommets that can be applied without special mold tooling onto the wiring harness.

Incorrect Method:
Unless required by the customer, do not used molded grommets.

Benefits/Effects:
When using a mold on-grommet there is a risk to damage the wire during the molding process. Rework is not practical, therefore it is highly recommended to use pre-molded style grommets. Additionally, if a wire harness has different bundle sizes, unique mold tooling will be required for each W/H variant.
Working in process of a molded-grommet needs floor space and time to cool the material, preventive maintenance, waste, etc are all considerations that cause MUDA (waste) in our plants.

Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

Design Develop Activities:	1
Item:	10

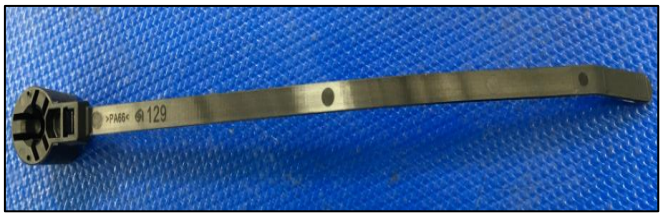
4.1.10 One-Piece Clip Selection

DFM Concept:

Avoid using two-piece clips. Using two-piece clips takes more time to assemble onto the harness.
When using two-piece clips, the mating process code is often not captured in the drawing nor associated to the labor assigned to the component.
Preassembled clips should be considered as one-piece clip and needs to be assured that they are not falling/disassemble during transportation.

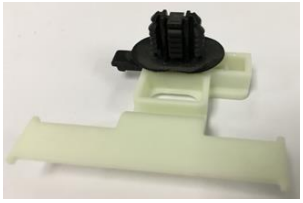
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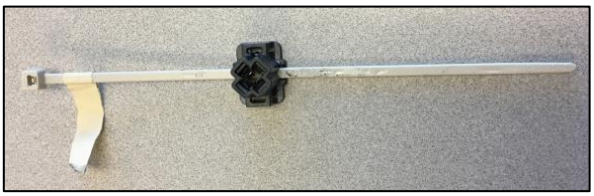




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Correct Method:
Using one-piece clips.

Incorrect Method:
Using two-piece clips.

Benefits/Effects:
During the handling process, two-piece clips can be un-assembled and may require unpaid extra time to re-assemble.

Takes more time to assemble onto harness.

Components are often received from the supplier in separate pieces and the drawing isn't calling for the mating process code to capture the labor.

Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

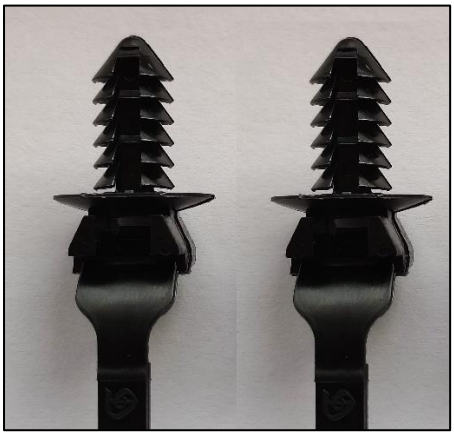
Design Develop Activities:	1
Item:	11

4.1.11 Avoid from using different stud diameter clips contained in the same W/H

DFM Concept:

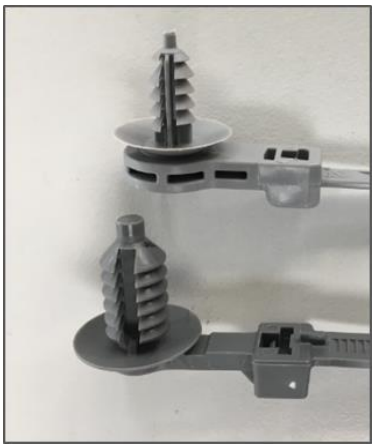
Commonized clips must be used when in the same W/H and in the same family. When proposing, the designer must consider the temperature class, material, color, and dimensional information.

OK



Correct Method:
Same stud diameter clips in the same W/H

NG



Incorrect Method:
Similar clips, but different stud diameter in the same W/H.

Benefits/Effects:
Have similar clip, same color but different stud diameter is very complex to detect and control the wrong installation.

Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

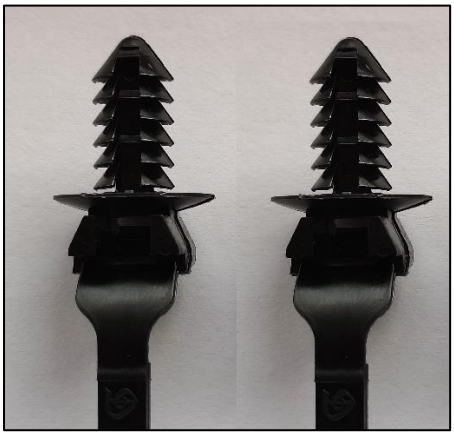
Design Develop Activities:	1
Item:	12

4.1.12 Avoid from using different stud length clips contained in the same W/H

DFM Concept:

Commonized clips must be used when in the same W/H and in the same family. When proposing, the designer must consider the temperature class, material, color, and dimensional information.

OK



Correct Method:
Same stud diameter clips in the same W/H

NG



Incorrect Method:
Similar clips, but different stud length in the same W/H.
Different part numbers with same hole and diameter.

Benefits/Effects:
Wrong clip installation.
Manufacturing does not prefer just changing color. Just changing color would require for manfuacturing to invest in color-detecting sensors. The designer must consider selecting the most cost effective clip unless function is compromised.

Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

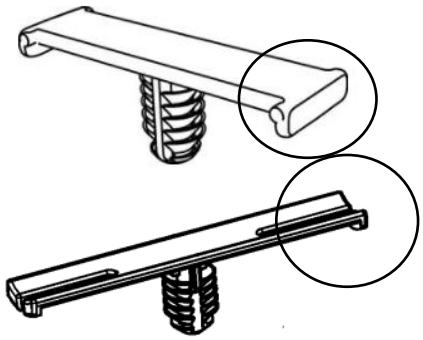
Design Develop Activities: 1
Item: 13

4.1.13 Use tape-on clips with raised ends ("wings")

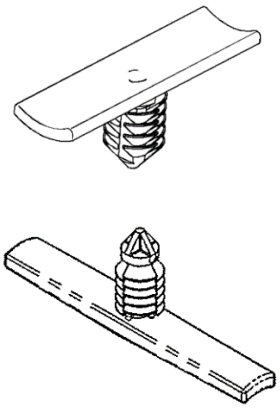
DFM Concept:

Avoid using tape-on clips without raised ends.
Clips with raised ends have been confirmed to comply with extraction forces. Clips without raised ends can potentially lead to a failure over time, due to tape deterioration and they can be detached from the fixing tape and from the W/H, causing loss of function.
If it's not possible to use a clip with raised ends the tape-on clip should be secured by cross tape instead of spot tapes to guarantee that the clip is properly fixed to the W/H.

OK



NG



Correct Method:

The spot tapes will be fixed by the clip's raised ends.
Retainers with raised ends are very robust and do not slip off the branch as the raised ends can catch on the tape.

Incorrect Method:

The clip can slip from the tape insulation over time as there are no raised ends to secure the spot tapes position.

Benefits/Effects:

Clips with raised ends are well retained in the W/H and therefore the risk of the clip detaching from the harness over time is very reduced.

Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

Design Develop Activities:	1
Item:	14

4.1.14 Washer tube selection

DFM Concept:

If washer tube is required, recommendation would be to select washer tube type/material as much rigid as possible.
Corrugated (instead of soft) washer tube should be recommended (better behavior to crushing /pinching).
Polyamide/Polypropylene (PA/PP) material should be important to improve behavior against twisted effect.
Also, the Design Engineer must always indicate problematic areas in the drawing and check that it cannot bend in the car with an angle shorter than 90 degrees.
Preferred case will always be to avoid the usage of washer tubes as part of the W/H if possible.

OK





NG





Correct Method:

Corrugated tube selected.

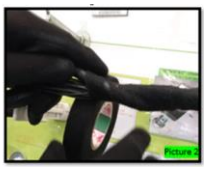
Incorrect Method:

Soft washer tube selected.

Benefits/Effects:

When Design Engineer select a washer tube with soft material taped together with long branches, we have the risk that tube can be twisted and squeezed. In certain areas can also bend and squeeze. If this cannot be avoided, the soft washer tube should not be attached together with wire harness circuitry wrapping.
This way with a rigid corrugated washer tube selected, any kind of taping method will not generate quality issues.







4.2 General

Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

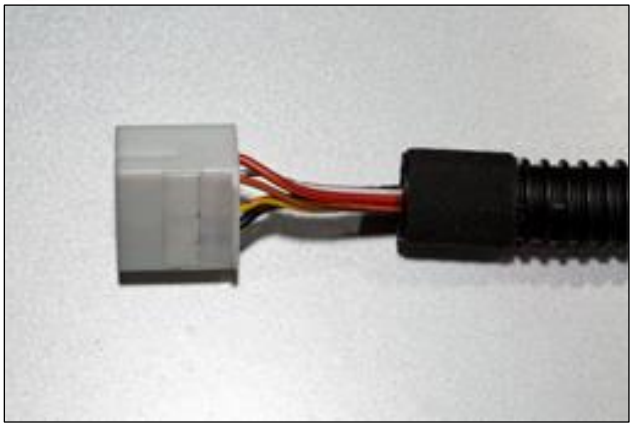
General: 2
Item: 01

4.2.1 Improper use of band clamps/tie straps

DFM Concept:

Tie straps/band clamps should not be used to fix any convolute. Use taping operations instead (spot taping/end wrap).

OK



NG



Correct Method:
Simple spot taping or end-wrapping should be used.

Incorrect Method:
Do not to use tie straps at the end of convolutes to fasten.

Benefits/Effects:
When you use tie straps instead of simple taping, you have a risk of dimensional problems due to tie straps/band clamps sliding during W/H shipment. You also reduce the quality "look" of the wiring due to the deformation of the covering causing it to slide during component application.

Assembly:		**General:**	2
Cost:		**Item:**	02
Environment & Safety:			
Quality:			
High Voltage:			

4.2.2 Improper fixing of clips through band clamp usage

DFM Concept:

a) Tape-on clips should be attached by using taping operations.

b) When customer still do want to have tie straps, the Designer should evaluate with Manufacturing if the Tape-on clip might need a covering underneath (i.e., taping, convolute, etc..) to avoid be either damaging the wires and/or sliding (see DFM 2.05).

OK



NG



Correct Method:
Tape is used for tape-on clips.

Incorrect Method:
Tie straps are used for tape-on clips. Damage to the wire is possible and dimensional stability is poor.

Benefits/Effects:
If you use tie straps to fix clips to a W/H you add assy operations that are not necessary. You also require a tool to secure and cut the clip. The goal is to pick the right clip and match it with the best labor operation. In most cases, tape-on clips are fixed with tape and strap clips are fixed with the fixing guns. Using tie straps could cause that the clip dimension to not be met due to loose fitting clip.

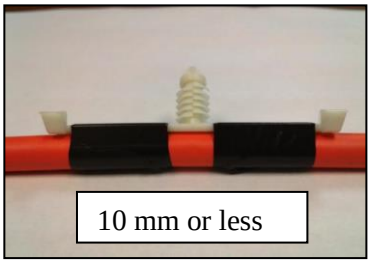
Assembly:		**General:**	2
Cost:		**Item:**	03
Environment & Safety:			
Quality:			
High Voltage:			

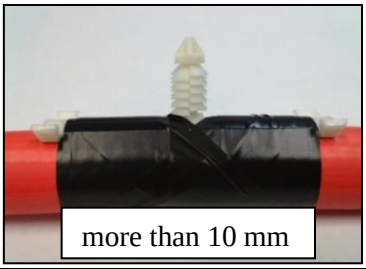
4.2.3 Taping method for clips / retainers

DFM Concept:

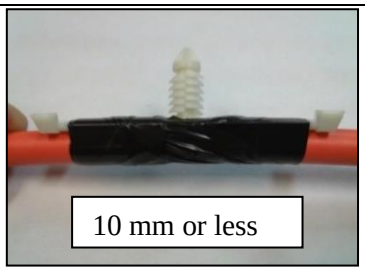
Use diaper/cross wraps on retainers as opposed to spot tape (provides better retention), depending on bundle size. Refer to the photos below for further detail.

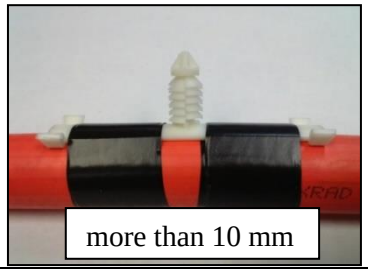
OK





NG





Correct Method:
Small bundles with spot tapes.
Large bundles with diaper/cross wraps.

Incorrect Method:
Small bundles with diaper/cross wraps.
Large bundles with spot tapes.

Benefits/Effects:
If diaper/cross wraps are used on small wire bundles to secure retainers, the tape will 'ride up' on the retention portion of the retainer, causing quality issues. For small bundles, individual tape ties (spot tape) should be used.

For large bundles, tape will provide better retention and have less total tape ends if a diaper/cross wrap is used. Also, for large bundles the tape will not ride up on the retention portion of the retainer, resulting in no quality risk.

Assembly:		**General:**	2
Cost:		**Item:**	04
Environment & Safety:			
Quality:			
High Voltage:			

4.2.4 Proper clip "tab" directions on strap clips for good workability

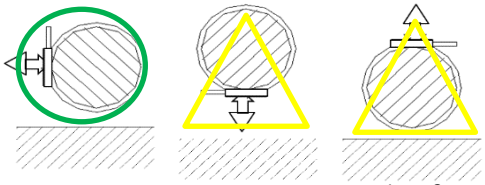
DFM Concept:
Please following the recommended guideline for clip tab directions:
Item 1 = Best
Item 2 = OK
Item 3 = OK
Item 4 = NG

The cross hatched area below the clips represents the assembly board.

For symmetrical clips, the designer should select the view in which the strap is facing towards the operator. For offset clips, the desiginer should select a clip which the strap is facing towards the operator. If this isn't possible, then the designer should select an offset clip which is fastened with tape.

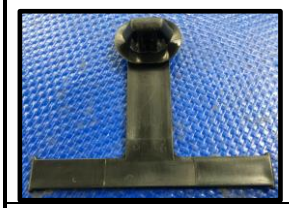
OK

For symmetrical strap clips:



Item 1 Item 2 Item 3

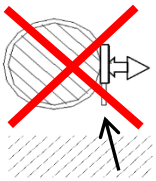
For offset clips:





NG

For symmetrical strap clips:



Item 4

For offset strap clips:



Correct Method:
Assure the clip tab direction to allow for easy "cut-off" during W/H assembly. In case for off-set strap clips, if the strap tail can be changed through a different clip selection, then, select an offset clip fastened with tape.

Incorrect Method:
Clip tab direction that are "under" the W/H or pointing in the "down" direction.

Benefits/Effects:
If the clip tabs are in the down direction or under the W/H bundle on the jig board, the operator has to remove the bundle to use the clip gun. This causes extra time to assemble the W/H and there is a risk every time you remove a portion of the harness from the jig holding fixtures.

Assembly:		**General:**	2
Cost:		**Item:**	05
Environment & Safety:			
Quality:			
High Voltage:			

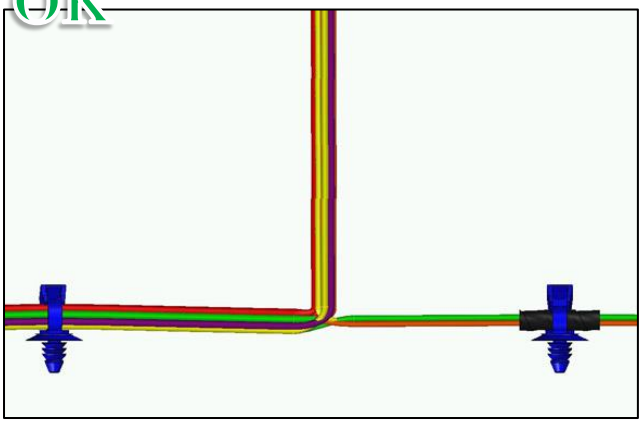
4.2.5 Strap clip fixing on harnesses and/or over Antenna/Specialty/Battery cables

DFM Concept:
Selected clip must be of the appropriate size so that it will be securely fastened to the wire harness bundle.

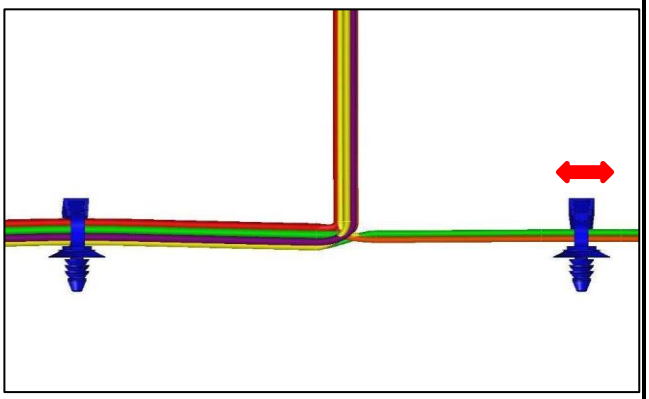
To avoid such unwanted movement, the engineer has to check the strap clip can be fixed appropriate. If this is not feasible, a spot tape needs to be added underneath.

In all unprotected wires, the Designer should check with Manufacturing if the Strap Clip might need a protection (as spot tape, etc) underneath to avoid be damaging the wires or that the clip be sliding.

OK



NG



Correct Method:
After manual testing the engineer added a spot tape to help the clip to stay in position.

Incorrect Method:
The strap clip can move along the small wire harness branch.

Benefits/Effects:
Using tape under critical strap fasteners will help to avoid quality issues caused by moving clips.
If the taping under the clip is performed right from the start of the harness development, then it will be part of the cost calculation.
Note: The underneath protection/taping has to be shown on the drawing in order to guarantee that it will be paid by the customer!

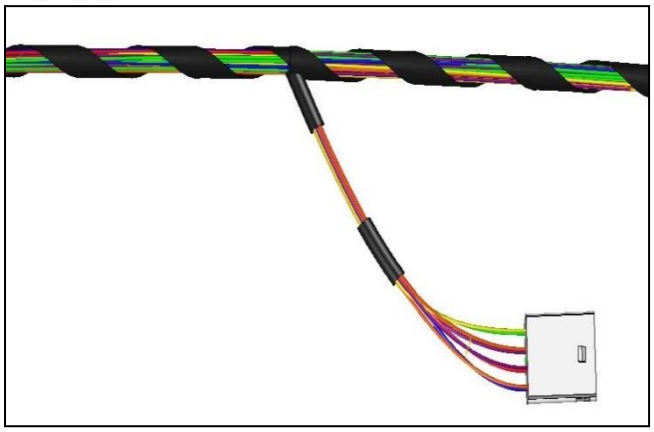
Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

General: 2
Item: 06

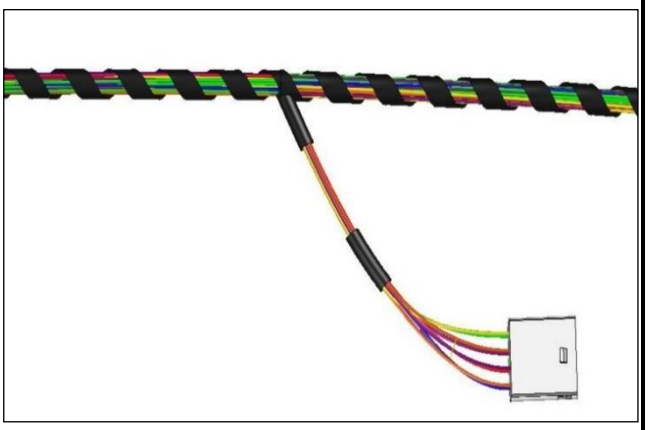
4.2.6 Preferred tape width

DFM Concept:
If applicable the design engineer should choose 19 mm tape.

OK



NG



Correct Method:
The engineer has chosen 19 mm tape.

Incorrect Method:
The engineer has chosen a smaller tape.

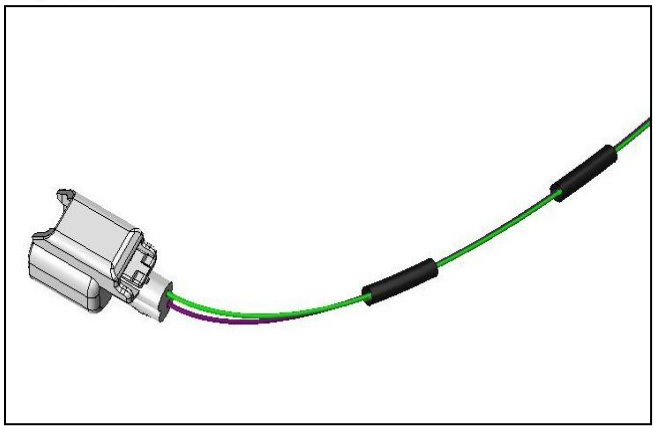
Benefits/Effects:
Complying to this DFM concept will help to ease the assembly process. Using just one type of tape will reduce the complexity and the space within the assembly area. Additionally, the handling for the operator is much more easier due to a reduced danger of picking up the wrong tape.

Assembly:		**General:**	2
Cost:		**Item:**	07
Environment & Safety:			
Quality:			
High Voltage:			

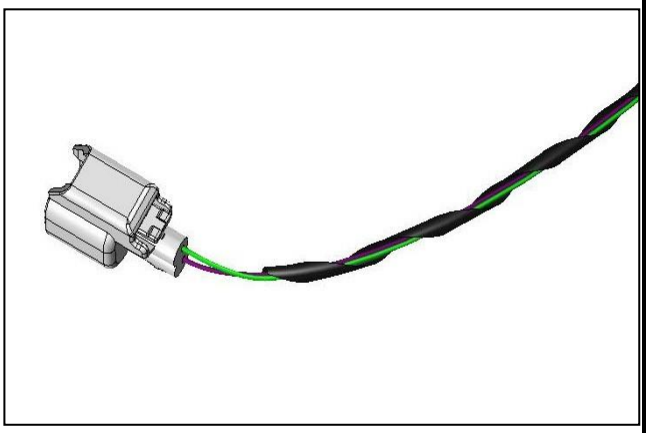
4.2.7 Taping on thin branches

DFM Concept:
On branches containing four or less thin wires the W/H design engineer should not specify spiral taping. Best solution in relation to manufacturing feasibility is to specify spot taping. If spot taping is not accepted by the customer, full taping has to be used.
Please note: Please review also DFM concept 2.08

OK



NG



Correct Method:
On a branch with two circuits the wiring harness design engineer has specified spot taping.

Incorrect Method:
On a branch with two circuits the design engineer has specified spiral taping. As visible on the picture the result of using spiral taping on thin branches is a poor quality finish. In most areas the tape sticks out.

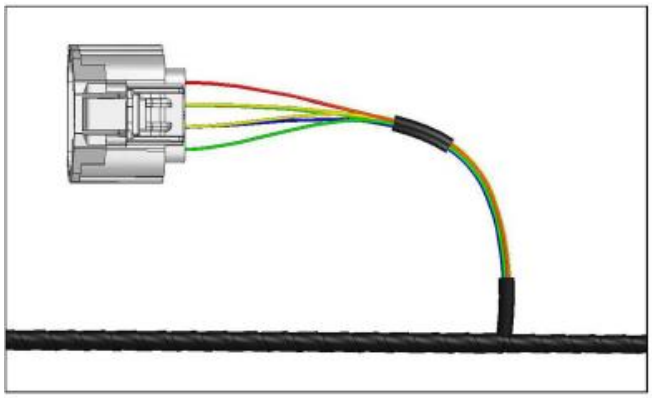
Benefits/Effects:
It is too difficult to wrap tape spirally around thin bundles, even more, if wider tapes are in use. When applying spot tape the process time will be decreased.
Additionally, in most cases it is not possible to meet the customers spiral taping specification. This could lead to potential quality issues because the part is not according to the drawing.

Assembly:		**General:**	2
Cost:		**Item:**	08
Environment & Safety:			
Quality:			
High Voltage:			

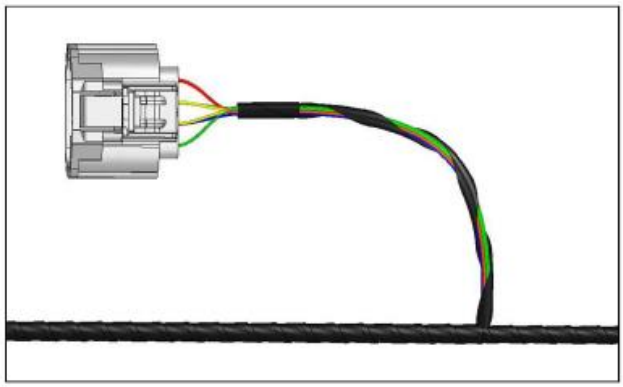
4.2.8 Taping on branches ≤ 150 mm

DFM Concept:
On branches ≤ 150 mm long the W/H design engineer must specify full taping or spot taping and not spiral taping. The preferred method is spot taping. When spot taping is not an option full wrap taping must be used. This is applicable for all types and widths of tape even when protection such as convolute and tube are used.

OK



NG



Correct Method:
The design engineer has chosen the best solution in relation to the manufacturing feasibility. On a branch not longer than or equal to 150 mm spot taping is used.

Incorrect Method:
On a branch < 150 mm the wiring harness design engineer has specified spiral taping. Some of the single wraps on the bundle are so close to each other that it looks like full taping.

Benefits/Effects:
In comparison to a spiral taping using spot tapes on branches < 150 mm is much more efficient. The choice would be to use spot tape if spot tape is not possible the next choice would be full taping. From the technical point of view, in most cases it is sufficient to use spot tapes on bundles < 150 mm to hold the wires together.
Additionally, in most cases it could be very difficult to meet the customer's spiral taping specification on short bundles. More so if wider tapes are used.

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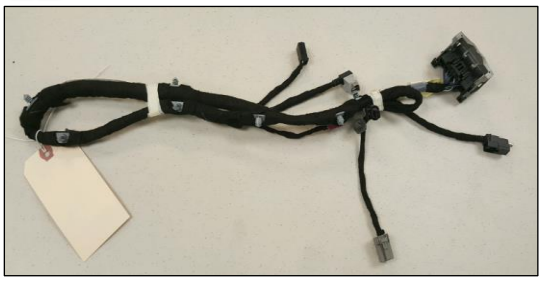
Assembly:		**General:**	2
Cost:		**Item:**	09
Environment & Safety:			
Quality:			
High Voltage:			

4.2.9 Tape Selection

DFM Concept:
1. Quantity of tape types: It is not efficient to change the tape when more than 2 types of tape are used in the drawing/same area. Especially to use only one tape per branch.

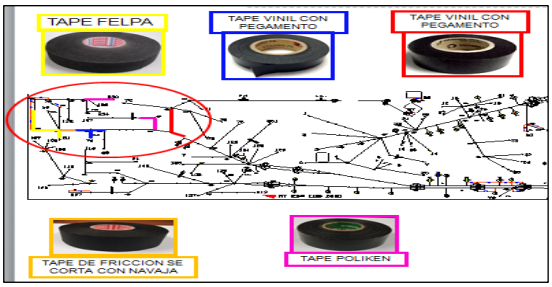
2. Tearable vs. non-tearable Tape: The type of tape manufacturing prefers is one in which can be torn by hand. Tape that requires tools (ring cutter) is not preferred.

OK



Use a maximum of 2 different tapes in the same drawing/area.

NG



Using 3 or more different tapes in the same drawing/area

Correct Method:
Maximum of 2 types of tapes in the drawing/area.

Incorrect Method:
More than 2 types of tapes in the drawing/area.

Benefits/Effects:
1. Compliance with this DFM concept will help to ease the manufacturing process. This DFM focuses on minimizing the tape usage is the same general area. Some regions are using a maximum of three (3) tape types: Standard tape, High Temperature tape, and Tape for marking.

2. Use of the ring cutter requires extra time and tool control.

Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

General: 2
Item: 10

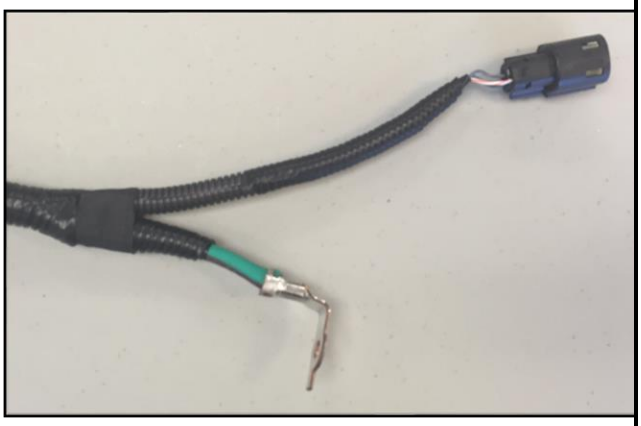
4.2.10 Use cross-taping (X, T, or P wrap) on all takeout points

DFM Concept:
Cross-taping must be used on all takeout points to secure the wires properly.

OK Prefered



NG Not Prefered



Correct Method:
Cross-taping on wire branch locations.

Incorrect Method:
Spot tape on wire branches.

Benefits/Effects:
Cross-taping is inherently stronger than a spot taping. When a spot tape is used to secure wire branches, those branches can move dimensionally during shipping and handling. Cross-taping will hold the wires more securely, preventing the wire movement during shipping, and avoiding potential quality issues.

Assembly:		**General:**	2
Cost:		**Item:**	11
Environment & Safety:			
Quality:			
High Voltage:			

4.2.11 Proper ID tag location

DFM Concept:
The position of an ID Tag on a W/H should be chosen under consideration of the following various points:
- The free space of the bundle should be wider than the width of the ID Tag
- ID Tag should not be assembled too close to connectors, fixing features or other harness parts.
- ID Tag should not be placed on branches which are optional content.
- ID Tag should not be placed on branches with a diameter which is too big to allow a proper fixation of the tag.
- Independently the ID Tag has to be placed according to customer requirements (e.g. non visible inside engine compartment)

OK



NG



Correct Method:
ID Tag with a width of 65mm positioned on a branch with an adequate length. Furthermore, there is no fixing feature located to close too the ID Tag.

Incorrect Method:
ID Tag with a width of 65 mm positioned on a branch with a length between clip and takeout of 60 mm! The ID tag is too close to the fixing feature.

Benefits/Effects:
It is much easier, efficient and requires a shorter process time for the manufacturing operators to assemble the ID Tag on a branch with an adequate length. It is important for the operator that the ID Tag is visible throughout the assembly process. Placing the ID Tag on a branch of the harness with an adequate space leads even to a reduced risk of damaging the ID Tag during shipping and handling processes. Moreover, if the ID Tag is too close to a takeout or fixing feature this situation leads to assembly issues at the customer side and therefore to customer complaints.

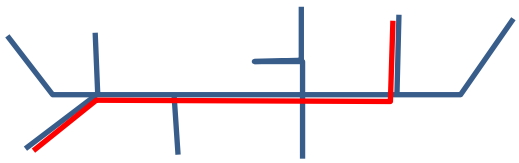
Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

General: 2
Item: 12

4.2.12 Commonize circuits names for the same functions across derivatives

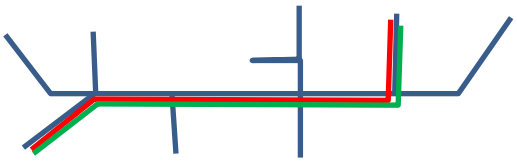
DFM Concept:
When a circuit has the same function and characteristics as length, terminals, seal, wire size, wire type they should be commonized across all derivatives.

OK



Circuit name	From	Cavity	To	Cavity	PT#1	PT#2
115	15139041_N7	14	TW46	1	O	O

NG



Circuit name	From	Cavity	To	Cavity	PT#1	PT#2
115	15139041_N7	14	TW46	1	O	-
115A	15139041_N7	14	TW46		-	O

Correct Method:
Having the same circuit (characteristics, routing, etc) in the drawings, with the same name for each part number.

Incorrect Method:
Having the same circuit (characteristics, routing, etc) in the drawings, but assigning different names for each part number.

Benefits/Effects:
It is best to have single work order for each name of circuit.

In MAE KOTEI process, this causes manufacturing to cut the same circuit with same terminals in different work instructions, having a lot of setups in the cutting machine, decreasing the efficiency in the production line.
In ATO KOTEI process, in sub-assy area, more space is needed to place every circuit according the work order assigned.

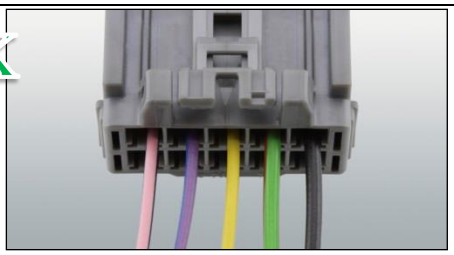
Assembly:		**General:**	2
Cost:		**Item:**	13
Environment & Safety:			
Quality:			
High Voltage:			

4.2.13 Proper circuit distribution within a connector

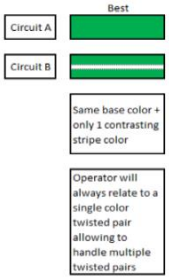
DFM Concept:
1) It is essential that identical base wire colors are not next to each other or above/below cavities in the same connector. Whenever possible, the wire color should be changed (as shown in the "OK" section below).
Exception: The same wire color is acceptable only when the wires are minimum of two wire sizes lower/higher from each other (e. g. 1.0mm² with 0.5 or 2.5mm² = OK; 1.0mm² with 0.75 or 1.5mm² = NG). If this isn't possible, then the designer should check with Manufacturing to add a marking/paint dot color to one end of the circuit.

2) It is essential that identical base wire colors with similar stripe colors are not next to each other or above/below cavities in the same connector. For twisted pairs, manufacturing prefers to have same base color plus only one contrasting tracer color (see diagram below).

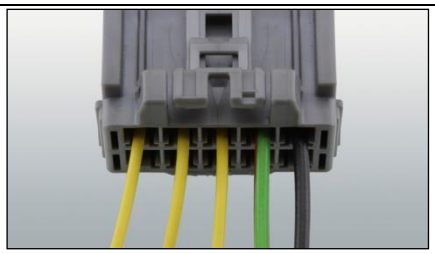
OK



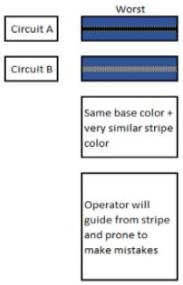
For twisted pairs only:



NG



For twisted pairs only:



Correct Method:
Wire colors are different in the same connector.

Incorrect Method:
Wires of same size and color in the same connector.

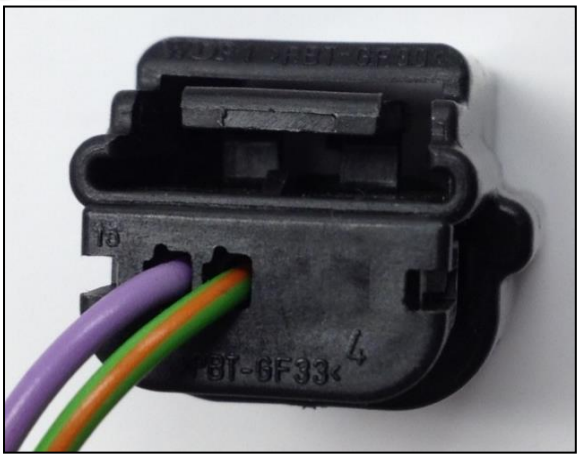
Benefits/Effects:
Having the same color wires in adjacent cavities requires special handling in W/H manufacturing either by process timing, process location or a physical marking of the circuit. Final electrical check should find the error, but the result is rework and corrective action plans to prevent/reduce the problem. All involve the reduction of an efficient W/H production process.

Assembly:		**General:**	2
Cost:		**Item:**	14
Environment & Safety:			
Quality:			
High Voltage:			

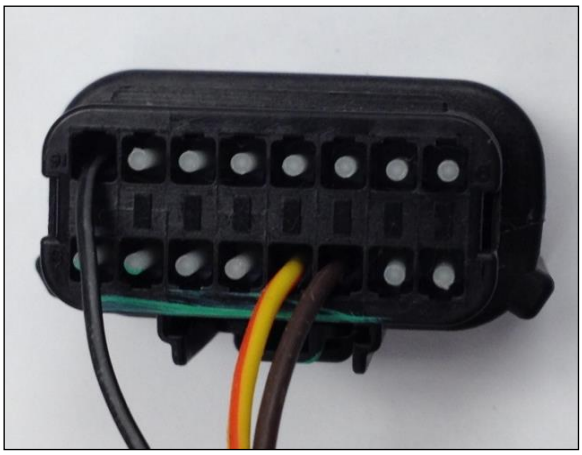
4.2.14 Use pre-plugged connector

DFM Concept:
For sealed connectors, all unused cavities need to be pre-plugged. Cavities that are not used in any part number for the same family should be pre-plugged from the supplier. Manufacturing requires to have a maximum of 3 flashed/stamped configurations for the same connector per branch, per production line of the same family wiring harness as it will be difficult to manage on the production shop floor. The designer must assure minimal plugs are used across all connector complexities and confirm with manufacturing.

OK



NG



Correct Method:
All non-used cavities are closed from the supplier.

Incorrect Method:
Open cavities are closed by plugs.

Benefits/Effects:
Having to use dummy circuits (plugs) could result in leaking problems.Extra process to install plugs (tooling operations, floor space).
Increased cycle time.Add new inspections points.
Best Practice and option it is to use pre-plugged connector but if this can't be done, avoid blunt cuts or dummy wires and revise only if it is an inline connector the possibility to use the same circuit ("jumping wire") and occupy 2 cavities at the same time in order to reduce the amount of this type of circuit. In this way, it can be eliminated the need to cut more wires and the usage of a shrink tube to seal the end.
Even if the Designer add the 'wire bridges' on inline connections, it is responsibility from him/her to check what circuits could potentially be on the mating side to avoid wrong electrical connections as well as to ask to Manufacturing if they do prefer to keep the dummy circuits since sometimes the "jumping wire" is very difficult to be inserted in certain connectors



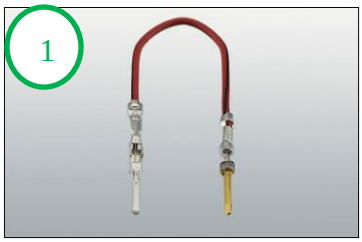


Assembly:		**General:**	2
Cost:		**Item:**	15
Environment & Safety:			
Quality:			
High Voltage:			

4.2.15 Avoid similar terminals on single circuits

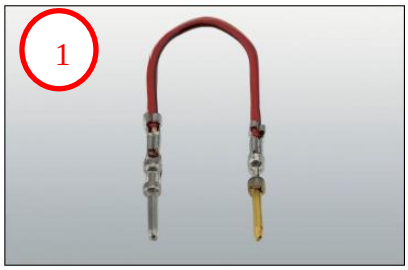
DFM Concept:
Circuits should have either identically configured and interchangeable terminals on both ends of the same circuit or two non-interchangeable terminals, to avoid the possibility of cross termination. Interchangeable terminals with different visual features like: plating, grease, solder, requirements (or not interchangeable terminal design like one piece terminals) could be incorrectly assembled without detection during electrical check. If this isn't possible, then the designer should add a visible check like a marking/paint dot color to one end of the circuit. However, this must be avoid through doing a redesign of the terminal or commonization at both ends.

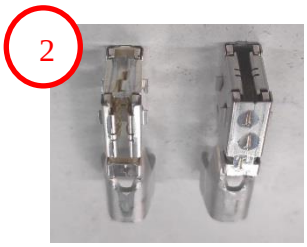
OK





NG







Correct Method:
The terminals with the different plating are not interchangeable. (1)

Incorrect Method:
For the same wire, the gold plated terminal on one end and tin plated terminal on the other end. The terminals are identical except the plating (1)
90º terminal is very similar to the 180º terminal. Since harness uses both terminals that are interchangeable, the terminal was inserted on the wrong connector (180º terminal inserted on the 90º connector) (2)

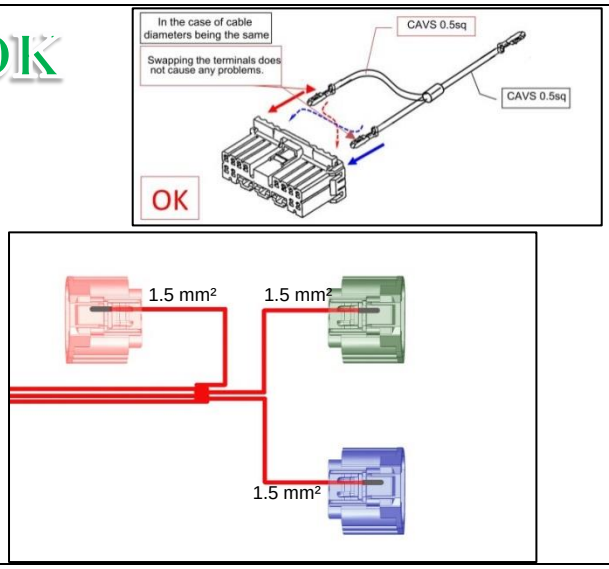
Benefits/Effects:
Using non-interchangeable terminals removes the "visual only" identification on circuits with different plating and distinguishes the proper ends of the circuit to the correct connector. Unique and non-interchangeable terminals create a full proof process that prevents the terminal from positively fitting into the wrong connector.
This error could not be identified during the electrical wire harness check because continuity testing will not detect incorrect plating, grease, solder or terminal type.

Assembly:		**General:**	2
Cost:		**Item:**	16
Environment & Safety:			
Quality:			
High Voltage:			

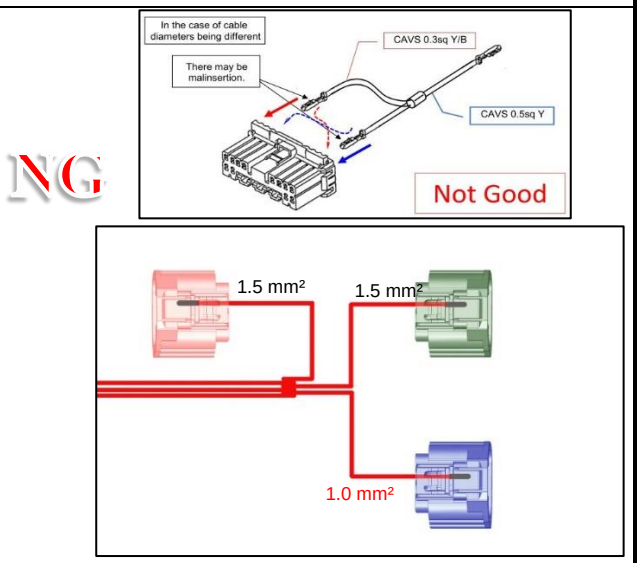
4.2.16 Same potential, similar wire size / different plating same terminal family

DFM Concept:
If multiple circuits are connected to the same potential (e.g. coming from the same splice, or double crimp) and use the same terminal type and wire color, those wires must be exactly the same.
This means wire sizes, platings must be the same as well as application of solder, or grease.
If wires have similar sizes or different platings that goes into connectors/cavities (e.g. at relay or busbar sockets), it is possible to swap the wires.
This kind of mistake cannot be detected during an electrical test.

OK



NG



Correct Method:
Analyze jointed wires to make sure to have the same terminal plating and same wire size.

Incorrect Method:
Use of same terminal type, but different plating and similar wire size.

Benefits/Effects:
Improved quality.
Terminals that are the same except for plating, solder, or grease may look the same. Neighboring wire sizes cannot be identified by the operator and can be inserted into the connector in the wrong location. There is no way to test for this condition which could lead to a quality issue for our customer.

Assembly:		**General:**	2
Cost:		**Item:**	17
Environment & Safety:			
Quality:			
High Voltage:			

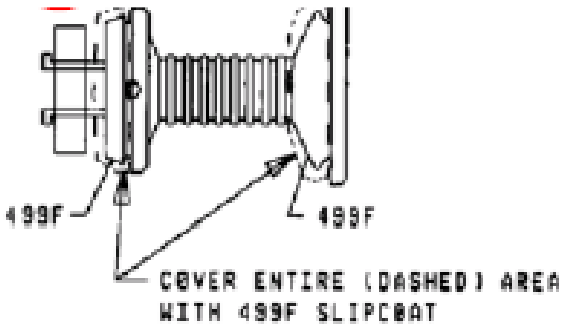
4.2.17 Avoid using grease and /or lubricants

DFM Concept:
Avoid using grease or lubricant. These can spread through the harness during handling and packaging.
If you must use lubricants, recommend using a "plastic bag" to keep the lubricants from spreading to other parts of W/H.
When this happens, the designer should add lubricant process to the drawing.

OK

Investigate components that do not require any lubricants/grease.

NG



Correct Method:
Connector, terminals and grommets that don't require lubricants to be present on the harness for installation.

Incorrect Method:
Adding lubricants as shown above that must be present for the customer.

Benefits/Effects:
Avoiding these secondary operations makes for a cleaner processing of the harness throughout assy. As these processes can only be visually checked, the chances of error by a missed operation, or insufficient application are quite high. Also, throughout the course of using these lubricants additional maintenance is required for the surrounding areas.

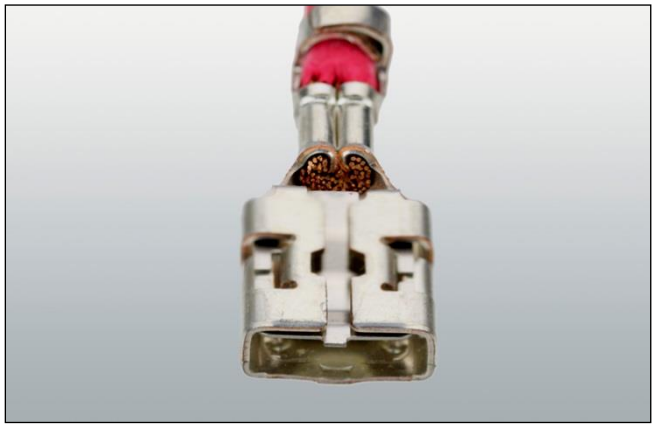
Assembly:		**General:**	2
Cost:		**Item:**	18
Environment & Safety:			
Quality:			
High Voltage:			

4.2.18 Avoid using solder on terminal crimps

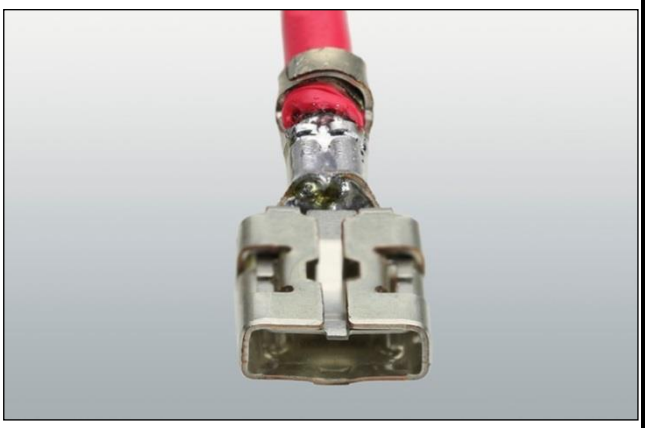
DFM Concept:
Choose only terminals where a crimping operation is sufficient. Check in detail if the proposed terminal is applicable for the current load, the used wire size, the maximum allowed pull out forces and if the Yazaki TCSS (Terminal Crimping Standard Specification) requires solder for the terminal.

Exception: Solder on terminal crimps is allowed on applications where a higher heat robustness is required (e. g. battery cable within exhaust area or in waterproof area).

OK



NG



Correct Method:
A terminal that is applicable for the current load and the used wire size. A crimping operation for this wire/terminal combination is sufficient.

Incorrect Method:
This is a solder to the wire/terminal combination

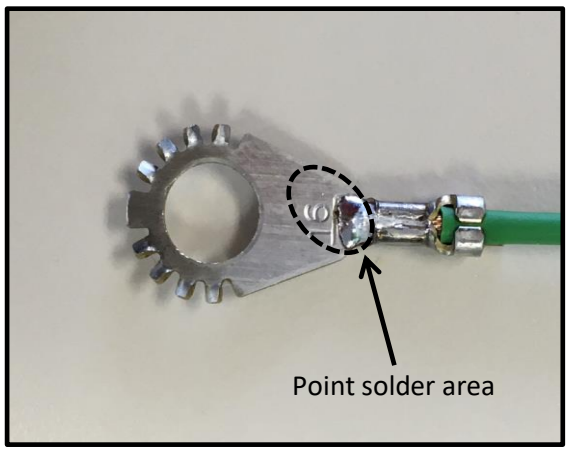
Benefits/Effects:
There is a cost benefit using terminals where a crimping operation is sufficient because reducing the process / labor time, equipment, inventory, material and storage space. The process monitoring is easier to handle. This will improve the quality of the final product. Finally the working environment will be better. Avoiding solder with hazardous material like lead will help to meet the EU directive 2000 / 53 / EC (End of Life Vehicles).

Assembly:		**General:**	2
Cost:		**Item:**	19
Environment & Safety:			
Quality:			
High Voltage:			

4.2.19 Avoid dip solder

DFM Concept:
Whenever possible or not required by the customer, avoid dip soldering as a post crimp operation.

OK



NG



Correct Method:
Select terminals qualified for use without solder for the wire type and gauge size specified. If required by the customer, point soldering may be necessary. However, only allow point solder in the area indicated above.

Incorrect Method:
Requires dip solder after crimping.

Benefits/Effects:
The elimination of a hazardous material and a cleaner work environment within the plant.
The use of proven/validated crimps will yield quality improvements through repeatability of solder free crimps. Crimps will be more robust if they are not required to be exposed to the high temperatures of the solder bath.

Reduce labor and material cost, cleaner work environment.

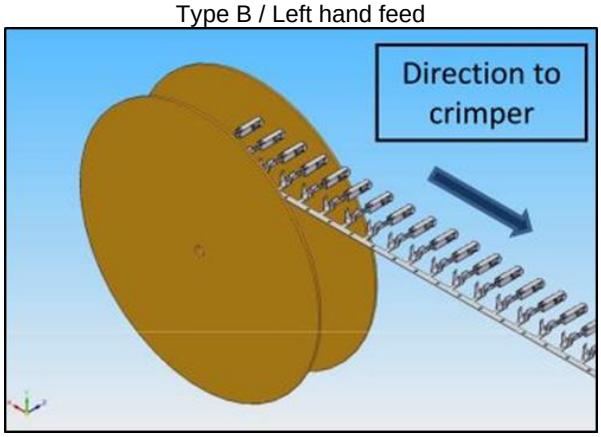
Assembly:		**General:**	2
Cost:		**Item:**	20
Environment & Safety:			
Quality:			
High Voltage:			

4.2.20 Terminal feeding from LHS

DFM Concept:
For an automatic crimping machine, terminals are needed which are coiled on a wheel. Many terminals are available in two different options (means two different part numbers): Coiled for left hand side feeding or right hand side feeding. Yazaki preferred is left hand side feeding and the version which is shown below: Terminals on top running to the right side of the wheel. Spin direction of the wheel is clockwise. Even the left hand side feeding with terminal running from below and spin direction of the counter clockwise is not preferred.
If the designer has the chance to decide between left or right hand side feeding, left hand side feeding has to be chosen.

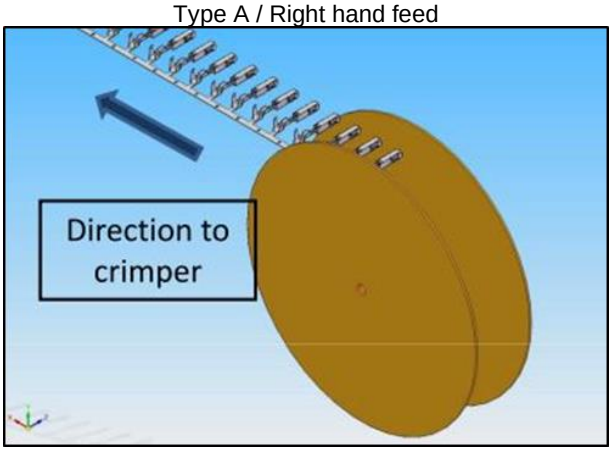
OK

Type B / Left hand feed



NG

Type A / Right hand feed



Correct Method:
The design engineer choses the terminal part number which stands for left hand side feeding.

Incorrect Method:
The design engineer chose the terminal part number which stands for right hand side feeding. This means that the terminals have to be uncoiled and then coiled again to be useable for left hand side feeding.

Benefits/Effects:
By choosing the correct terminal part number which allows left hand side feeding, there is no need of changing the terminal direction by first uncoil and then recoil them.
This saves labor time as well as a potential damage of the terminals during coiling can be avoided.
Please note: Not all terminals have different part numbers for left or right hand side feeding available.

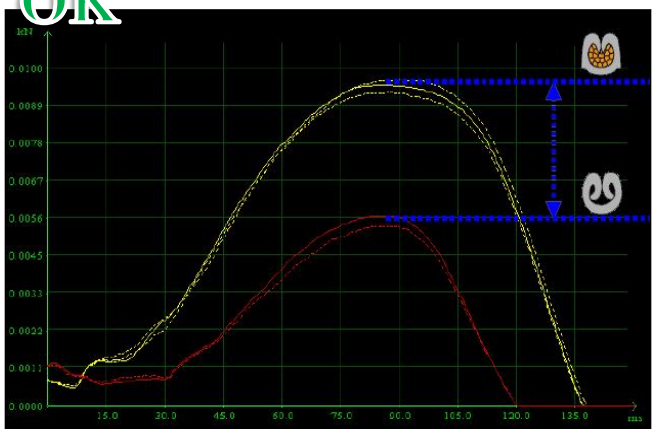
Assembly:		**General:**	2
Cost:		**Item:**	21
Environment & Safety:			
Quality:			
High Voltage:			

4.2.21 Crimp force monitoring: Phantom errors based on small headroom

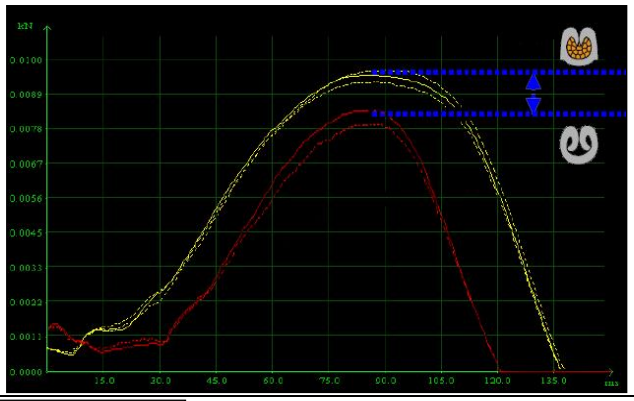
DFM Concept:
Check with the plant if during the first prototype builds extraordinary numbers of phantom defects occurred. These phantom defects have to be based on a small Headroom and high relative spread. If yes, then the design center and the plant have to check the possibility to change the affected terminal wire combination (change wire or terminal).

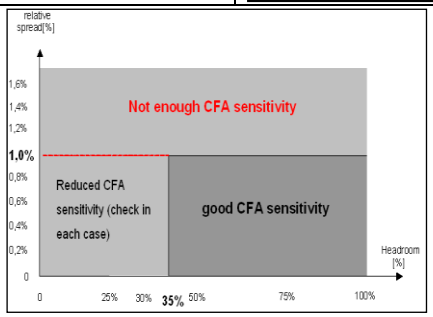
Phantom defects: Good crimps rejected by the CFM system.

OK



NG





Correct Method:
Good CFA sensitivity: The monitoring of crimping process is high-reliable, very good CFM sensitivity; tighter tolerances can be used to monitor very sensitive crimping defects. In order to identify this characteristic, these 2 conditions need to be observed:
Headroom: ≥ 35 % & Relative Spread: ≤ 1.0 %

Incorrect Method:
Reduced CFA sensitivity:
Headroom: < 35 % & Relative Spread: ≤ 1.0 %

Not enough CFA sensitivity:
Relative Spread: > 1.0 %

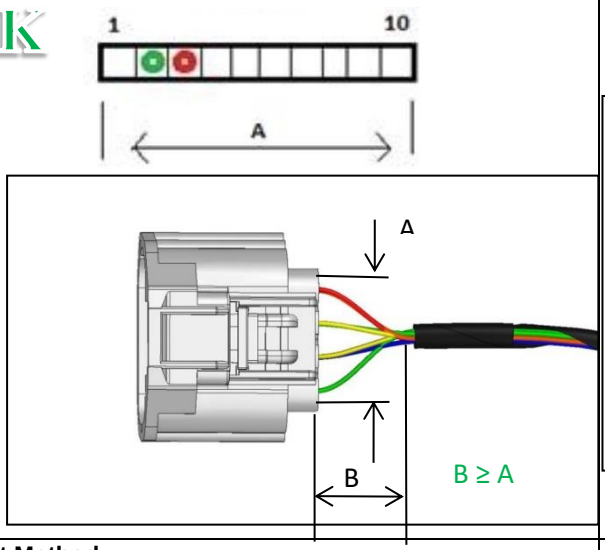
Benefits/Effects:
By changing to crimp combinations with an accurate headroom the number of phantom defects can be reduced. The reduced number of phantom errors will lead to a more robust and lean W/H production.

Assembly:		**General:**	2
Cost:		**Item:**	22
Environment & Safety:			
Quality:			
High Voltage:			

4.2.22 Minimum distance between tube / taping and the back of a connector

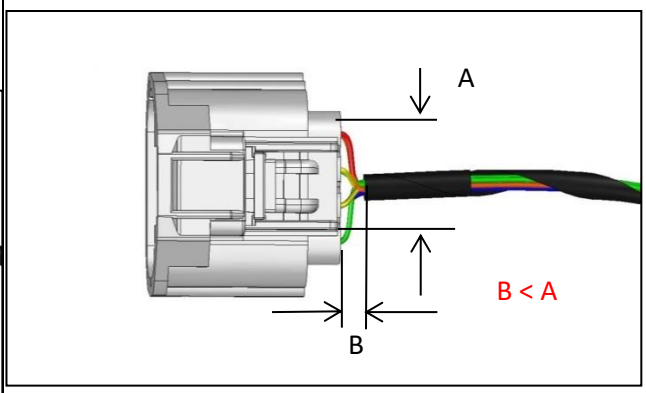
DFM Concept:
Whenever it is not required by the customer (customer beautification requirements) the minimum distance between the back of a connector and a tube or taping (distance B) should be at least greater or similar than the maximum spread of the wires across the back of the connector (distance A). If connector has a cover, it is not applicable.

OK



Correct Method:
Distance between connector and taping is according to the relation B ≥ A.

NG



Incorrect Method:
Distance between connector and taping is not according to the relation B < A.

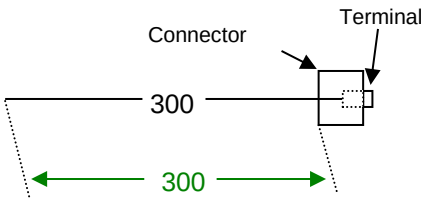
Benefits/Effects:
A sufficient distance between a connector and a tube/taping eases the assembly process. Furthermore, a proper gap prevents especially the outermost circuits from unwanted forces and prevents stretched seals in the back of sealed connectors. Stretched seals can also be the cause of non-waterproofness.

Assembly:		**General:**	2
Cost:		**Item:**	23
Environment & Safety:			
Quality:			
High Voltage:			

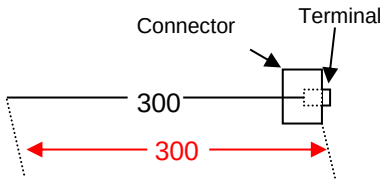
4.2.23 Define a clear measuring point at the connector

DFM Concept:
If it is in accordance with the used CAD system and with the customer requirements, the measuring point of a connector has to be chosen at the back or at the front of a connector, but avoid a mix of measuring points.
If the back of the connector is not clearly defined (e.g. eyelets or connectors with a protection cover) the measurement point has to be clearly defined (e.g. with a detail at the drawing) and the plant has to be informed.

OK



NG



Correct Method:
The chosen measuring point is at the back of the connector.

Incorrect Method:
The measuring point was chosen once at the front of the connector and once at the beginning of the terminal.

Benefits/Effects:
Following this DFM concept simplifies the handling for the manufacturing process. Choosing e.g. the crimp area of the terminal as the measuring point makes it difficult for final inspection.

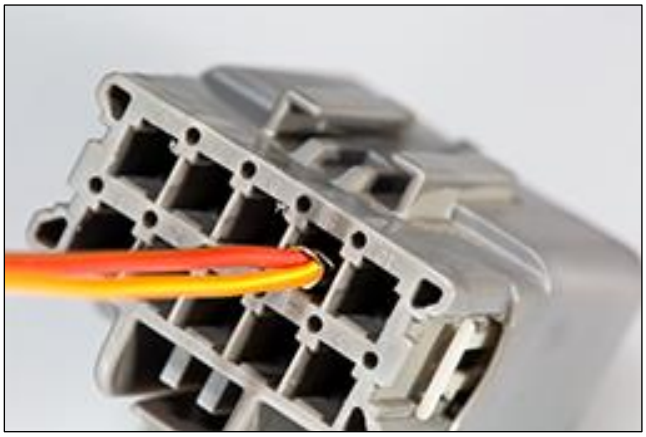
Assembly:		**General:**	2
Cost:		**Item:**	24
Environment & Safety:			
Quality:			
High Voltage:			

4.2.24 Appropriate terminal-wire-housing selection

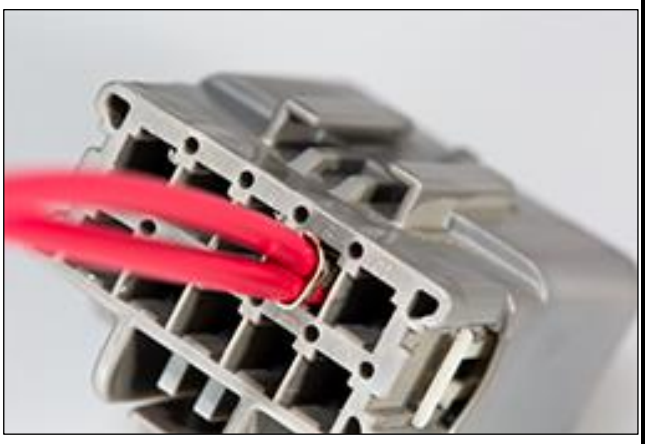
DFM Concept:
Only choose terminal-wire combinations that will fit into the connector cavity properly when crimped.
Especially when using a double crimp design, please verify if the terminal-wire combination will fit properly into the connector cavity housing.

Avoid the need to use a special tool to ensure the terminal is fully seated which increases the risk of Terminal Push-Out (TPO), increases the takt time and additional tool investment not being paid.

OK



NG



Correct Method:
Choosing a terminal-wire that fits properly into the connector when crimped.

Incorrect Method:
Choosing a terminal-wire combination that doesn't fit properly into the connector cavity when crimped.

Benefits/Effects:
Verifying the terminal-wire combination to insure that it will fit into the connector cavity properly, will prevent damage to the wire and terminal. Reduce terminal tool investment and exceed the standard takt time of circuit insertion process.

Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

General:	2
Item:	25

4.2.25 Delivering condition of Eyelets

DFM Concept:
Avoid using Eyelets as single part, use instead Eyelets on reel when both are available. This applies only to Low Voltage which it is wire size equal or less than 6mm2. It is not applicable for tubular eyelet terminals.

OK



NG



Correct Method:
The preferred method is to use Eyelets with the delivering condition on reels.

Incorrect Method:
The use of Eyelets as single part causes quality problems.

Benefits/Effects:
Eyelets on reels will assure everytime the same position of Eyelets, so the production area will have less quality problems and will save time, because not every single Eyelet has to be put in correct direction. The automatic crimp process with Eyelets assures an higher quality.

4.3 Layout

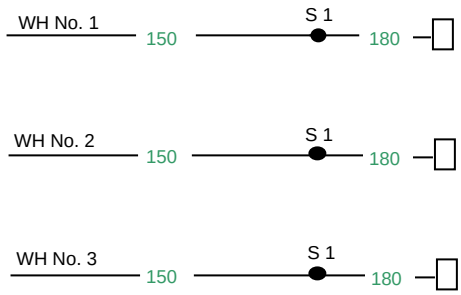
Assembly:		**Layout:**	3
Cost:		**Item:**	01
Environment & Safety:			
Quality:			
High Voltage:			

4.3.1 Commonization of splice location (same splice within different part numbers)

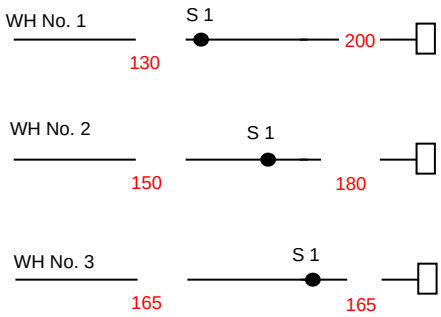
DFM Concept:

A splice that is common to any or all variants must be located in the same place for all variants across all drawings for the same family on the same vehicle line.

OK



NG



Correct Method:
The splice is in the same place on all variants.

Incorrect Method:
The splice is in different places on each variant.

Benefits/Effects:
Having the splices in the same place on all variants will reduce the number of cut leads and commonizes the board design and work instructions.

Assembly:
Cost:
Environment & Safety:
Quality:
High Voltage:

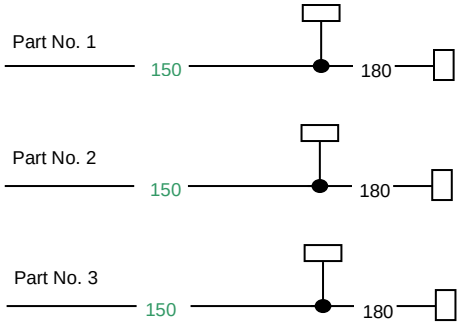
Layout: 3
Item: 02

4.3.2 Branch length commonization between part numbers within the same family

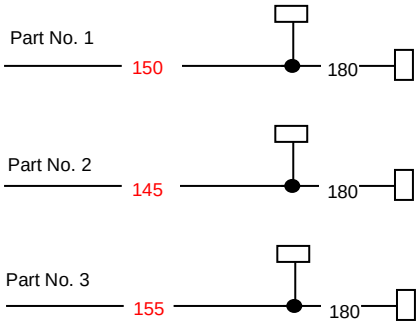
DFM Concept:

Branches across W/H families should not have unique part numbers. (e.g. window motor takeout should be the same length across all door W/H part numbers drawings)

OK



NG



Correct Method:
Branch is same for every part number in a family

Incorrect Method:
Unique lengths of a branch based on W/H part number.

Benefits/Effects:
It is important to be able to use a single jig fixture for connectors on a jig board that can build multiple W/Hs. The risk of misbuilds caused by different part numbers on the same jig board is significantly reduced by following this requirement.

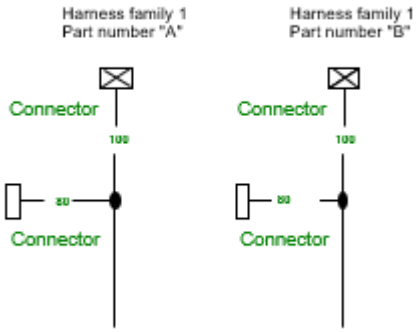
Assembly:		**Layout:**	3
Cost:		**Item:**	03
Environment & Safety:			
Quality:			
High Voltage:			

4.3.3 Use same drawing layout within one harness family

DFM Concept:

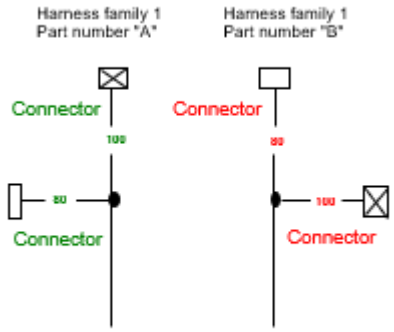
The position and arrangement of components (connectors, junction, or fuse boxes, etc.), takeout directions and fixing devices has to be the same on each drawing of one harness family.

OK



Correct Method:
Arrangement of takeouts and connectors is the same on both part numbers!

NG



Incorrect Method:
Even though there is no technical requirement given, the arrangement of the takeouts and connectors are different in both part numbers.

Benefits/Effects:
Using the same position and arrangements of components, takeouts etc. eases and increases the efficiency of the CAE work of creating the jig board layout. Worst effect of different arrangements on drawings can be the creation and usage of different jig boards where it is not necessary from a technical point of view!

Assembly:
Cost:
Environment & Safety:
Quality:
High Voltage:

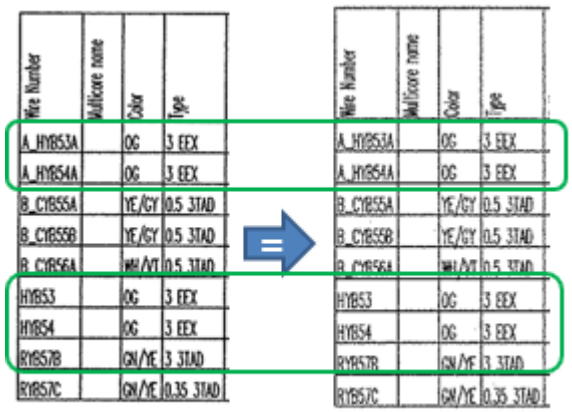
Layout: 3
Item: 04

4.3.4 Commonize wire type within families

DFM Concept:

Try to commonize wire type and wire size, wire colors of the same circuit with in a family.

OK



IP Harness A IP Harness B

NG



IP Harness A IP Harness B

Correct Method:
Same wires all versions

Incorrect Method:
Different wires all versions

Benefits/Effects:
The different wire types make manufacturing inefficient because the wire cutting machine must be set up differently for the different wire types. This also introduces the possibility that the wrong wire type will be used on the wrong harness level (a potential quality risk).

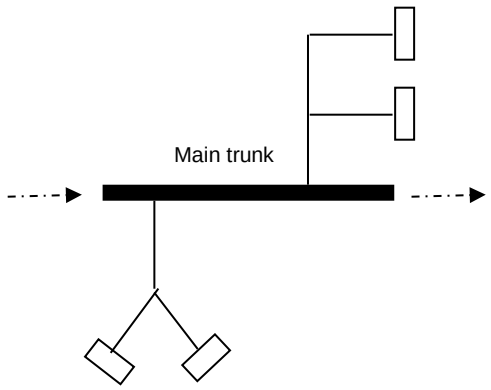
Assembly:		**Layout:**	3
Cost:		**Item:**	05
Environment & Safety:			
Quality:			
High Voltage:			

4.3.5 Position of main trunk on W/H drawing

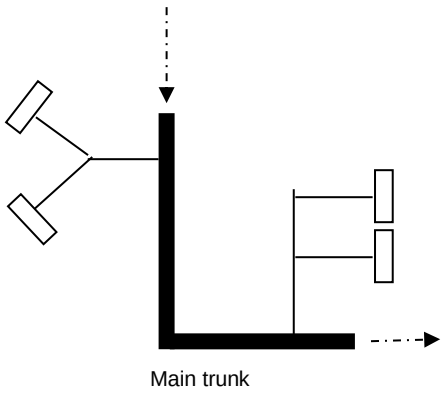
DFM Concept:

If there is no technical requirement, the position of the main trunk on a W/H drawing has to be located in the center from the left side to the right side of the drawing. Additionally, the component with the highest amount of circuits may also be placed on the left side of the drawing.
Definition "main trunk": Branch within a W/H with the most circuits.

OK



NG



Correct Method:
Main trunk was positioned straight from the left side to the right side of the drawing!

Incorrect Method:
Main trunk was positioned without any concept on the drawing.

Benefits/Effects:
One working step during the preparation of the jig board layout is to arrange the main trunk on the jig board drawing. This branch is laid into the center of the drawing starting from the left side. Reason for this is to assure a sufficient "working height" for the operator at the jig board. Therefore, the taping process is more efficient and ergonomically starting from the left side! This concept helps reducing time for preparing the jig board layout.

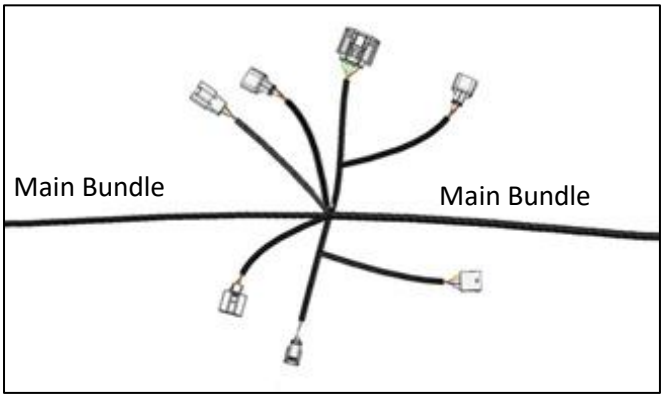
Assembly:		**Layout:**	3
Cost:		**Item:**	06
Environment & Safety:			
Quality:			
High Voltage:			

4.3.6 Maximum number of branches at a single branch point

DFM Concept:

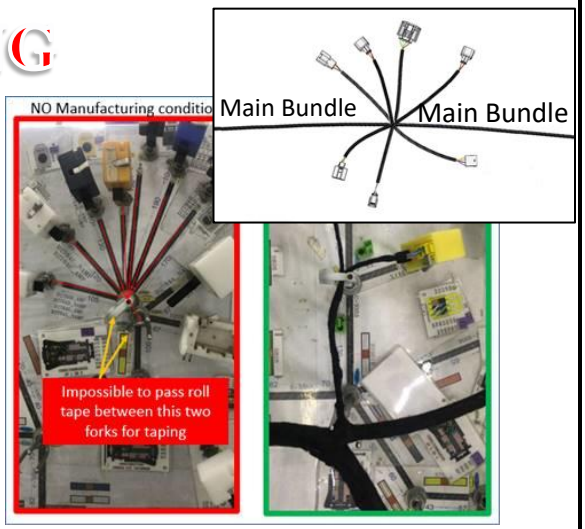
Target the maximum number of takeouts (branches) at a single point to be 5 or less branches additional to the main bundle. Additional to this, there cannot be more than 3 branches on one side of the main bundle.

OK



Correct Method:
Limiting the number of branches at a single point to 5 or less.

NG



Incorrect Method:
Having more than 5 branches at a single point.

Benefits/Effects:
In the event you have the excessive number of branches at one point, manufacturing has a difficult time with the density of build "fixtures" all in one area. Hand space, and jig board fixture space are critical points to high efficiency W/H manufacturing. Also, jig boards that have densely populated areas of branches require extra time to route, tape and apply components to. The basic goal is to keep as much room as possible between branches and components.

IMPORTANT NOTE: There are times, that there are taping areas in the harness in which we only have to tape very short areas that not even the tape roll passes through. In this case, the proposal is to change the taping node onto the main branch and not segmented into 2 branches and then the “V”. Minimum length range recommendable in which can tape without any issues is 135mm.

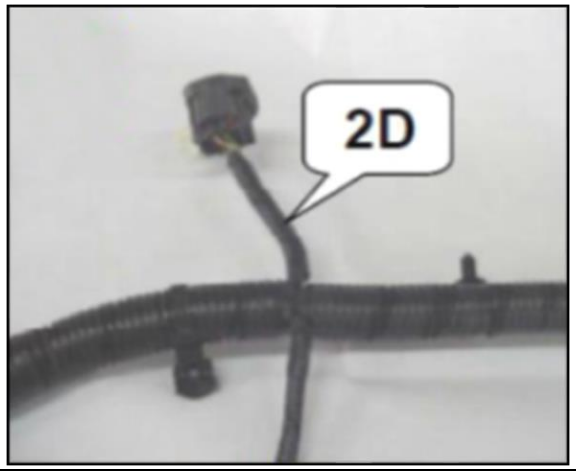
Assembly:		**Layout:**	3
Cost:		**Item:**	07
Environment & Safety:			
Quality:			
High Voltage:			

4.3.7 Proper orientations for takeouts (keep in 2 dimensions)

DFM Concept:

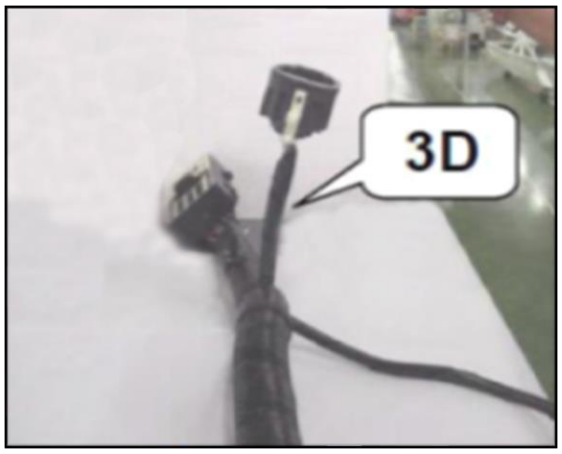
Avoid the use of takeouts in 3 dimensions. Complexity of the board design becomes too complicated to have simple W/H manufacturing. The designer should include additional length to the takeout or change the branch location to avoid the 3D.

OK



Correct Method:
Design W/H in 2 dimensions so that it fits in the 2D shape of a jig board.

NG



Incorrect Method:
Branches that come out in the Z-direction (3D).

Benefits/Effects:
Since assy boards are basically in only 2 dimensions, adding a 3rd dimensional requirement reduces efficiency by causing excessive handling of the harness taping and finishing.

Packaging is also a concern since the harnesses tend to "settle" in the box, deforming the shape/direction of that 3D branch.

Assembly:		**Layout:**	3
Cost:		**Item:**	08
Environment & Safety:			
Quality:			
High Voltage:			

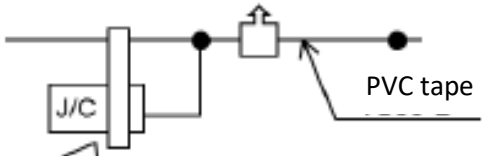
4.3.8 Proper use and location of branch tape-backs

DFM Concept:

Be sure to package connector branches like the one shown below in such a manner that does not move during the finishing process (adding covering) after circuit check process.

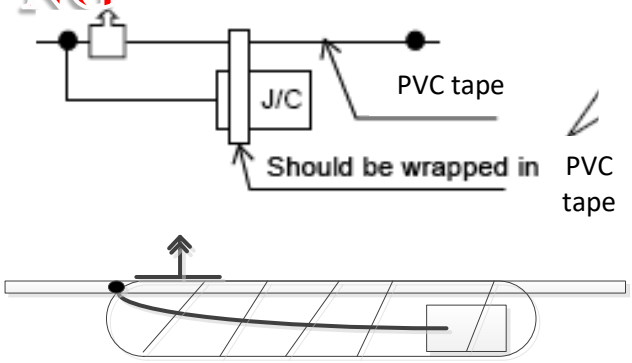
In this case, the connector is moved to be outside the PVC tape area.

OK



PVC tape is added and does not effect connectors processing due to electrical check requirements.

NG



PVC tape must be added after electrical check

Correct Method:
Putting connectors on branches that don't require a finishing process at stages when the harness is not on the jig board.

Incorrect Method:
In the above case, circuit checking has to be done before overtaping the connector is done.

Benefits/Effects:
Anytime you ask for a covering to be added, tape back to be placed, etc. after the continuity test, you decrease the efficiency of the W/H assy process. It is easier to complete all the covering/taping applications during the initial assy of the W/H rather than after continuity testing. During the W/H build, you should review all the processes that have to be done after the harness comes off the jig board and try to make design change that will reduce these processes.

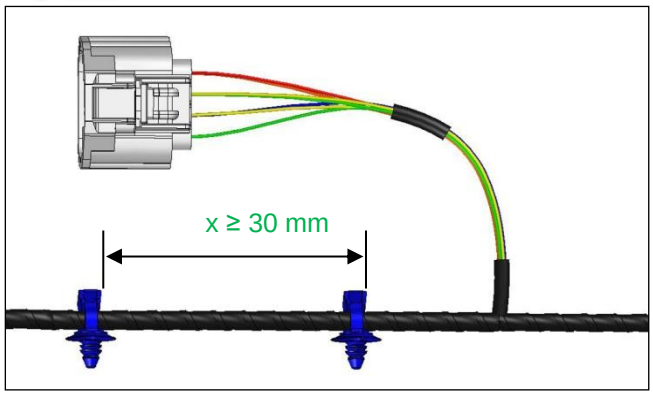
Assembly:		**Layout:**	3
Cost:		**Item:**	09
Environment & Safety:			
Quality:			
High Voltage:			

4.3.9 Distance of fixing parts to components (≥ 30 mm)

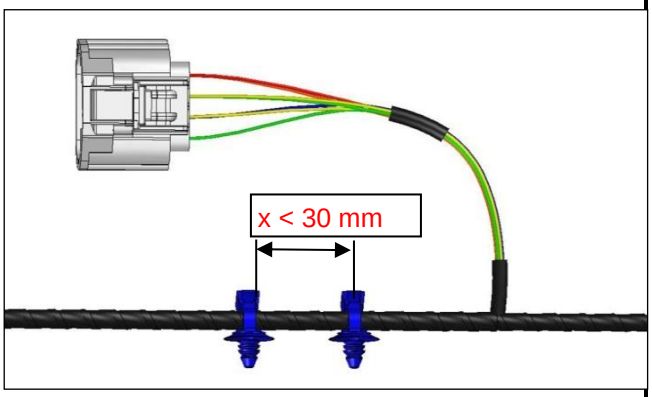
DFM Concept:

A fixing part (clip, retainer, strap, etc.) within a W/H should have a minimum distance of 30 mm to other fixing parts, grommets or other harness parts.
Reference measuring point is the center of the fixing part and the beginning of the component (if no fixing part).

OK



NG



Correct Method:
Fixing feature does have a minimum distance of 30 mm to the component.

Incorrect Method:
Fixing feature does not have a minimum distance of 30 mm to the component.

Benefits/Effects:
A sufficient space between fixing features and a component helps the manufacturing operator to assemble the clip at its designated location.

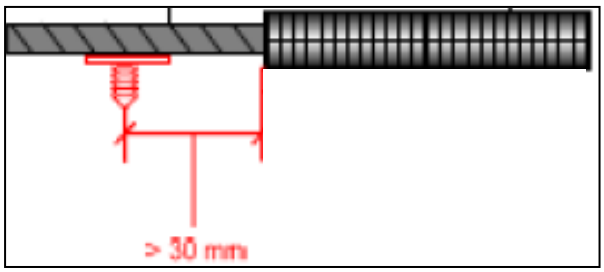
Assembly:		**Layout:**	3
Cost:		**Item:**	10
Environment & Safety:			
Quality:			
High Voltage:			

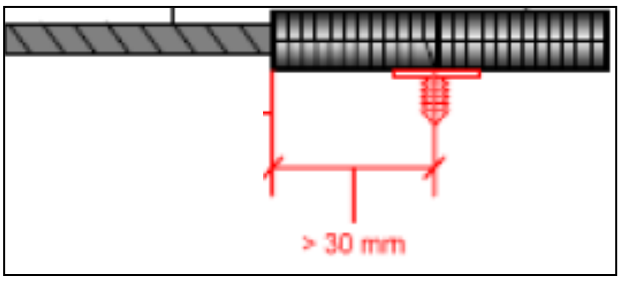
4.3.10 Proper location of clips in relation to applied coverings

DFM Concept:

Clips on Tubes need to be positioned a minimum distance of 30 mm from the end of the tubing and/or branch/node point. This dimension is measured from the end of the tube to the center of the clip. This applies to both tie strap and tape on clips.

OK





Correct Method:
Minimum distance between the clip and end of tubing is 30 mm from the end of the tubing.

Incorrect Method:

Benefits/Effects:
A proper/efficient process is assured without any restrictions when placing clips with adequate distance to the edge of tubes. There is no risk of a clip sliding off the edge of a tube.
Important: When you put clips at the ends of covering or protective components, jig board layout becomes much more difficult. Too many fixtures have to occupy the same space on the jig board.
Avoid placing clips on the branch/node points.

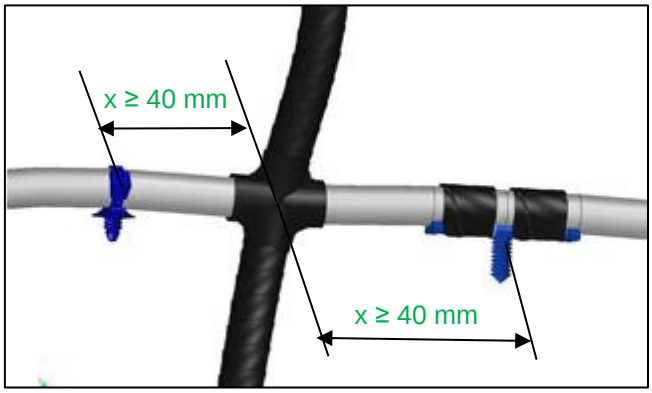
Assembly:		**Layout:**	3
Cost:		**Item:**	11
Environment & Safety:			
Quality:			
High Voltage:			

4.3.11 Distance of fixing parts to takeout (≥ 40 mm)

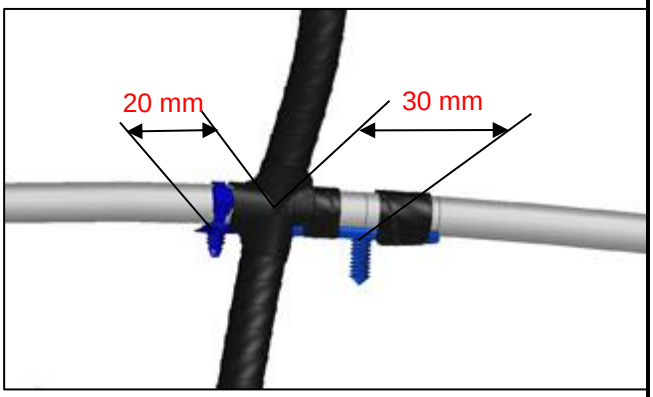
DFM Concept:

A fixing part (clip, retainer, strap, etc.) within a wiring harness should have a minimum distance of 40 mm to takeout. The designer should strictly avoid placing a fixing part directly on a takeout.

OK



NG



Correct Method:
Fixing feature does have a minimum distance of 40mm to the takeout.

Incorrect Method:
Fixing feature does not have a minimum distance of 40 mm to the takeout.

Benefits/Effects:
A sufficient space between fixing features and a takeout helps the manufacturing operator to assemble the clip at its allocated location. Placing a fixing feature directly on a takeout require a special jig fork. The price of those special jig forks is about two times higher than the price for standard jig forks.

Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

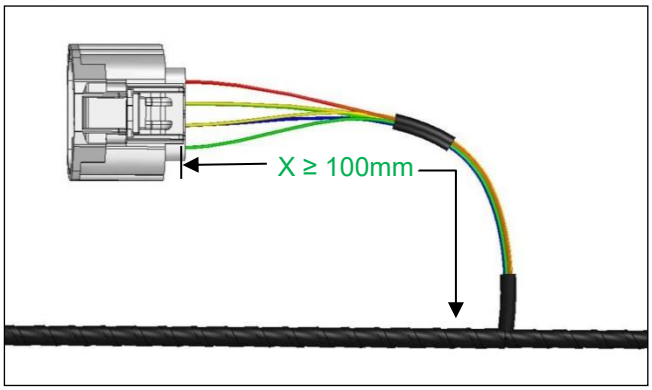
Layout:	3
Item:	12

4.3.12 Minimum clearance from takeout point to connector (≥ 100 mm)

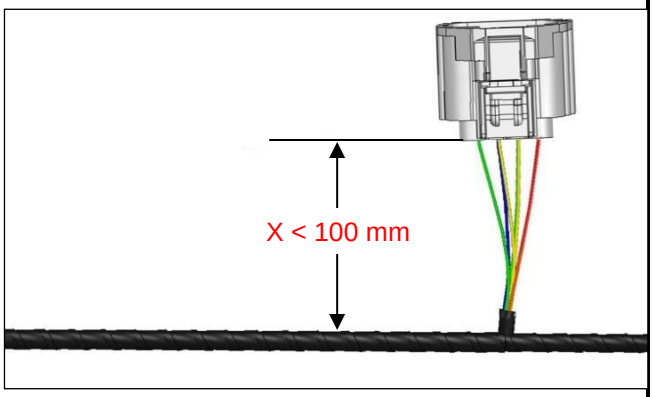
DFM Concept:

The clearance from a harness takeout point to a connector should be at least 100 mm. Measuring point is the back of the connector. The design engineer should check to move the takeout point when the necessary takeout length is smaller than 100 mm.

OK



NG



Correct Method:
The clearance from the takeout point to the back of the connector is 100 mm.

Incorrect Method:
The distance from the harness takeout point to the back of the connector is less than 100 mm.

Benefits/Effects:
Complying with this concept will lead to a better board layout, as there will be enough space between the assembly fixtures (jigs) to simplify the board layout. This eases the taping, routing and terminal insertion operations. Additionally, takeouts with an adequate length improve also the assembly process within the customer manufacturing plant.

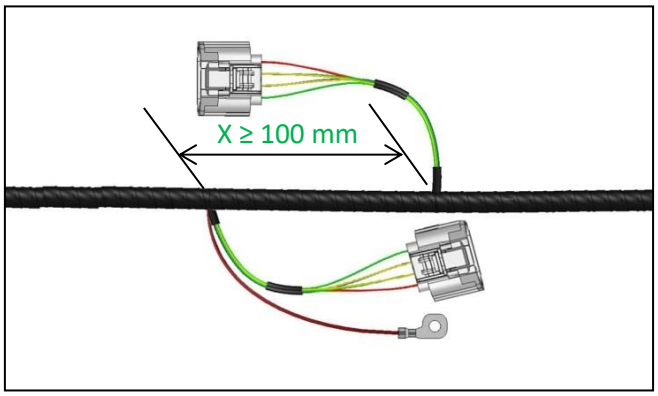
Assembly:		**Layout:**	3
Cost:		**Item:**	13
Environment & Safety:			
Quality:			
High Voltage:			

4.3.13 Minimum clearance between two takeout points (≥ 100 mm)

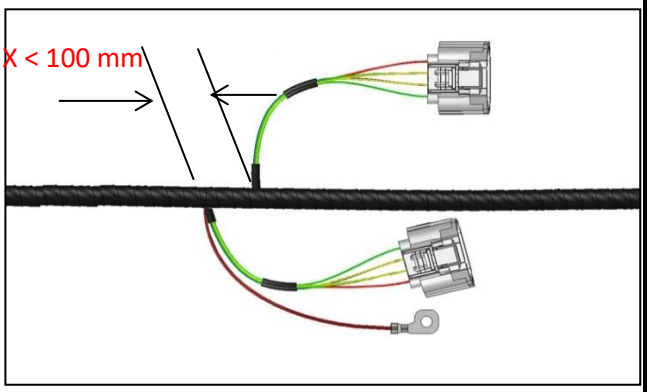
DFM Concept:

The clearance between two takeout points should be at least 100mm. The design engineer should check:
To move one of the takeout points or to combine both takeouts, when the clearance between the two takeouts is smaller than 100 mm.

OK



NG



Correct Method:
The distance between two harness takeout points is 100 mm.

Incorrect Method:
The distance between two harness takeout points is less than 100 mm.

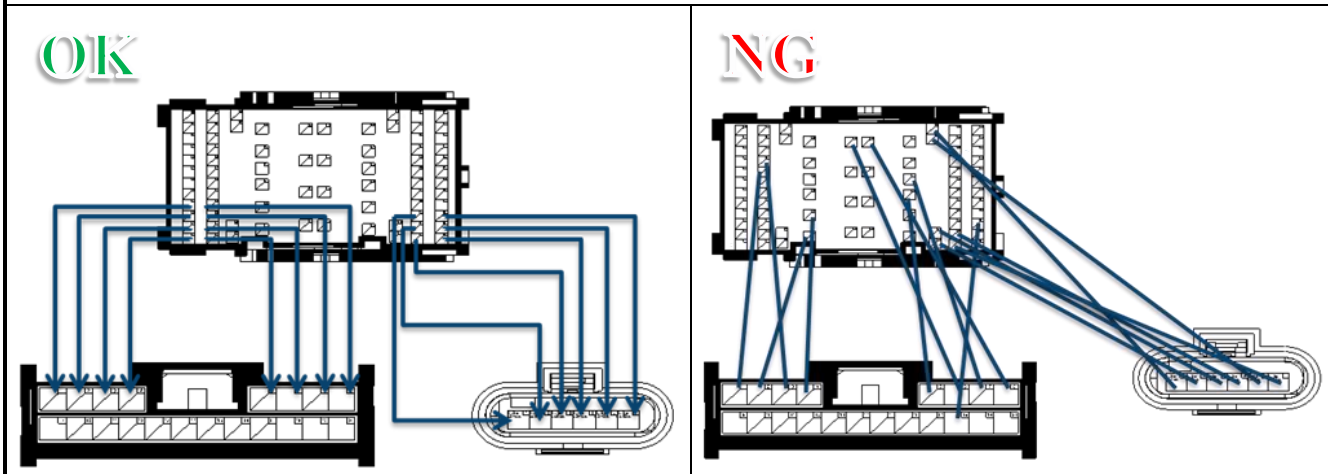
Benefits/Effects:
A clearance between two takeout points with more than 100 mm will improve the board layout regarding the alignment of the assembly fixtures (Jigs), the taping operation and the routing of the branches. Takeouts with an adequate clearance between each other will ease the assembly process within the customer manufacturing plant.

Assembly:		**Layout:**	3
Cost:		**Item:**	14
Environment & Safety:			
Quality:			
High Voltage:			

4.3.14 Pin-out of in-line connectors should match the process insertion sequence

DFM Concept:
This DFM Item must be incorporated as first step and then continue with the attachment about the circuitry pinout design for Connector in-line. The design engineer should check the best location of the circuits in the pin-out of connectors in order to match the process insertion sequence.
Arrange the pins configuration by group or module. Priorities are as follows:
1) Identify inline connectors that can changed internally.
2) Large Families, example Body should be the first priority.
3) Size of the connector. The big connector should be the higher priority.

When relocating twisted pairs, please verify that the circuits are located in an adjacent cavity and not so far away from each other to avoid problems in the untwisted wire.



Correct Method:	**Incorrect Method:**
The location of circuits in the connector match the sequence of insertion	Circuits location in the connector do not follow a sequence of insertion

Benefits/Effects:
No need to search, avoiding the counting of cavities to locate the insertion point.
Reduce wire tangling.
Make easy insertion (no need to move other wires apart to make room for insertion)
Reduce mis-insertions.
Reduce fluctuations and have a stable insertion cycle.
Release operators of making decisions, reducing fatigue.

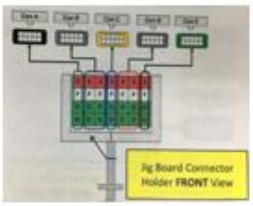
Layout:	3
Item:	14
Attachment:	14a

4.3.14 Attachment: Circuitry Pinout design for Connector in-line

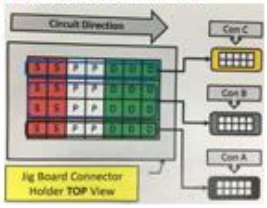
DFM Concept:
This should be the second step after revising the Pin-out of connectors that should match the process insertion sequence. Looking for an empty cavity on the moving jig board often causes quality issues. The design engineer should assign, locate the circuitry in a such manner that the circuit insertion will be more efficient. In a straight connector, ground and splice circuits are preferred to be in front, followed by pre-sealed cavities (plugs), then all direct circuits. In a dressed connector, ground and splice circuits are preferred to be in the far back, followed by pre-sealed cavities (plugs), then all direct circuits. Note that for both straight and dressed connectors, as shown in the photos below, they should be 3D customer connector installation view.

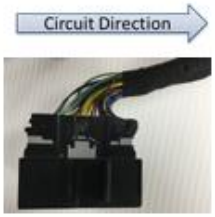
OK

Straight Connector



Dressed Connector





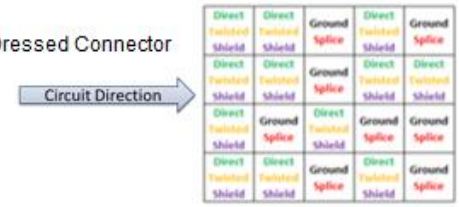
NG

Random Pin-out

Straight Connector



Dressed Connector



Correct Method:
Optimum design. Circuits distributed by groups by connector (A/B/C). Grounds in corners (almost always inserted in last station, so best to leave for corners). Splices on top view ease insertion.

Incorrect Method:
Random pinout will often cause tangling and/or crossed circuits. Under this condition it is unlikely all circuits will be closed with Navigation System.

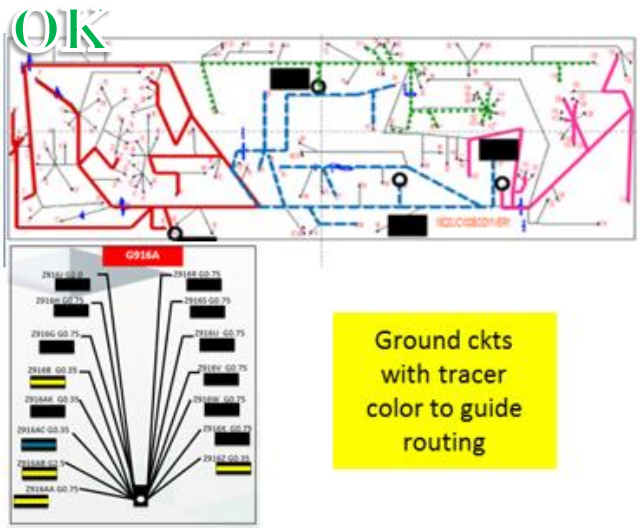
Benefits/Effects:
Splices and Grounds are almost always assembled in the conveyor. Looking for an empty cavity on the moving jig board often causes quality issues.
Direct Circuits, and twisted pairs are almost always assembled in subassembly stations.
Pinouts should be designed and approved by Mfg in concurrence with BU allowing the operator to assemble free from clearing room among tangled wires.

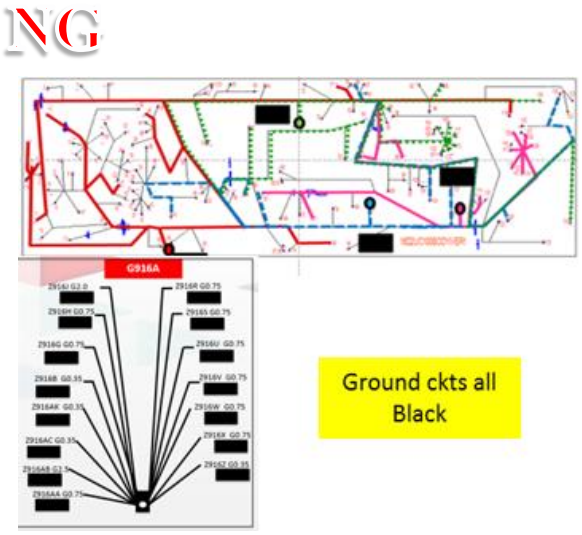
Assembly:		**Layout:**	3
Cost:		**Item:**	15
Environment & Safety:			
Quality:			
High Voltage:			

4.3.15 Ground subassemblies should be in sections & different trace color to guide routing

DFM Concept:
The design engineer should assign the ground circuits to the best location section across the assembly board. If this condition cant be accomplished, at least avoid ground circuits that go along the entire W/H (routed within a ground eyelet terminal) and add a tracer color to differentiate.

If this isn't possible, then the designer should check with Manufacturing to add a marking/paint dot color to one end of the circuit.





Correct Method:
Ground circuits are distributed in sections and with tracer color to guide the routing

Incorrect Method:
Circuits for each ground cross the assy board from left to right

Benefits/Effects:
Eliminate complexity
Have easy routing
Reduce wire length
Simple and easy subassembly

4.4 Coverings & Components

Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

Coverings & Components: 4
Item: 01

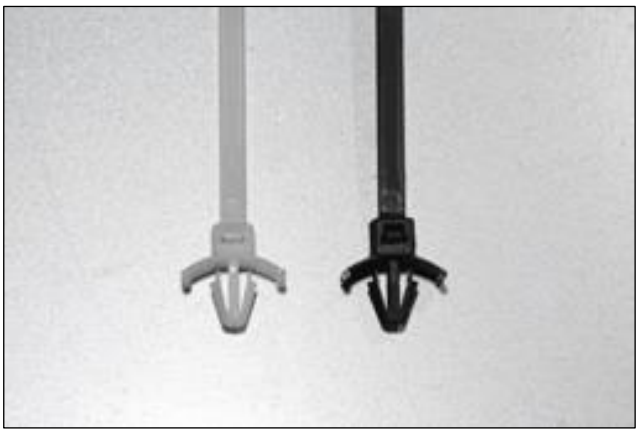
4.4.1 Abstain from using similar clips contained in the same W/H

DFM Concept:
Commonized clips must be used when in the same W/H and in the same family. When proposing, the designer must consider the temperature class, material and color. The communization include strap length, color, offset and similar dimensions. When this is not possible, then add a spot color tape to differentiate the branch and keep the same clip.

OK



NG



Correct Method:
Same clips in the same W/H

Incorrect Method:
Similar clips, but not exactly the same in the same W/H.
Different p/n but can handle same hole and diameter.

Benefits/Effects:
Wrong clip installation.
Manufacturing does not prefer just changing color. Just changing color would require for manufacturing to invest in color-detecting sensors.

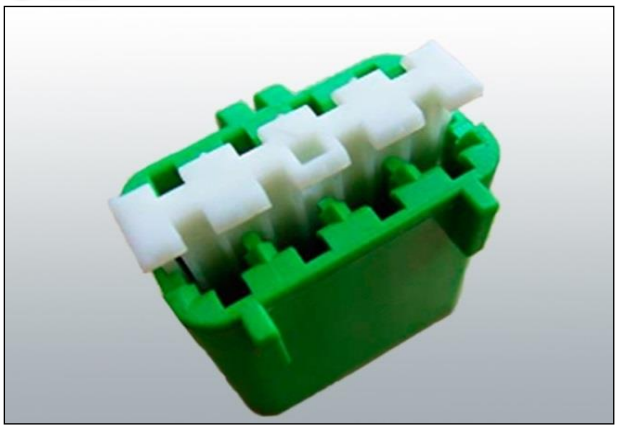
Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

Coverings & Components:	4
Item:	02

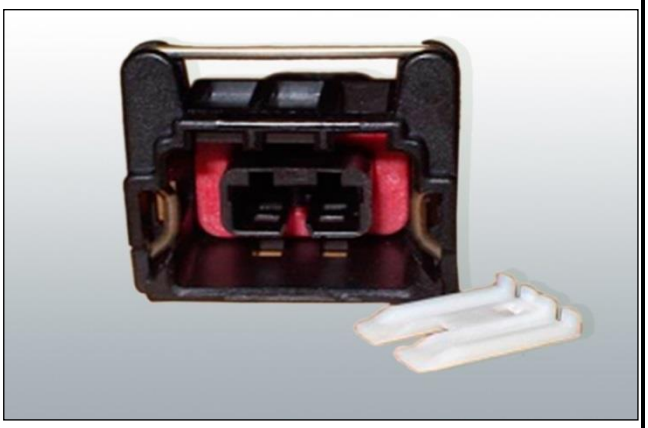
4.4.2 Integrated connector spacers

DFM Concept:
The design engineer must avoid using connectors that require the usage of separate spacers which have to be applied after terminal insertion. Pre-loaded front loading spacers or rear holder wedges must be chosen if possible.

OK



NG



Correct Method:
2 piece connectors with "preset" wedges.
Connectors with "hinged" rear holders.

Incorrect Method:
Connectors with loose piece terminal locks.
This includes front spacers, rear caps/holders, and all others parts that secure the terminals.

Benefits/Effects:
Using connectors that require the use of separate spacers requires the need for additional operations, moreover, a spacer detection test is essential to ensure presence and terminal locking.

Main Benefit: Efficiency improvement.

Assembly:		**Coverings & Components:**	4
Cost:		**Item:**	03
Environment & Safety:			
Quality:			
High Voltage:			

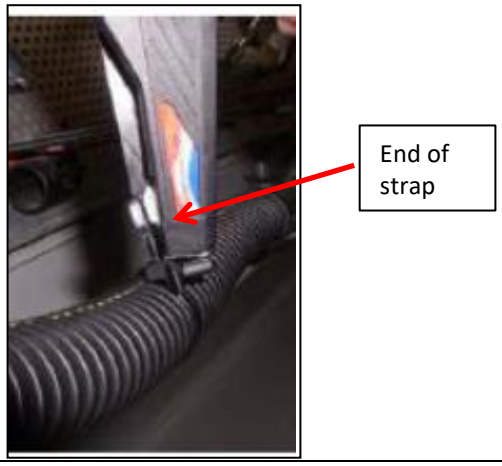
4.4.3 Tie straps or strap clips need to have sufficient length for tool usage

DFM Concept:
By using cable ties on branches it must be pointed out that after fixation of the cable tie, the strap length must be long enough to fasten and cut the strap using the proper tool. Select the strap length according to bundle circumference plus 80 mm minimum. To reduce complexity try to communize the strap in regards to total strap length.

OK



NG



Correct Method:
Tie strap has sufficient length to be easily grasped and perform the necessary cut off/torque process.

Incorrect Method:
Tie strap is barely long enough to engage the tool.

Benefits/Effects:
Having the ability to fasten and cut the cable tie with the proper tool leads to more accurate fixing of the clip; having enough length to easily see the tab and set the tool improves efficiency by shortening process time.

Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

Coverings & Components:	4
Item:	04

4.4.4 Lever position clearly defined

DFM Concept:
Lever should be shipped to customer plant in a defined position to perform connector installation.
Designers need to clarify how the lever should be shipped to the final customer.

OK

View on Drawing



Lock Lever must be in this position for harness shipping

Correct Method:
Lock Lever in the correct position.

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No view on drawing showing lever position

Incorrect Method:
No notes in the drawing and the lever lock is in the wrong position

Benefits/Effects:
Difficult operation to install circuits.
Added labor not included in manufacturing.
Added penalty labor in the GCT component registration.

Assembly:		**Coverings & Components:**	4
Cost:		**Item:**	05
Environment & Safety:			
Quality:			
High Voltage:			

4.4.5 Foam tape on connectors

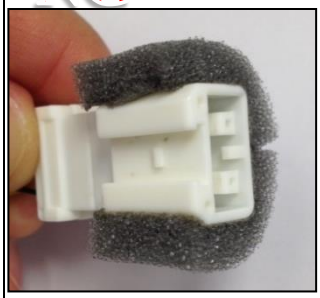
DFM Concept:
Must use precut foam tape and specify precut length to avoid addional time to cut.

OK

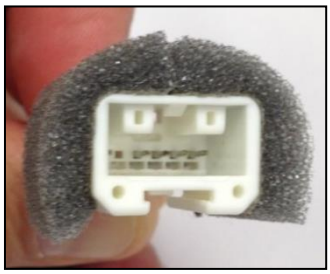




NG



Not Preferred



Correct Method:
Full foam taped connector. W/H Manufacturing does not need additional cuts the foam tape.

Incorrect Method:
W/H Manufacturing has to cut and place foam tape on connector.

Benefits/Effects:
Extra processing time to perform additional cutting of the foam tape.
Increase Tact time due to additional cutting process.
Extra (unpaid) labor.

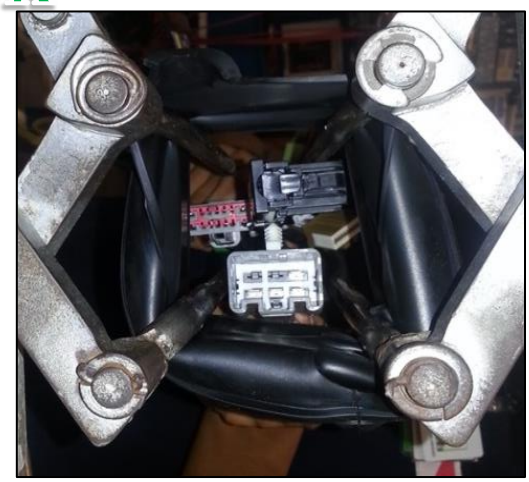
Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

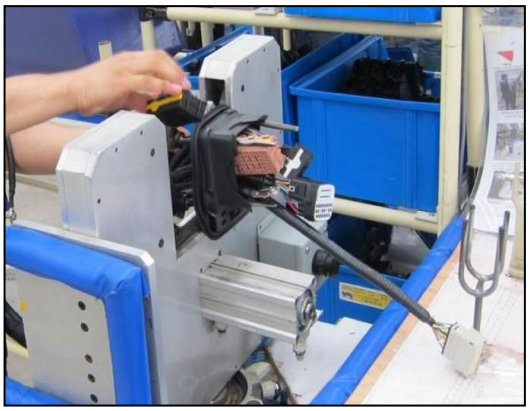
Coverings & Components:	4
Item:	06

4.4.6 Select a grommet that can expand to necessary size during assy process

DFM Concept:
When specifying a grommet, the engineer must ensure that the maximum expandable limit of the grommet is appropriate for the components that need to pass through it during W/H assy. See below.

OK





Correct Method:
In the above case, all the necessary components can pass through stretched grommet.

Incorrect Method:

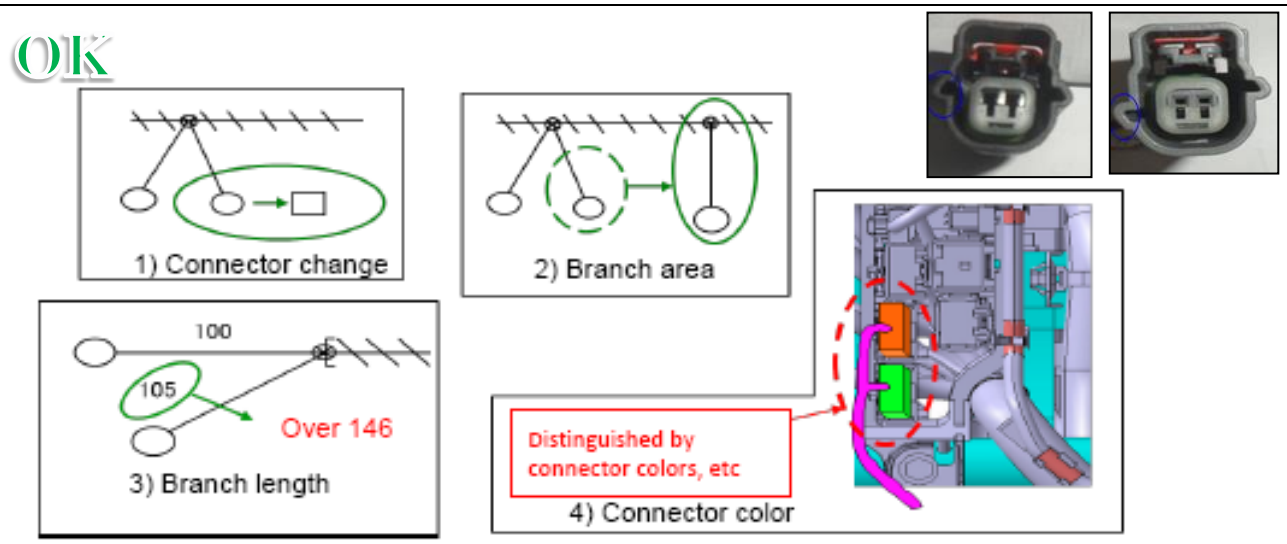
Benefits/Effects:
By assuring that the necessary harness components will pass through the grommet during the stretching process, there will be less scrap and fewer quality issues associated with the assy process. If the grommet cannot be stretched, then the harness may have to be assembled through the grommet a few circuits at a time. This would substantially increase labor time and assembly complexity.

Additionally, if the grommet can't be stretched, ensure that the Global Unit Man-Hour assigned to the component fulfill the required time to pass through all circuits and "muda time"/wasted or if necessary to create a process code to include that extra time and to be included into the drawing in order that can be quoted with the required tack time to install it.

Assembly:		**Coverings & Components:**	4
Cost:		**Item:**	07
Environment & Safety:			
Quality:			
High Voltage:			

4.4.7 Same connector usage at same branch point

DFM Concept:
Do not make W/H designs that have multiple branches at the same point using the same connector and the same length.



Correct Method:
If you must have the same connector, vary the lengths by approx. 50 mm, change colors, have different keyways or try to move the branch.

Incorrect Method:

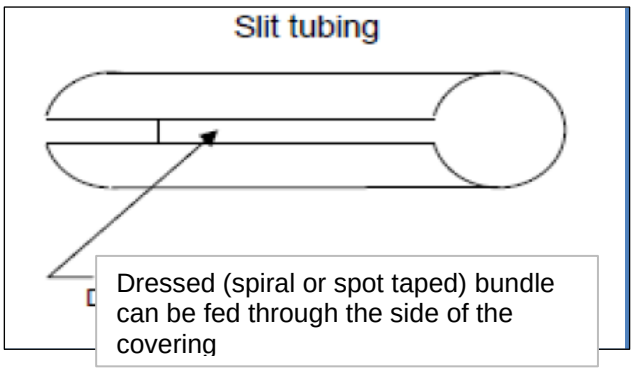
Benefits/Effects:
When you have too many similarities within a W/H area, misbuilds can occur by easily building the W/H opposite to what is expected by the W/H engineer. Even though the problem might be found with an electrical check, the result would be rework of the W/H causing un-necessary work on the W/H.
The goal is to make these areas fool-proof for routing and installing circuits. Pay careful attention to the process in the plant that are required if you have same connectors in nearby areas and look for opportunity to fool proof the design to prevent W/H manufacturing mistakes.

Assembly:		**Coverings & Components:**	4
Cost:		**Item:**	08
Environment & Safety:			
Quality:			
High Voltage:			

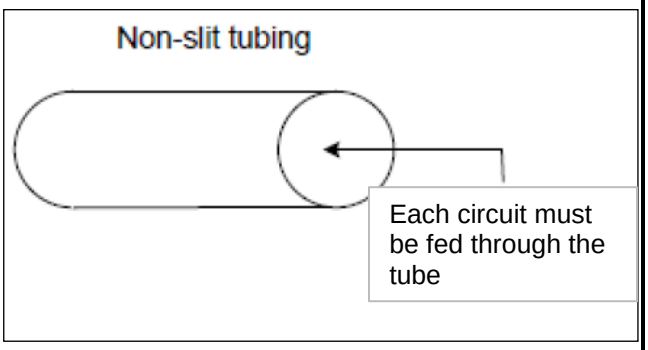
4.4.8 Slit tubing vs. non-slit tubing

DFM Concept:
Slit coverings should be used on main bundles of wire harness. Minimize the use of non-slit.

OK



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Correct Method:
Slit tubing allows the covering to be added after all wires are routed on the board.

Incorrect Method:
Non-Slit tubing requires individual wires to be feed through the covering individually.

Benefits/Effects:
Use of slit covering avoids the need of extra labor for passing all of the circuits one by one by the operator, easing harness assembly. Avoiding passing them through the covering will result in less risk of damage to the terminals and quality will improve.

Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

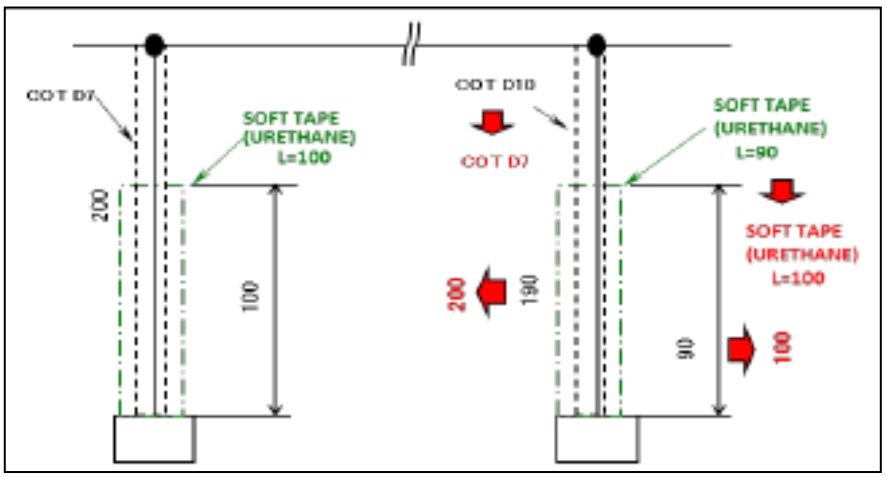
Coverings & Components:	4
Item:	09

4.4.9 Commonization of coverings/sleeves

DFM Concept:
If there are similar coverings of similar length and size within the harness, the responsible engineer must work to commonize the coverings/sleeves.

1) Use the same cover size if the diameters of branch lines are slightly different.
2) Branch dimensions and the cover length should be the same.
3) Avoid having different types of coverings/sleeves within the same area.

OK



Correct Method:
Coverings on adjacent branches are changed to be the same length, diameter and/or size.

Incorrect Method:

Benefits/Effects:
If there are pre-cut coverings of similar size and length, it creates a situation where the operator can choose the wrong covering for an area. This can lead to a quality issue due to improper coverage of the harness, inefficient to process, or rework.

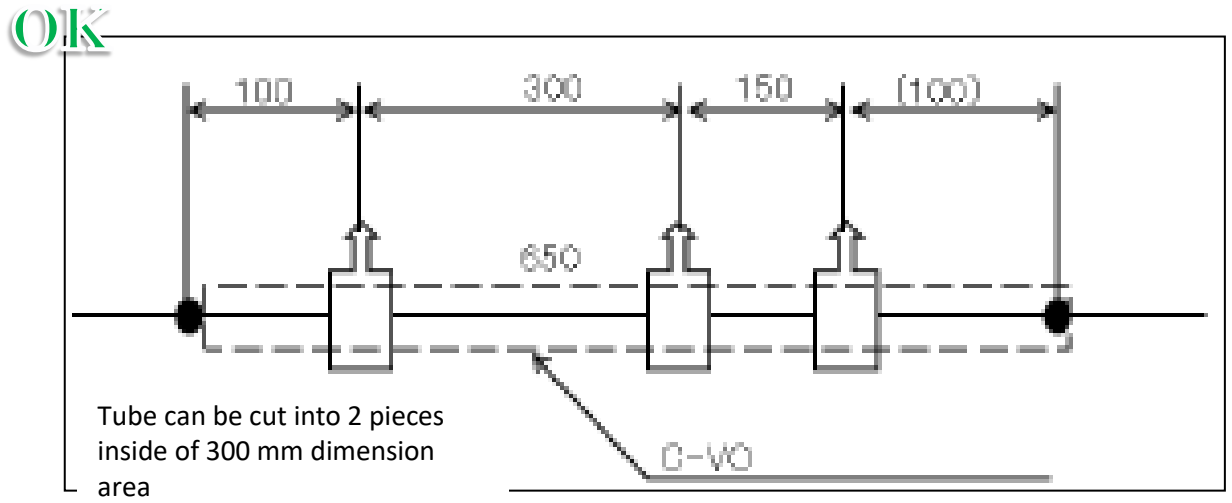
Having various types of coverings/sleeves increases the complexity of component handling and the potential risk of making a mistake during installation.

Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

Coverings & Components:	4
Item:	10

4.4.10 Dividing coverings

DFM Concept:
Consider allowing manufacturing to divide long runs of covering into smaller pieces, the responsible engineer must indicate on the drawing if the covering can be divided and at what point(s). If in the area selected, slitted coverings are used, then an additional straight tape must be used on the location where the coverings join.



Correct Method:
Allow W/H manufacturing to "split" coverings of 500 mm or longer.

Incorrect Method:

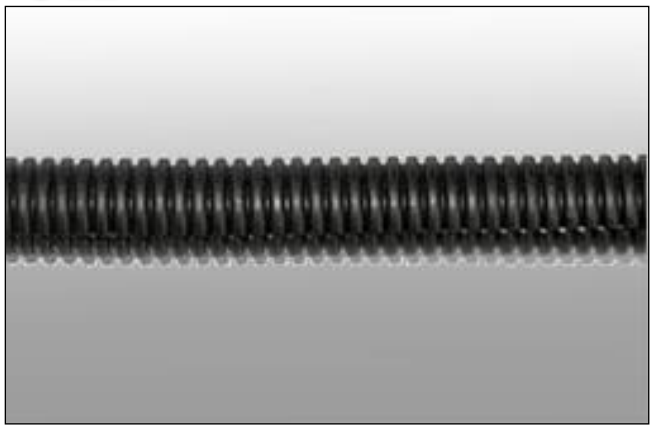
Benefits/Effects:
It is difficult and inefficient for manufacturing to install covering longer than 500 mm. It was determined that 500 mm should be the cutoff length for continuous coverings. If you can split the covering, you allow W/H factory to split the pieces to make for quicker/easier installation onto the W/H.

Assembly:		**Coverings & Components:**	4
Cost:		**Item:**	11
Environment & Safety:			
Quality:			
High Voltage:			

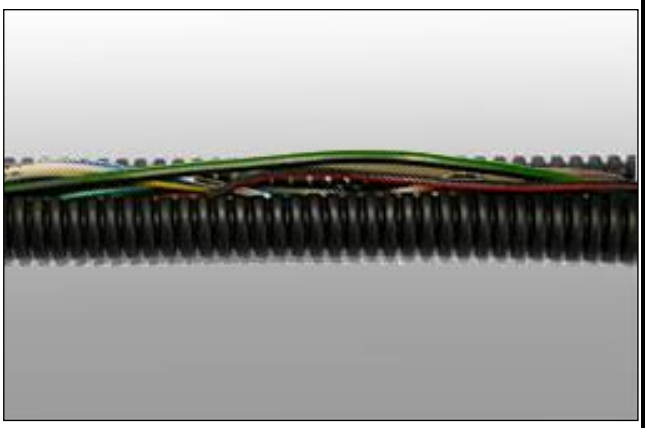
4.4.11 Align covering dimension with W/H bundle diameter

DFM Concept:
By using protection like PVC tubes, convoluted tubes, shields etc. on the W/H, the designer should take into consideration to choose a protector diameter that fits with the branch diameter. Maximum filling rate should be equal or less than 80% of the inner tube diameter. This concept is applicable for unslitted and slitted protectors. Particular attention has to be paid when placing splices, specially waterproofed splices, under the protection. Due to shrink tubes or taping around the welding knot, splices are bigger than the same amount of single wires. Also, the size of the used terminals has to be taken into consideration.

OK



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Correct Method:
The chosen covering does have a sufficient inner diameter for the wiring harness bundle (e. g. ≤ 80% of the inner diameter).

Incorrect Method:
The chosen covering does not have a sufficient inner diameter for the W/H bundle. Additional taping around the tube is necessary for closing the protection properly.

Benefits/Effects:
Following this DFM concept simplifies the handling for the manufacturing process. There is enough space left to lead the circuits through the protection. The risk of loop backs and circuits getting stuck is reduced. Additional taping operations for closing the protection properly can be deleted. All in all this measure increases the quality of the product and therefore the customers satisfaction.

Assembly:		**Coverings & Components:**	4
Cost:		**Item:**	12
Environment & Safety:			
Quality:			
High Voltage:			

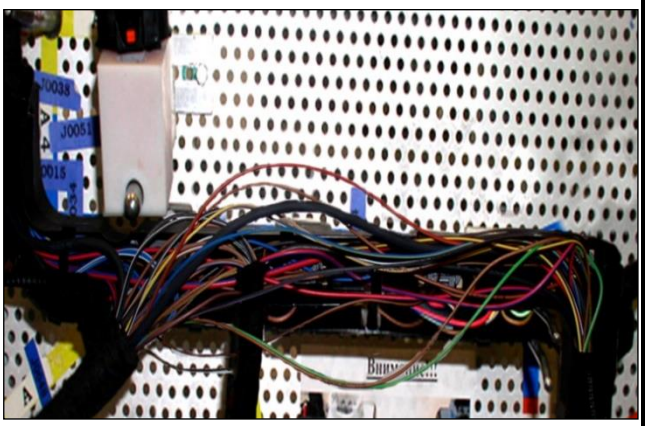
4.4.12 Using taping inside troughs / protectors

DFM Concept:
Wire/circuit management inside troughs and protectors. Depending on the shape/cross-section of the shield this may or may not apply. This will not apply to protectors with low-height areas.

OK



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Correct Method:
The harness bundle inside the channel is bundled with tape. This will vary depending on protector shape/cross-section.

Incorrect Method:
No bundling within the channel. This will vary depending on protector shape/cross-section. Manufacturing has to assure appropriate circuit lengths are optimized.

Benefits/Effects:
Compliance with this DFM will ease assembly of the channel onto the harness. It will help to retain the wires within the channel keeping them from getting trapped or damaged during final harness assembly (trapped in trough cover) or during vehicle assembly.

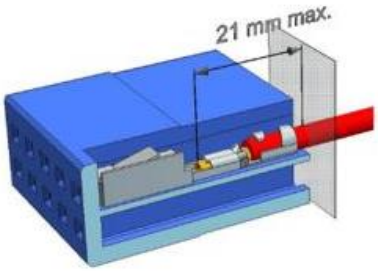
Assembly:		**Coverings & Components:**	4
Cost:		**Item:**	13
Environment & Safety:			
Quality:			
High Voltage:			

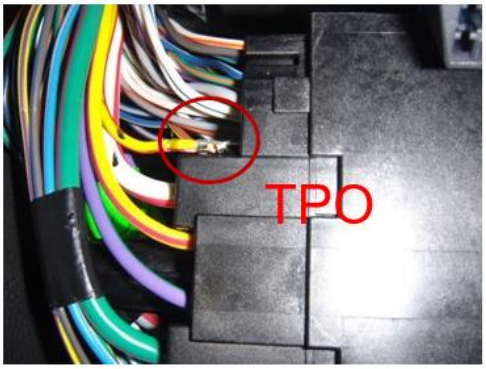
4.4.13 Avoid Connectors that may cause Terminal Push Out (TPO)

DFM Concept:

The Designer should avoid select any connector in which the depth of insertion is large. It should be a maximum of 21 mm so that the wire does not bend while inserting. This happens more often when the circuit have a wire size 0.35mm2 or less, but it could happen as well for bigger wire sizes.

OK





Correct Method:	**Incorrect Method:**
The wiring harness designer selected a connector which the maximum depth of insertion is 21mm.	

Benefits/Effects:

If the connector has an insertion depth greater than 21mm, it increases the risk that the circuit be not well inserted and have a terminal push out that would cause the vehicle be not functioning and/or causing waterleakage. Also, this may cause inefficieny into production line due to rework or extra time required trying to solve the issue and more drastically, to replace completely the circuit since the circuit was already damaged and bent causing that be not possible to be inserted.

4.5 Wire

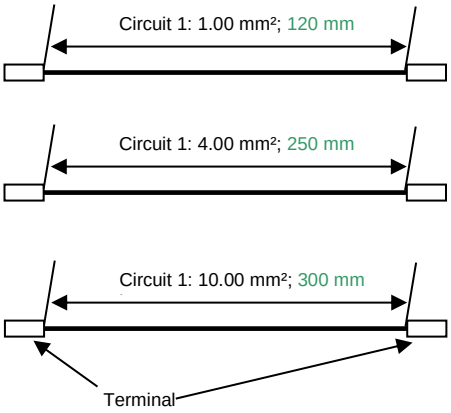
Assembly:		**Wire:**	5
Cost:		**Item:**	01
Environment & Safety:			
Quality:			
High Voltage:			

4.5.1 Minimum circuit length (100 mm / 200 mm / 300 mm)

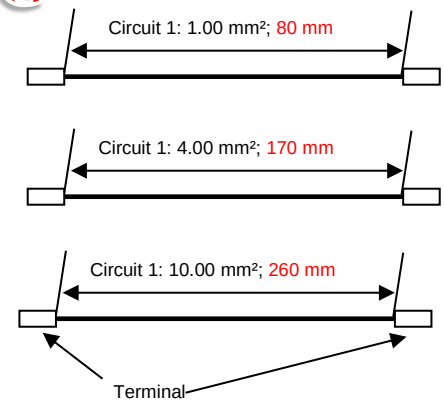
DFM Concept:
The minimum wire length is recommended as follows:
- For wire sizes < 4.00 mm² the minimum circuit length should be 100 mm.
- For wire sizes 4.00 mm² ≤ x < 10.00 mm² the minimum circuit length should be 200 mm.
- For wire sizes ≥ 10.00 mm² the minimum circuit length should be 300 mm.

Please note: The above mentioned lengths refer to the circuit length up to the beginning of the terminal.
Please note: The minimum cutting lengths depends on the safety cover of the cutting machine and not on the minmum cutting lengths of the automatic crimping machine.

OK



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Correct Method:
All circuits have an adequate length.

Incorrect Method:
Chosen circuit lengths are shorter than recommended within this concept.

Benefits/Effects:
Using the minimum circuit length as recommended above, improves the handling for assembly processes like assembling circuits on the jig board or producing splices. This simplified process leads to a decreased overall process time. In addition, when performing manual operations a higher safety standard is given for the operator as a consequence of adequate circuit lengths.

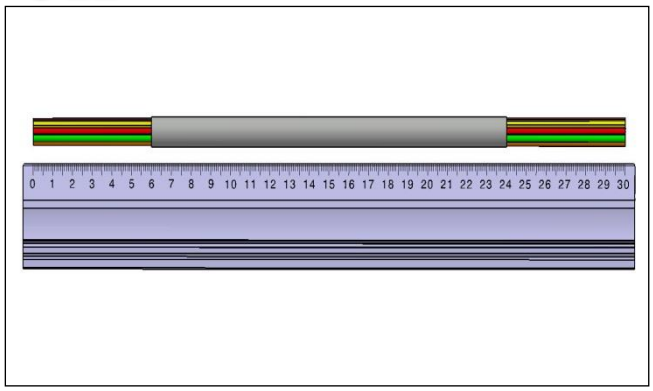
Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

Wire:	5
Item:	02

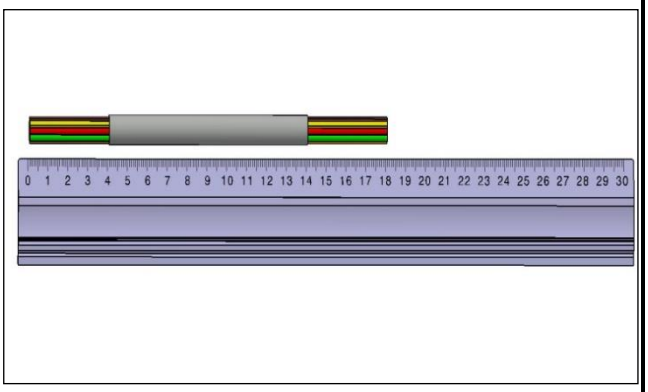
4.5.2 Minimum length of shielded wires (≥ 180 mm)

DFM Concept:
When using shielded wires the design engineer should ensure that the remaining insulation on the shielded wires has a minimum length of 180 mm and the ends of the wire should still have a wire length of 60 mm.

OK



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Correct Method:
Length of remaining insulation on the shielded wires is ≥ 180 mm.

Incorrect Method:
Length of remaining insulation on the shielded wires is < 180 mm. Insulation can be easily moved along the wire.

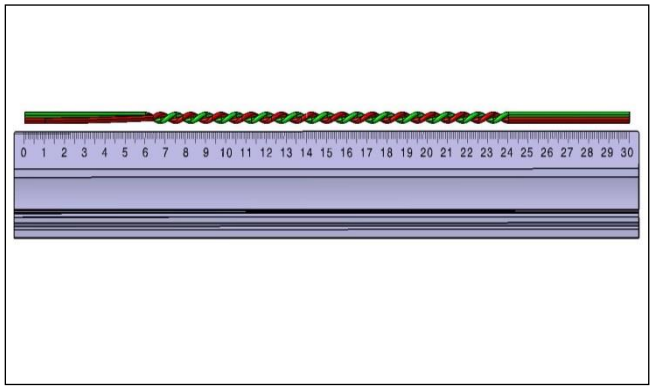
Benefits/Effects:
The engineer should consider that a minimum stripping length of 60 mm on each side of the wire is necessary for other processing steps like crimping or splicing. This stripping length cannot be shielded. So the minimum common wire length of shielded wires with 60 mm stripped on each side should be 300 mm. Following this DFM concept improves the quality of the end product and reduces additional processes like taping to fix the insulation to avoid a self movement.

Assembly:		**Wire:**	5
Cost:		**Item:**	03
Environment & Safety:			
Quality:			
High Voltage:			

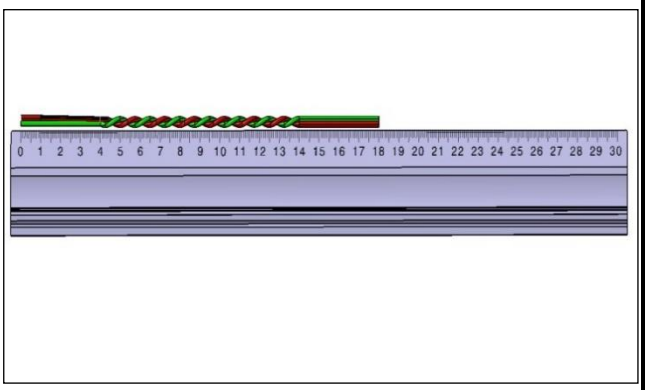
4.5.3 Minimum length of twisted wire (≥ 180 mm)

DFM Concept:
When using twisted wires the design engineer should ensure that the common wire route length has a minimum length of 300 mm. The effective twisted circuit length should have a minimum length of 180 mm.
Note: The designer should use 60 mm as untwisted wire lengths at the ends includes the wire lengths inside the connector and the crimp area. However, if the untwisted area for connectors it is required in less dimension, the minimum one should be 50 mm and if so, then the minimum length would be 280 mm. For splice, it should be considered an untwist area of 75 mm as minimum to the splice center location. If OEM Customer specify something different, you MUST follow what they are requiring and confirm there is an equipment that can do it.

OK



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Correct Method:
Length of twisted circuits within the W/H assembly exceeds the minimum effective twisted circuit length of 180 mm.

Incorrect Method:
The effective length of the twisted circuits within the W/H assembly is shorter than 180 mm.

Benefits/Effects:
The engineer should consider that a minimum stripping length of 60mm on each side of the wire is necessary for other processing steps like crimping or splicing. This stripping length cannot be twisted. So, the effective twisted circuit length of a 300 mm wire is 180 mm. This is an adequate length to keep them twisted. There is a higher risk that the circuits untwisting by themselves if the twisted circuit length is shorter than 180 mm. Following this DFM concept improves the quality of the harness.

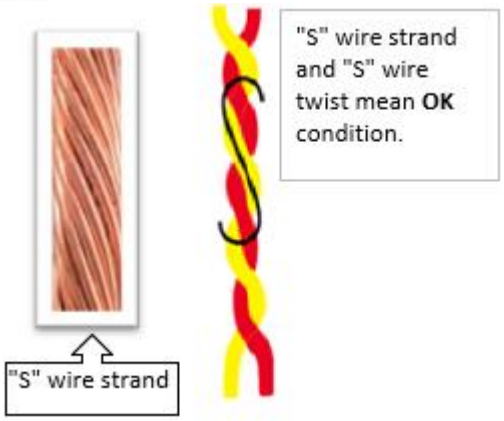
Assembly:		**Wire:**	5
Cost:		**Item:**	04
Environment & Safety:			
Quality:			
High Voltage:			

4.5.4 Same twist direction for strands and wire

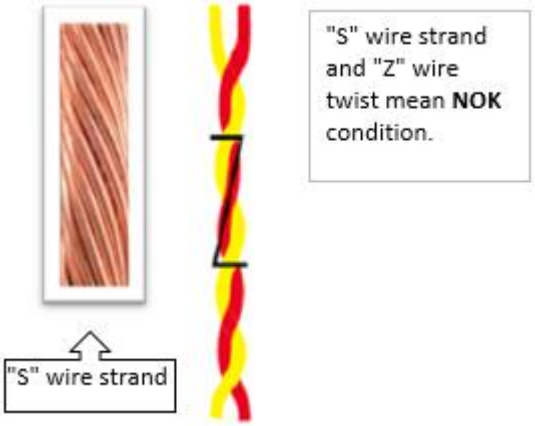
DFM Concept:
Twisting direction(S/Z):

S or Z indicates the direction in which the wires are twisted. The designation S/Z is used, as they can always be determined in the end product.
The conductors should always be twisted in the same direction like the stranded wires to avoid burst wires.

OK



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Correct Method:
The wires are twisted in same direction as stranded single wire.

Incorrect Method:
The wires are twisted in opposite direction as twisted strands of single wire.

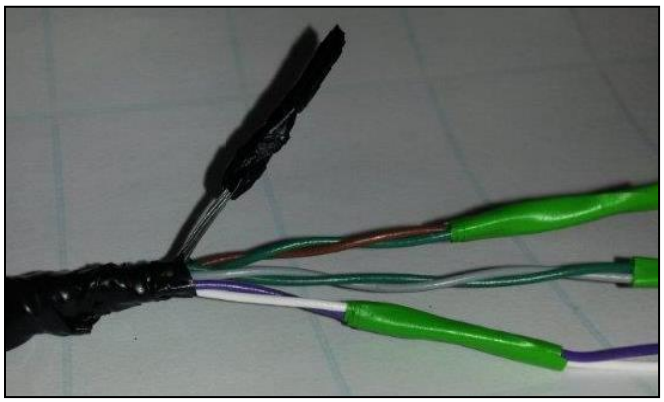
Benefits/Effects:
It is advantageous to twist the wires in the same direction as the twisted strands inside of wires. If you take two pair identical wires and twist one with S-direction and the other with Z-direction they will have different length and quality. This is why the wires shall be twisted in same direction as the strands inside the wire.

Assembly:		**Wire:**	5
Cost:		**Item:**	05
Environment & Safety:			
Quality:			
High Voltage:			

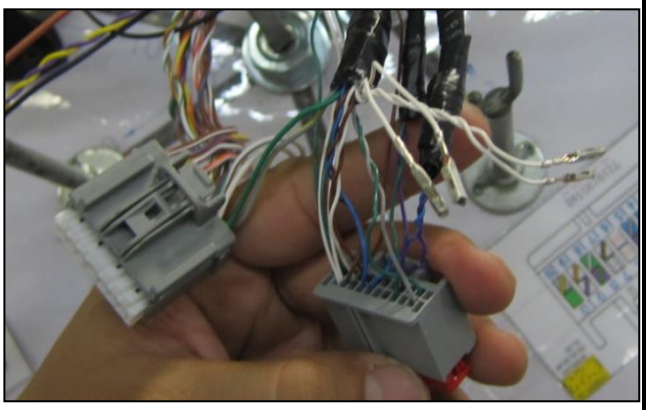
4.5.5 Shielded/Twisted wire must include different color wires

DFM Concept:
Assure all circuits contained in a shielded/twisted wire assy have different colors.

OK



NG



Correct Method:
All circuits in the shielded wires have different colors (including twisted pairs within the shielded assy).

Incorrect Method:
Shielded wire assy with same circuit colors. This also applies to twisted pairs within the shielded wire.

Benefits/Effects:
Wrong terminal insertion by the operators can result in reworks as well as quality issues.

For this type of process the insulation wire 1C7 or 3TAD is not a manufacturable condition, the property of this insulation is too sticky and during the shield process the drain wire is not able to slip through the cables with type insulation, causing damage and requiring an additional man power. Example, man power to high pod electrical test and man power to coil and feed the equipment and avoid tangle up.

Assembly:		**Wire:**	5
Cost:		**Item:**	06
Environment & Safety:			
Quality:			
High Voltage:			

4.5.6 No Heat shrink process for wire type T1 / Class A (85°C)

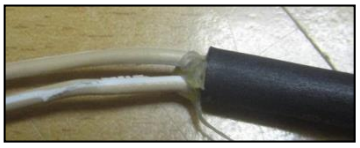
DFM Concept:
On processes such as shield wire or splice process in which a waterproof shrink tube is required, it's very important to compare the "wire temperature class" Vs "shrink tube temperatures".
Typically, waterpoof shrink tubes need 135 °C of shrink temperature to reach shrink ratio, during short cycle time, ex. 8-20 seconds(example for NR 1-4). We need to ensure that wire temperature class can handle such working temperatures during shrink process.
Class "A" temperature (85 °C) is not applicable for waterproof shrink tube.

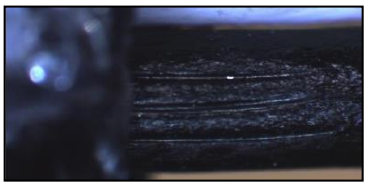
OK



NG







Correct Method:
Wire class is proper to applicable shrink temperature. During shrinking process the heat which is absorbed by wire is not sufficient to cause chemical modification and consequently melting.

Incorrect Method:
Wire insulation start to melt due to absorbed heat during shrinking process.
In cases such as shield wire application the effect is even more visible, due to the effect of hot shield in contact with wire insulation can cause that shield touch the copper strands.

Benefits/Effects:
No risk to have field claim issue due to short circuit.

Assembly:		**Wire:**	5
Cost:		**Item:**	07
Environment & Safety:			
Quality:			
High Voltage:			

4.5.7 Limit number of wire colors (only valid for YEL)

DFM Concept:
To limit the complexity of wire colors and the consequential number of terminal wire color combination a "positive list" was established to define the preferred wire colors in dependency to the wire size and temperature class (T2 means 100 °C; T3 125 °C and T4 150 °C).
If the W/H engineer has the free choice to define the wire color, he has to use the "positive list" and define only wire colors which are mentioned on the list.

OK

wire temp class	wire size	color
T2	1.0	red/blue
T2	0.5	blue/black
T3	1.5	red
T4	4.0	red/green

NG

wire temp class	wire size	color
T2	1.0	red/orange
T2	0.5	blue/brown
T3	1.5	grey
T4	4.0	red/violet

Correct Method:
The engineer did not have to follow any customer wire color regulations and defined the wire colors according to the positive list.

Incorrect Method:
The engineer did not have to follow any customer wire color regulations, but he defined wire colors which are not mentioned on the positive list.

Benefits/Effects:
Following the positive list will help to reduce the number of different wire colors.
This can provide more warehouse space, less wire crimp combination and less set up times.
In addition, less scrap material, because it is not allowed to use wires older than 6 month for ultrasonic splice welding.

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Assembly:		**Wire:**	5
Cost:		**Item:**	07
Environment & Safety:		**Attachment:**	5.07a
Quality:			
High Voltage:			

4.5.7 Attachment a: Limit number of wire colors: T2 (100 °C) positive list

0.13 mm²

Base Color	blue	brown	yellow	grey	green	orange	violet	red	white	black
blue	X									
brown		X								
yellow	X		X							
grey				X						
green			X		X					
orange						X				
violet			X				X			
red	X		X					X		
white	X	X			X		X	X	X	
black			X						X	X

0.17 mm²

Base Color	blue	brown	yellow	grey	green	orange	violet	red	white	black
blue	X									
brown		X								
yellow	X	X	X							
grey				X						
green			X		X					
orange						X				
violet			X				X			
red	X		X					X		
white	X	X			X		X	X	X	
black			X						X	X

0.35 mm²

Base Color	blue	brown	yellow	grey	green	orange	violet	red	white	black
blue	X	X	X	X	X		X	X	X	X
brown	X	X	X	X	X	X	X	X	X	X
yellow	X	X	X	X	X	X	X	X	X	X
grey	X	X	X	X	X	X	X	X	X	X
green	X	X	X	X	X		X	X	X	X
orange	X	X				X				
violet	X	X	X	X	X	X	X	X	X	X
red	X	X	X	X	X			X	X	X
white	X	X	X	X	X		X	X	X	X
black	X	X	X	X	X		X	X	X	X

0.5 mm²

Base Color	blue	brown	yellow	grey	green	orange	violet	red	white	black
blue	X		X	X	X		X	X	X	X
brown	X	X	X	X	X	X	X	X	X	X
yellow	X	X	X	X			X	X	X	X
grey	X	X	X	X	X	X	X	X	X	X
green	X	X	X	X	X		X		X	X
orange						X				
violet		X	X	X	X		X		X	
red	X		X	X	X		X	X	X	X
white	X	X	X		X	X	X	X	X	X
black	X	X	X	X	X		X	X	X	X

0.75 mm²

Base Color	blue	brown	yellow	grey	green	orange	violet	red	white	black
blue	X	X	X	X	X			X	X	X
brown	X	X	X		X		X		X	
yellow		X	X				X	X		X
grey	X	X	X	X			X	X	X	X
green	X		X	X	X		X	X	X	X
orange						X				
violet	X	X	X	X	X	X	X		X	
red	X	X	X	X	X			X	X	X
white	X	X		X	X		X	X	X	X
black	X	X	X	X	X		X	X	X	X

1.0 mm²

Base Color	blue	brown	yellow	grey	green	orange	violet	red	white	black
blue	X	X	X	X	X		X	X	X	X
brown	X	X	X	X	X		X	X	X	X
yellow			X					X		X
grey	X			X			X	X	X	X
green	X				X		X	X	X	X
orange						X				
violet							X	X	X	X
red	X	X	X	X	X		X	X	X	X
white	X			X	X		X	X	X	X
black	X	X	X					X	X	X

1.5 mm²

Base Color	blue	brown	yellow	grey	green	orange	violet	red	white	black
blue	X		X		X			X	X	
brown		X	X	X	X		X		X	X
yellow	X	X	X					X	X	X
grey		X		X				X		X
green		X	X		X			X		X
orange										
violet	X						X	X	X	X
red	X	X	X	X	X		X	X	X	X
white	X	X	X		X		X	X	X	X
black	X	X	X	X	X		X	X	X	X

2.5 mm²

Base Color	blue	brown	yellow	grey	green	orange	violet	red	white	black
blue	X			X	X		X	X	X	
brown		X	X				X	X	X	X
yellow			X		X			X		X
grey			X	X	X			X		X
green			X		X					
orange										
violet							X		X	X
red	X	X	X	X	X		X	X	X	X
white					X				X	
black	X		X	X	X		X	X	X	X

4.0 mm²

Base Color	blue	brown	yellow	grey	green	orange	violet	red	white	black
blue	X									
brown		X					X		X	X
yellow			X							
grey				X						
green					X					
orange										
violet										
red	X	X	X	X	X		X	X	X	X
white	X							X	X	X
black								X	X	X

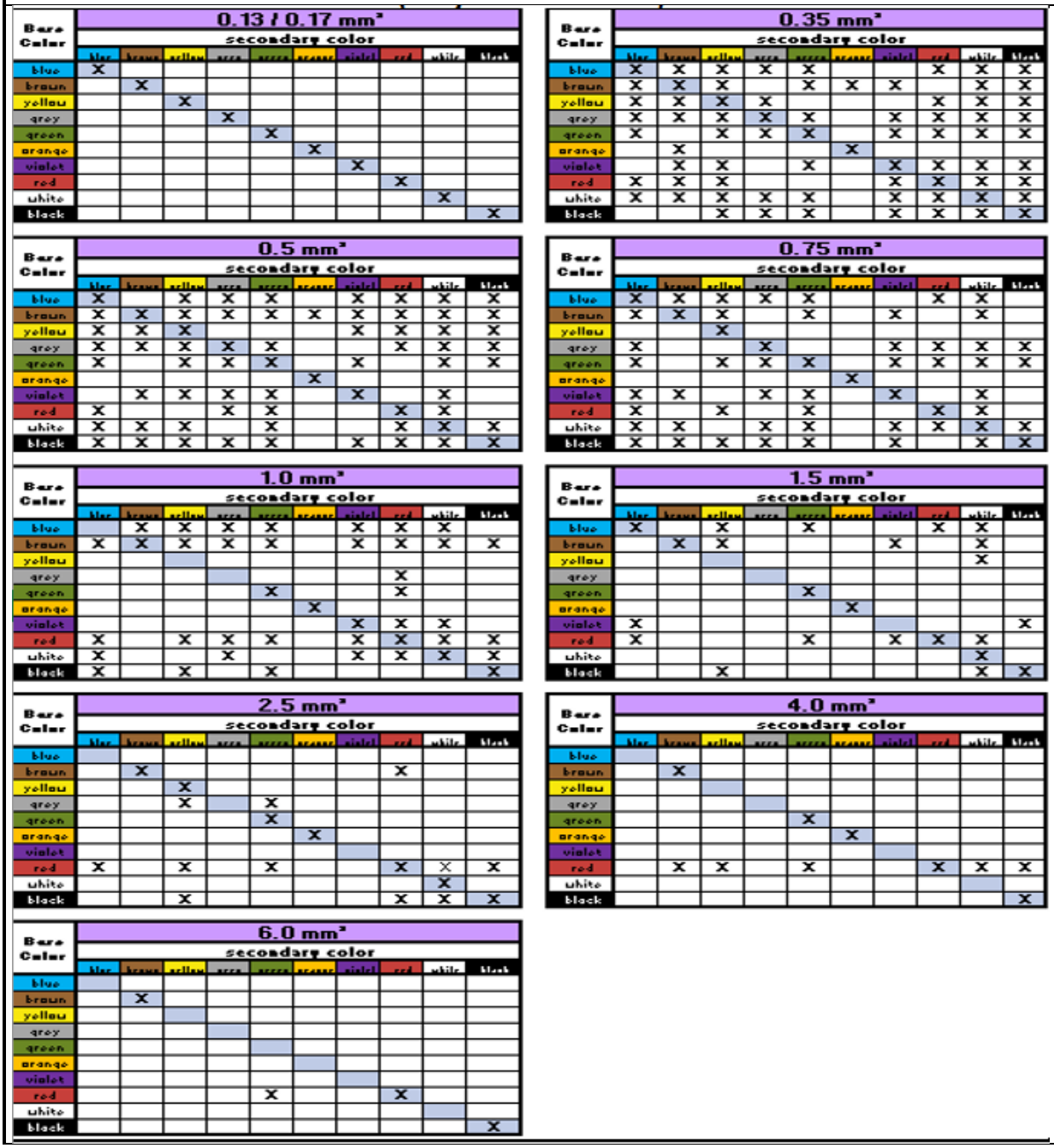
6.0 mm²

Base Color	blue	brown	yellow	grey	green	orange	violet	red	white	black
blue										
brown		X			X					
yellow			X							
grey										
green					X					
orange										
violet										
red	X		X		X			X	X	X
white									X	
black			X							X

Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

Wire:	5
Item:	07
Attachment:	5.07b

4.5.7 Attachment b: Limit number of wire colors: T3 / T4 (125 °C / 150 °C) positive list



0.13 / 0.17 mm²

Base Color \ secondary color	blue	braun	yellow	grey	green	orange	violet	red	white	black
blue	X									
braun		X								
yellow			X							
grey				X						
green					X					
orange						X				
violet							X			
red								X		
white									X	
black										X

0.35 mm²

Base Color \ secondary color	blue	braun	yellow	grey	green	orange	violet	red	white	black
blue	X	X	X	X	X			X	X	X
braun	X	X	X		X	X	X		X	X
yellow	X	X	X	X				X	X	X
grey	X	X	X	X	X		X	X	X	X
green	X		X	X	X		X	X	X	X
orange		X				X				
violet		X	X		X		X	X	X	X
red	X	X	X				X	X	X	X
white	X	X	X	X	X		X	X	X	X
black			X	X	X		X	X	X	X

0.5 mm²

Base Color \ secondary color	blue	braun	yellow	grey	green	orange	violet	red	white	black
blue	X		X	X	X		X	X	X	X
braun	X	X	X	X	X	X	X	X	X	X
yellow	X	X	X				X	X	X	X
grey	X	X	X	X	X			X	X	X
green	X		X	X	X		X		X	X
orange						X				
violet		X	X	X	X		X		X	
red	X			X	X			X	X	
white	X	X	X		X			X	X	X
black	X	X	X	X	X		X	X	X	X

0.75 mm²

Base Color \ secondary color	blue	braun	yellow	grey	green	orange	violet	red	white	black
blue	X	X	X	X	X			X	X	
braun	X	X	X		X		X		X	
yellow			X							
grey	X			X			X	X	X	X
green	X		X	X	X		X	X	X	X
orange						X				
violet	X	X		X	X		X		X	
red	X		X		X			X	X	
white	X	X		X	X		X	X	X	X
black	X	X	X	X	X		X		X	X

1.0 mm²

Base Color \ secondary color	blue	braun	yellow	grey	green	orange	violet	red	white	black
blue		X	X	X	X		X	X	X	
braun	X	X	X	X	X		X	X	X	X
yellow										
grey								X		
green					X			X		
orange						X				
violet							X	X	X	
red	X		X	X	X		X	X	X	X
white	X			X			X	X	X	X
black	X		X		X					X

1.5 mm²

Base Color \ secondary color	blue	braun	yellow	grey	green	orange	violet	red	white	black
blue	X		X		X			X	X	
braun		X	X				X		X	
yellow									X	
grey										
green					X					
orange						X				
violet	X									X
red	X				X		X	X	X	
white									X	
black			X						X	X

2.5 mm²

Base Color \ secondary color	blue	braun	yellow	grey	green	orange	violet	red	white	black
blue										
braun		X						X		
yellow			X							
grey			X		X					
green					X					
orange						X				
violet										
red	X		X		X			X	X	X
white									X	
black			X					X	X	X

4.0 mm²

Base Color \ secondary color	blue	braun	yellow	grey	green	orange	violet	red	white	black
blue										
braun		X								
yellow										
grey										
green					X					
orange						X				
violet										
red		X	X		X			X	X	X
white										
black										X

6.0 mm²

Base Color \ secondary color	blue	braun	yellow	grey	green	orange	violet	red	white	black
blue										
braun		X								
yellow										
grey										
green										
orange										
violet										
red					X			X		
white										
black										X

Assembly:		**Wire:**	5
Cost:		**Item:**	08
Environment & Safety:			
Quality:			
High Voltage:			

4.5.8 Minimum wire size for sealed circuits (0.30 mm²)

DFM Concept:
On a terminal or connector systems that requires the usage of seals, mat seals or grommets the wiring harness design engineer should avoid specifying circuits with a diameter smaller than 0.30 mm².

OK	NG
Usage of wire seals for a wire size with a diameter of ≥ 0.30 mm².	Usage of wire seals for a wire size with a diameter of < 0.30 mm².

Correct Method:
The wiring harness design engineer specified seals for a wire size with a diameter ≥ 0.30 mm².

Incorrect Method:
The wiring harness design engineer specified seals for a wire size with a diameter < 0.30 mm².

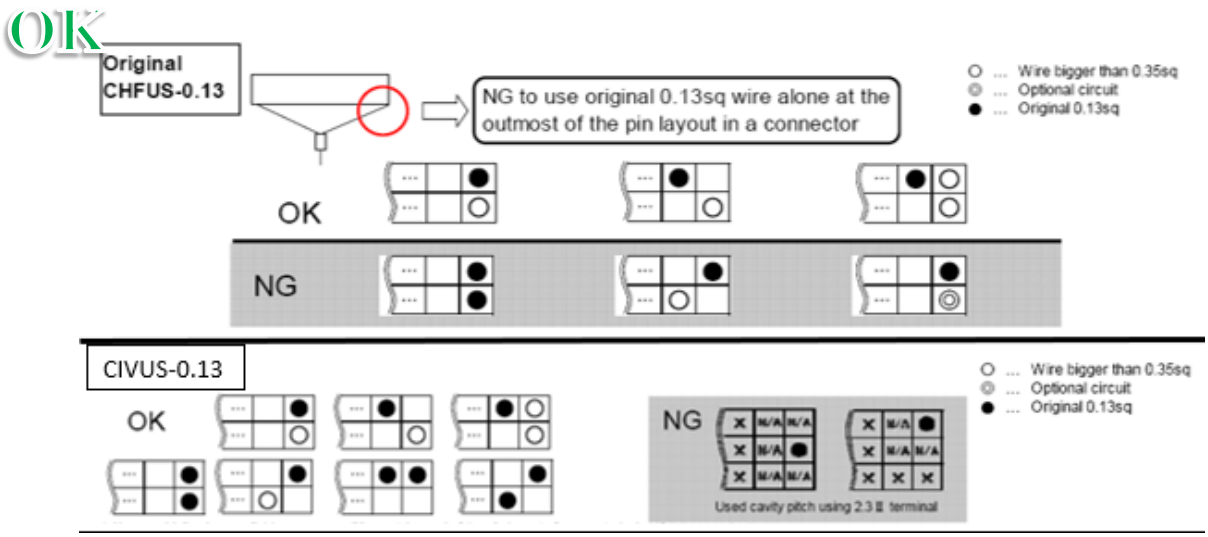
Benefits/Effects:
The insertion of circuits with a diameter smaller than 0.30 mm² into a terminal or connector systems that requires the usage of seals, mat seals or grommets leads to assembly problems due to a higher risk of snapping or violating the stripped circuit ends. Additionally, it is an essential higher effort to straighten the wires to ease the assembly of seals. Also, a smaller wire size means more elastic wires, which makes the seal installation more difficult. Circuits with a diameter bigger or equal than 0.30 mm² are more robust and less susceptible for snapping. Therefore, those circuits simplify the handling for the manufacturing operator.

Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

Wire: 5
Item: 09

4.5.9 Proper location for wires 0.13 mm2 in a connector (only for CIVUS and CHFUS wire)

DFM Concept:
Do not make W/H designs that the 0.13 mm² wire be alone at the outmost of the pin layout in a connector.
In general, avoid placing 0.13 mm² cable on the outside of the connector layout.
This concept is only valid for CIVUS and CHFUS wire, other 0.13 mm² material types need verification according to OEM and wire supplier specification.



Correct Method:
See above figures.

Incorrect Method:
See above figures.

Benefits/Effects:
When you have placed the 0.13 mm² wire size in the outmost location, the wire has a high risk to break the copper threads.
If unsure of whether it's CIVUS or CHFUS, refer to the requirements of CHFUS to assure no concerns.

Assembly:		**Wire:**	5
Cost:		**Item:**	10
Environment & Safety:			
Quality:			
High Voltage:			

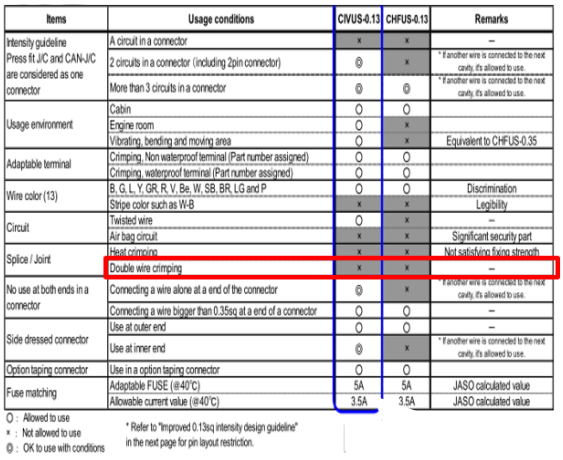
4.5.10 Double crimp for 0.13 mm2 wire

DFM Concept:
Until proven by the use of Double Crimp machine, Double crimping for 0.13 mm2 wire is prohibited. This DFM item will be completed upon approval of 0.13 mm2 Double Crimp wire.

OK

The designer shouldn't assign double crimping for 0.13 mm²

NG



Items	Usage conditions	CIVUS-0.13	CHFUS-0.13	Remarks
Intensity guideline Press fit J/C and CAN-J/C are considered as one connector	A circuit in a connector	x	x	–
	2 circuits in a connector (including 2pin connector)	◎	x	* If another wire is connected to the next cavity, it's allowed to use.
	More than 3 circuits in a connector	◎	◎	* If another wire is connected to the next cavity, it's allowed to use.
Usage environment	Cabin	○	○	
	Engine room	○	x	
	Vibrating, bending and moving area	○	x	Equivalent to CHFUS-0.35
Adaptable terminal	Crimping, Non waterproof terminal (Part number assigned)	○	○	
	Crimping, waterproof terminal (Part number assigned)	○	○	
Wire color (13)	B, G, L, Y, GR, R, V, Be, W, SB, BR, LG and P	○	○	Discrimination
	Stripe color such as W-B	x	x	Legibility
Circuit	Twisted wire	○	x	–
	Air bag circuit	x	x	Significant security part
Splice / Joint	Heat crimping	x	x	Not satisfying fixing strength
	Double wire crimping	x	x	–
No use at both ends in a connector	Connecting a wire alone at a end of the connector	◎	x	* If another wire is connected to the next cavity, it's allowed to use.
	Connecting a wire bigger than 0.35sq at a end of a connector	○	○	–
Side dressed connector	Use at outer end	○	○	–
	Use at inner end	◎	x	* If another wire is connected to the next cavity, it's allowed to use.
Option taping connector	Use in a option taping connector	○	○	
Fuse matching	Adaptable FUSE (@40°C)	5A	5A	JASO calculated value
	Allowable current value (@40°C)	3.5A	3.5A	JASO calculated value

○ : Allowed to use
× : Not allowed to use
◎ : OK to use with conditions

* Refer to "Improved 0.13sq intensity design guideline" in the next page for pin layout restriction.

Correct Method:
The designer shouldn't assign double crimping for 0.13 mm²

Incorrect Method:
It does not pass.

Benefits/Effects:
According to YC Japan, testing the results of double crimping 0.13 mm² doesn't meet the C_{pK} requirements.

Assembly:		**Wire:**	5
Cost:		**Item:**	11
Environment & Safety:			
Quality:			
High Voltage:			

4.5.11 Twisting for 0.13 mm² wire

DFM Concept:
Before implementing on 0.13 mm² wire a twist operation, the wire and the twist parameter (twist process) has to be tested and approved. Only after approval the design can be implemented into mass production.

OK

NG

Items	Usage conditions	CIVUS-0.13	CHFUS-0.13	Remarks
Intensity guideline Press fit J/C and CAN-J/C are considered as one connector	A circuit in a connector	x	x	–
	2 circuits in a connector (including 2pin connector)	◎	x	* If another wire is connected to the next cavity, it's allowed to use.
	More than 3 circuits in a connector	◎	◎	* If another wire is connected to the next cavity, it's allowed to use.
Usage environment	Cabin	○	○	
	Engine room	○	x	
	Vibrating, bending and moving area	○	x	Equivalent to CHFUS-0.35
Adaptable terminal	Crimping, Non waterproof terminal (Part number assigned)	○	○	
	Crimping, waterproof terminal (Part number assigned)	○	○	
Wire color (13)	B, G, L, Y, GR, R, V, Be, W, SB, BR, LG and P	○	○	Discrimination
	Stripe color such as W-B	x	x	Legibility
Circuit	Twisted wire	○	x	–
	Air bag circuit	x	x	Significant security part
Splice / Joint	Heat crimping	x	x	Not satisfying fixing strength
	Double wire crimping	x	x	–
No use at both ends in a connector	Connecting a wire alone at a end of the connector	◎	x	* If another wire is connected to the next cavity, it's allowed to use.
	Connecting a wire bigger than 0.35sq at a end of a connector	○	○	–
Side dressed connector	Use at outer end	○	○	–
	Use at inner end	◎	x	* If another wire is connected to the next cavity, it's allowed to use.
Option taping connector	Use in a option taping connector	○	○	
Fuse matching	Adaptable FUSE (@40°C)	5A	5A	JASO calculated value
	Allowable current value (@40°C)	3.5A	3.5A	JASO calculated value

○ : Allowed to use
× : Not allowed to use
◎ : OK to use with conditions

* Refer to "Improved 0.13sq intensity design guideline" in the next page for pin layout restriction.

Correct Method:
Using CIVUS wire type or similar, it can be twisted.

Incorrect Method:
When using CHFUS wire type or similar, the twisting is prohibited.

Benefits/Effects:
Due to different insulation and conductor material of 0.13 mm² wire, the wire has a different twisting behavior and a different behavior after twisting. For this reason the wire and the twist parameter (twist per meter and space between the twisted wire) needs to be tested in advanced.

Assembly:		**Wire:**	5
Cost:		**Item:**	12
Environment & Safety:			
Quality:			
High Voltage:			

4.5.12 No splice joints for 0.13 mm² wire (only valid for YNCA)

DFM Concept:
Until proven by the use of splice joints machine, splice joints for 0.13 mm² wire is prohibited. This DFM item will be completed upon approval of 0.13 mm² splice joints wire.

OK

The designer shouldn't assign splice joints for 0.13 mm².

NG

Items	Usage conditions	CIVUS-0.13	CHFUS-0.13	Remarks
Intensity guideline Press fit J/C and CAN-J/C are considered as one connector	A circuit in a connector	x	x	–
	2 circuits in a connector (including 2pin connector)	◎	x	* If another wire is connected to the next cavity, it's allowed to use.
	More than 3 circuits in a connector	◎	◎	* If another wire is connected to the next cavity, it's allowed to use.
Usage environment	Cabin	○	○	
	Engine room	○	x	
	Vibrating, bending and moving area	○	x	Equivalent to CHFUS-0.35
Adaptable terminal	Crimping, Non waterproof terminal (Part number assigned)	○	○	
	Crimping, waterproof terminal (Part number assigned)	○	○	
Wire color (13)	B, G, L, Y, GR, R, V, Be, W, SB, BR, LG and P	○	○	Discrimination
	Stripe color such as W-B	x	x	Legibility
Circuit	Twisted wire	○	x	–
	Air bag circuit	x	x	Significant security part
Splice / Joint	Heat crimping	x	x	Not satisfying fixing strength
	Double wire crimping	x	x	–
No use at both ends in a connector	Connecting a wire alone at a end of the connector	◎	x	* If another wire is connected to the next cavity, it's allowed to use.
	Connecting a wire bigger than 0.35sq at a end of a connector	○	○	–
Side dressed connector	Use at outer end	○	○	–
	Use at inner end	◎	x	* If another wire is connected to the next cavity, it's allowed to use.
Option taping connector	Use in a option taping connector	○	○	
Fuse matching	Adaptable FUSE (@40°C)	5A	5A	JASO calculated value
	Allowable current value (@40°C)	3.5A	3.5A	JASO calculated value

○ : Allowed to use
× : Not allowed to use
◎ : OK to use with conditions

* Refer to "Improved 0.13sq intensity design guideline" in the next page for pin layout restriction.

Correct Method:
The designer shouldn't assign splice joints for 0.13 mm².

Incorrect Method:
0.13 mm² does not pass the requirements.

Benefits/Effects:
According to YC Japan, testing the results of slicing 0.13 mm² with CIVUS and CHFUS doesn't meet the C_{pK} requirements.

Assembly:		**Wire:**	5
Cost:		**Item:**	13
Environment & Safety:			
Quality:			
High Voltage:			

4.5.13 Untwist length for Ethernet Cables

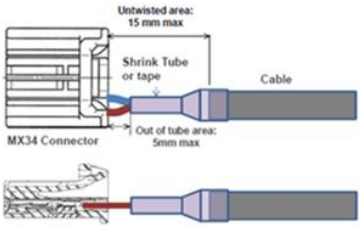
DFM Concept:

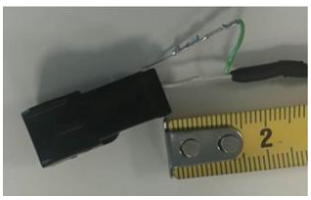
For Ethernet cables there is a strong possibility of broken terminal or broken circuit during the insertion process if untwisted length is less than 50mm from the terminal to the middle of the spot tape.
If OEM Customer specify something different, you MUST follow what they are requiring. In those cases, it's recommended to insert all terminals at same time in order to not being damaged.

OK



NG





Correct Method:

Untwisted length from terminal to the middle of the spot tape is 50mm.

Incorrect Method:

Untwisted length from terminal to the middle of the spot tape is less than 50mm.

Benefits/Effects:

It allows less wire bending during the manual crimping of the terminals and also, the insertion of the wire alone in the connector.

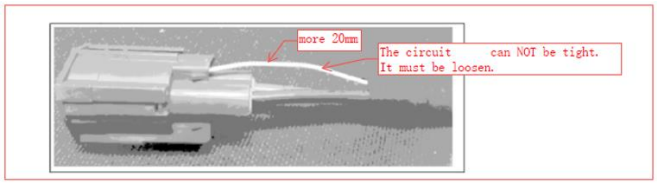
Assembly:		**Wire:**	5
Cost:		**Item:**	14
Environment & Safety:			
Quality:			
High Voltage:			

4.5.14 Adding extra length wire to avoid damage to circuit

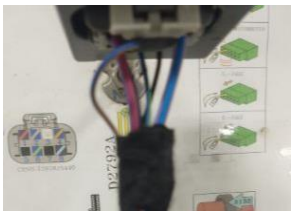
DFM Concept:

The Designer should evaluate to add additional extra length to circuits that are 0.35mm^2 or less wire size to avoid be broken during the handling process or wiring harness vehicle routing.

OK







NG

The circuit can be damaged or broken during bending the wiring harness



Correct Method:

The designer evaluated the wiring harness routing and added an extra length to the circuit for small wire sizes 0.35mm^2 or less.

Incorrect Method:

During the wiring harness vehicle the connector has a bending that breaks the circuit and/or terminal push out.

Benefits/Effects:

If it is selected small wire sizes to be used in the wiring harness and not evaluated the bending areas, the circuit can be broken and damage. This might represent over inspection, inneficiencies into production line due to internal quality claims or even worst, external customer claims.

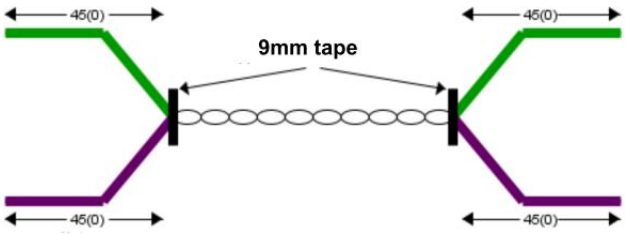
Assembly:		**Wire:**	5
Cost:		**Item:**	15
Environment & Safety:			
Quality:			
High Voltage:			

4.5.15 Use of 9mm tape on twist pairs open ends

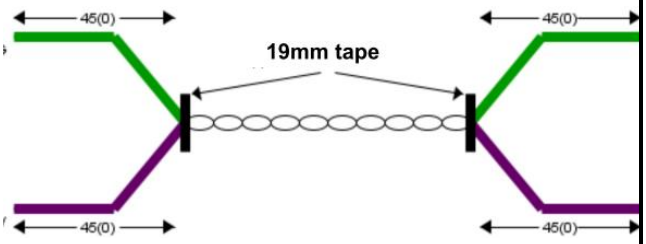
DFM Concept:

The Designer must specify 9mm spot tapes in open ends in case of customer request spot taping.
Only PVC or textile tapes are allowed.

OK



NG



Correct Method:

The spot tapes that fix the twisted wire pair have 9mm width.

Incorrect Method:

The spot tapes that fix the twisted wire pair have 19mm width.

Benefits/Effects:

The automatic twisted wire machines are not capable to handle 19mm tape width.
Through applying this concept, it will allow automation in the cutting area by using equipment with spot tape function, which will eliminate the manual processes and saving time.

4.6 Splicing

Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

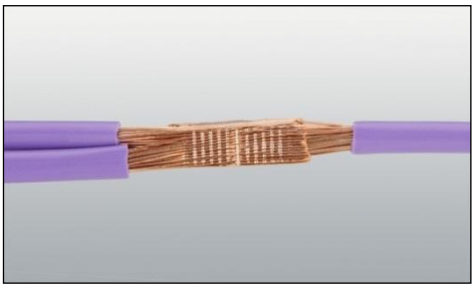
Splicing: 6
Item: 01

4.6.1 Preferred welding process

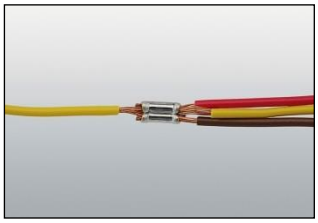
DFM Concept:

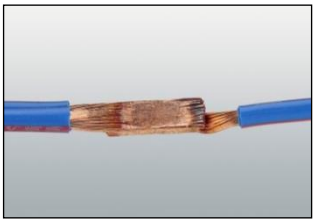
The preferred process for manufacturing splices is the ultrasonic welding method.
Therefore, the wiring harness design engineer should investigate the appropriate manufacturing plant's capabilities for splicing operations..

OK



NG





Correct Method:

As preferred welding method to produce splices the ultrasonic welding method was chosen.

Incorrect Method:

To produce splices the center strip splices or resistance welding method was chosen.

Benefits/Effects:

Producing splices with the ultrasonic welding method makes it possible for the assembly plant to use standardized equipment and processes. Using ultrasonic welding reduces the amount of splices due to the fact that it allows the use of up to 10 circuits on an ultrasonic splice compared to center strip splices where only 3 circuits are allowed. Further advantage is an integrated process monitoring and more exact calibration of parameters (e.g. energy). Due to the higher heat, that is necessary when using the resistance welding method, a higher risk of damaging the insulation occur. Another advantage is the possibility to have more than one setup for one welding machine.

Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

Splicing: 6
Item: 02

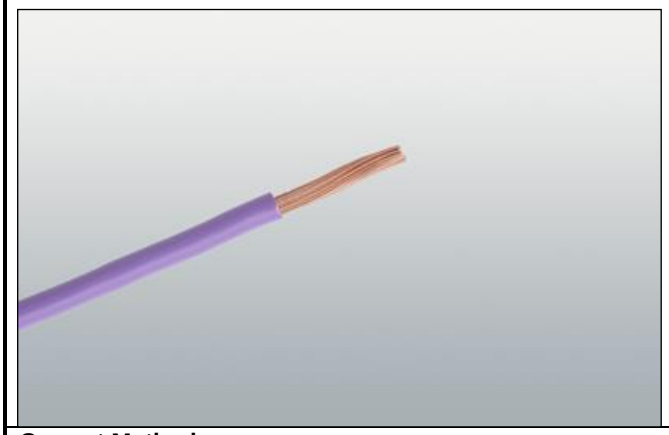
4.6.2 No tinned wires for ultrasonic welding

DFM Concept:

The designer has to avoid choosing tinned wires for ultrasonic welding and should furthermore check in each case if the tinned wires can be replaced with non-tinned wires.

Caution: Wire combination copper/tin and tin/tin is not possible with the ultrasonic welding method.

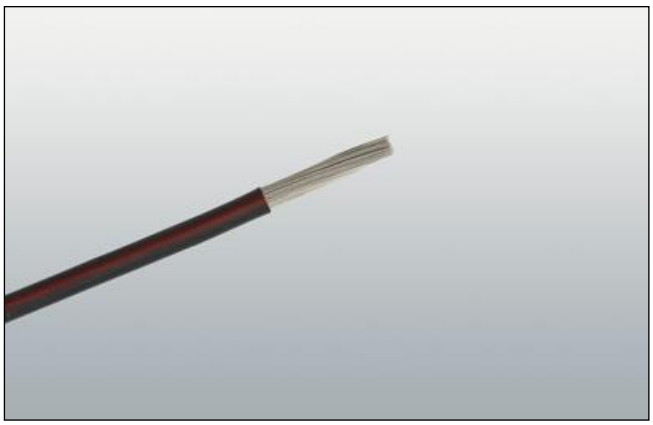
OK



Correct Method:

The wiring harness designer specifies only copper wires.

NG



Incorrect Method:

The wiring harness designer specifies tinned wires.

Benefits/Effects:

Splicing tinned circuits requires the resistance welding method. But, the preferred welding process is the ultrasonic method. Compared to the resistance welding method the ultrasonic welding machines contain an automatic integrated process monitoring system. Calibrating the welding parameters is easier.
(Please review also DFM concept 6.01).

Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

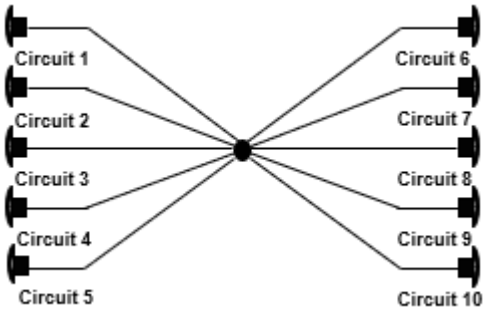
Splicing: 6
Item: 03

4.6.3 Total quantity of wires on a splice (only valid for Ultrasonic welding and Resistance welding)

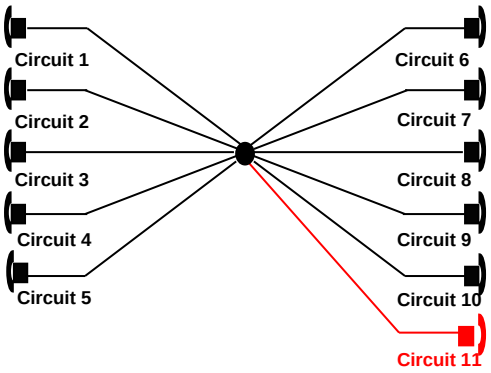
DFM Concept:
The total quantity of circuits on an splice is depending on the way how the circuits are connected to each other.
1.) End splice with a maximum of 6 individual wires.
2.) Go through splice with a maximum of 10 individual wires (not more than 5 on each side).

Please note that this concept is valid for waterproofed and non-waterproofed splices!
(According to "WIRE WELDING PROCESS" EA-MF-WH-S-029)

OK



NG



Correct Method:

The wiring harness design engineer specified up to 10 circuits on the go through splice.

Incorrect Method:

The wiring harness design engineer specified 11 or more circuits on the go through splice.

Benefits/Effects:

Having more than 6 circuits on an end connection splice or 10 circuits on a go through splice leads to handling problems during the welding process. It is difficult for the operator to insert a higher amount of circuits in an adequate way into the welding machine. The risk of several single wire strands sticking out of the welding knot is much higher. To following this DFM concept leads to a higher quality, productivity and to a lower error rate.
For welded eyelet terminals, revise the supplier recommendation maximum amount of circuits during the welding process.

Assembly:		**Splicing:**	6
Cost:		**Item:**	04
Environment & Safety:			
Quality:			
High Voltage:			

4.6.4 Maximum total cross section on splices (Produced with Ultrasonic Welding Technology)

DFM Concept:

The total cross section of circuits spliced together with the ultrasonic welding technology should not exceed 25 mm².
It is possible to weld splices above 25 mm² with UW technology. However first, this must be confirmed by the OEM. This cross section can be extended up to 45 mm² (or more) with notification of the manufacturing plant and a special machine. (According to "WIRE WELDING PROCESS" EA-MF-WH-S-029)

OK	NG
The total cross section is ≤ 25 mm² OR The total cross section is > 25 mm² ≤ 45 mm² and the plant agrees to this.	The total cross section is > 25 mm² and the plant is NOT aware of this.
Correct Method: The design engineer specified a splice with a total cross section of ≤ 25 mm². The total cross section is between 25 mm² and 45 mm² and the customer and manufacturing plant is informed.	**Incorrect Method:** The design engineer specified a splice with a total cross section between 25 mm² and 45 mm² without plant notification. The design engineer specified a splice with a total cross section of > 45 mm².

Benefits/Effects:

If the total cross section of a splice is not bigger than 25 mm² a standard ultrasonic welding machine can be used. For cross sections between 25 mm² up to 45 mm² (or more) special equipment is necessary.

Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

Splicing: 6
Item: 05

4.6.5 Maximum total cross section on splices (only valid for YEL and YNCA)

Produced with Resistance Welding Technology

DFM Concept:
The total cross section of circuits spliced together with the resistance welding technology should not exceed 16 mm².
This cross section can be extended up to 35mm² (or more) with notification of the manufacturing plant and a special machine.

(According to "WIRE WELDING PROCESS" EA-MF-WH-S-029)

OK	NG
The total cross section is ≤ 16 mm² OR The total cross section is > 16 mm² ≤ 35 mm² and the plant agrees to this.	The total cross section is > 16 mm² and the plant is NOT aware of this.

Correct Method:	Incorrect Method:
The wiring harness design engineer specified a splice with a total cross section of ≤ 16 mm². The total cross section is between 16 mm² and 35 mm² and the manufacturing plant is informed.	The total cross section is > 16 mm² and the plant is NOT aware of this.

Benefits/Effects:

If the total cross section of a splice is not bigger than 16 mm² a standard resistance welding machine can be used. For cross sections between 16 mm² up to 35 mm² special equipment is necessary.

Assembly:
Cost:
Environment & Safety:
Quality:
High Voltage:

Splicing: 6
Item: 06

4.6.6 Large splice cross-sections > 26 mm²: Maximum quantity of wires

DFM Concept:

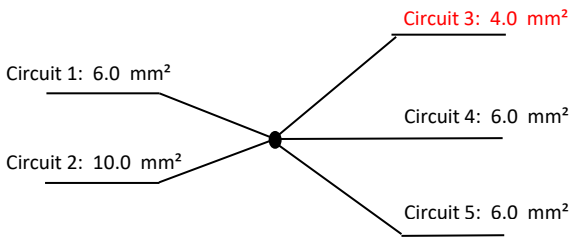
For splice cross-sections > 26 mm² the engineer has to follow the below mentioned table.
This design recommendation is based on existing experiences and has to be always validated by relevant department (CSBU and manufacturing plants) in advance and has to be approved by the customer.

(According to "WIRE WELDING PROCESS" EA-MF-WH-S-029)

OK

maximum quantity of wires	used wire cross-section which can be combined for creation of splice combinations between 26 mm² and 50 mm²
3	25; 16; 10; 7
4	16; 10; 7; 6
5	10; 6
5	10; 7; 6; 5
6	7; 6; 5
7	7; 6; 5; 4

NG



Correct Method:

The rule for large splice cross-sections is applied in accordance to table shown above.

Incorrect Method:

The wiring harness engineer deviates from the rule.

Benefits/Effects:

A better quality of the splices in terms of strength characteristic.

Assembly:
Cost:
Environment & Safety:
Quality:
High Voltage:

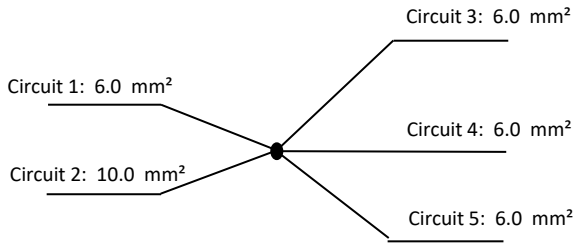
Splicing: 6
Item: 07

4.6.7 Large splice cross-sections > 26 mm²: Unsymmetrically wire arrangement

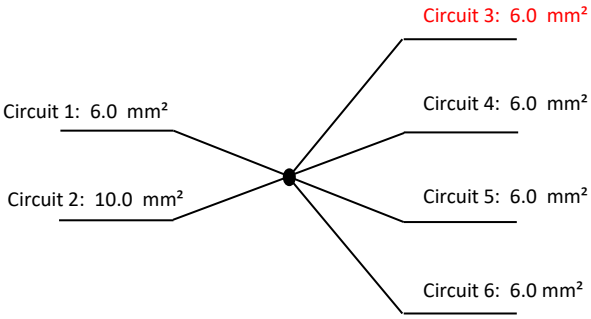
DFM Concept:

For splice cross-sections > 26 mm² and more than 3 wires, the number of wires with the smallest cross-section coming from one side needs to be odd (1,3,5,..). Otherwise a symmetrical arrangement appears which will reduce the strength characteristic of the splice.
This design recommendation is based on existing experiences and has to be always validated by relevant department (CSBU and manufacturing plants) in advance and has to be approved by the customer.
(According to "WIRE WELDING PROCESS" EA-MF-WH-S-029)

OK



NG



Correct Method:

The number of the smallest wire coming from the right side is odd.

Incorrect Method:

The number of the smallest wires coming from the right side is even.

Benefits/Effects:

A better quality of the splices in terms of strength characteristic.

Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

Splicing: 6
Item: 08

4.6.8 Protection of splices containing wires with low cross-section

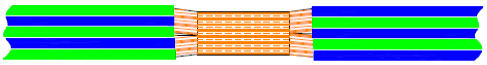
DFM Concept:

The practice of low cross-section wires such as 0.13 mm² or 0.22 mm² has shown, that CpK for the peel and pull forces cannot be achieved. This means that an extra mechanical protection (heat shrink tube) must be installed on the splice. The validation and the approval of such cross-sections have to be agreed always by the customer due to the high risk. (According to "WIRE WELDING PROCESS" EA-MF-WH-S-029)
Additionally, the splice tape is not allowed for splices with 22 gauge (0.35mm2) wire or smaller, or wires utilizing ultra-thin insulation, must only use heat shrink tubing to cover the joint.

OK



NG



Correct Method:

The design engineer decided to add a shrink tube because a 0.13 mm² wire is part of the splice.

Incorrect Method:

Although a 0.13 mm² is part of the splice, no shrink tube has been added.

Benefits/Effects:

Due to the additional mechanical robustness of the splice forced by the heat shrink tube, better C_{pK} for peel and pull forces can be achieved.

Assembly:
Cost:
Environment & Safety:
Quality:
High Voltage:

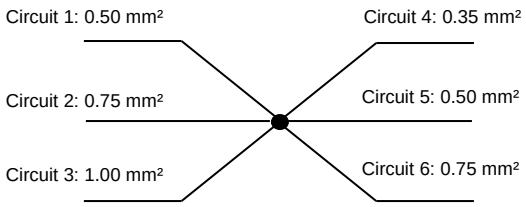
Splicing: 6
Item: 09

4.6.9 Nominal cross section combination on splices

DFM Concept:

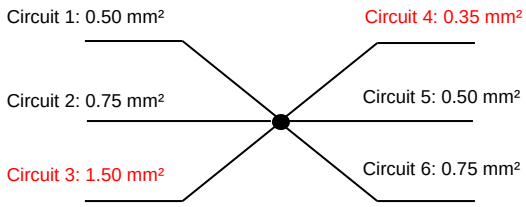
The circuit cross section on a single splice should be specified in the way, that the maximum ratio between the smallest and biggest circuit cross section is not bigger than 3:1.
For further information please review attachment 6.09a.

OK



1.00 mm²/0.35 mm²= 2.85

NG



1.50 mm²/0.35 mm²= 4.28

Correct Method:

The wiring harness design engineer specified only circuits on the splice where the maximum difference between each circuit is not greater than 3:1.

Incorrect Method:

The wiring harness design engineer specified two circuits on the splice (circuit 3 and circuit 4) where the ratio of the cross section between both circuits is bigger than three.

Benefits/Effects:

If the ratio between biggest and smallest circuit cross section is higher than 3:1 a reliable process monitoring cannot be guaranteed anymore. The ultrasonic welding machine will not be able to detect missing strands or even missing small wires. This results in a potential quality risk. The lack for not respect 3:1 ratio demands extra labor and cause downtime.

In case there is no other technical solution, additional manual checks need to be initiated and performed. Therefore, these cases should be limited to the absolute minimum and a written statement/deviation should be obtained from the OEM that they are aware of this potential quality risk.

Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

Splicing:	6
Item:	09
Attachment:	6.09a

4.6.9 Attachment a

All dimensions are in mm²

0.13 mm²: Due to varying conductor materials, based on different suppliers, detailed investigations are mandatory.

		Allowed nominal cross section of remaining circuits																	
		0.13	0.22	0.30	0.35	0.50	0.75	0.85	1.00	1.25	1.50	2.00	2.50	3.00	4.00	5.00	6.00	8.00	10.00
Thinnest nominal cross section	0.13	X	X	X	X														
	0.22		X	X	X	X													
	0.30			X	X	X	X	X											
	0.35				X	X	X	X	X										
	0.50					X	X	X	X	X	X								
	0.75						X	X	X	X	X	X							
	0.85							X	X	X	X	X	X						
	1.00								X	X	X	X	X	X					
	1.25									X	X	X	X	X					
	1.50										X	X	X	X	X				
	2.00											X	X	X	X	X	X		
	2.50												X	X	X	X	X		
	3.00													X	X	X	X	X	
	4.00														X	X	X	X	X
	5.00															X	X	X	X
	6.00																X	X	X
	8.00																	X	X
	10.00																		X

Assembly:
Cost:
Environment & Safety:
Quality:
High Voltage:

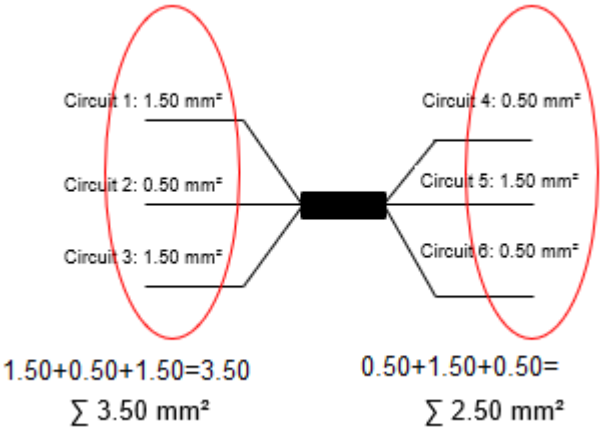
Splicing: 6
Item: 10

4.6.10 Maximum total cross section ratio between opposite wires on splices
Only applicable for waterproof splices

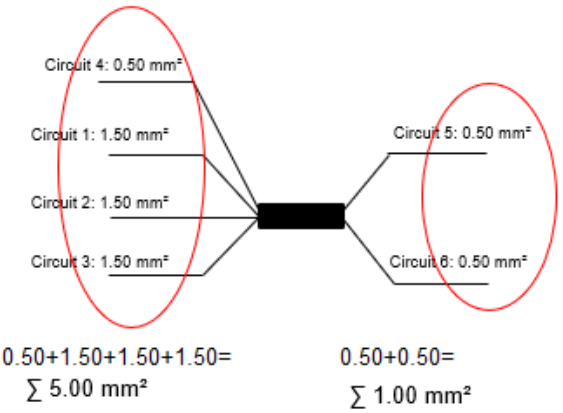
DFM Concept:

The ratio between the total cross-sections of the wires coming from left side and the wires coming from the right side of the go through splice may not exceed 3:1.
(According to "WIRE WELDING PROCESS" EA-MF-WH-S-029).
If the Customer or Quality Department permits to have a different ratio, then you may follow it, but not to be bigger than 4.5 to 1.

OK



NG



Correct Method:

The wiring harness design engineer specified a splice with a ratio not exceeding 3:1 for the total cross section of opposite wires.

Incorrect Method:

The wiring harness design engineer specified a splice with a ratio exceeding 3:1 for the total cross section of opposite wires.

Benefits/Effects:

The handling during the splice operation will be easier. Furthermore, the quality of waterproof splices will improve due to a more consistent shrinking of the shrink tube.

Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

Splicing: 6
Item: 11

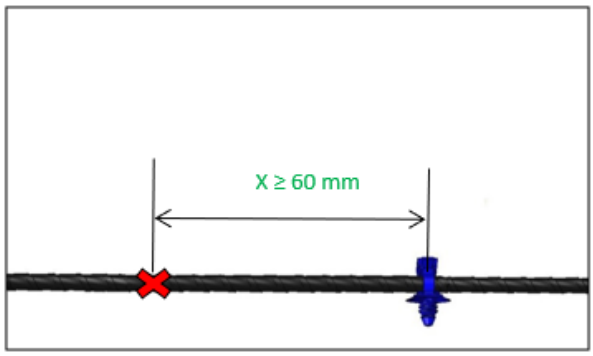
4.6.11 Position of splices in relation to components

DFM Concept:
The position of a splice (waterproof and non-waterproof) in relation to clips, grommets and other components assembled on a W/H should have a minimum distance of 60 mm.

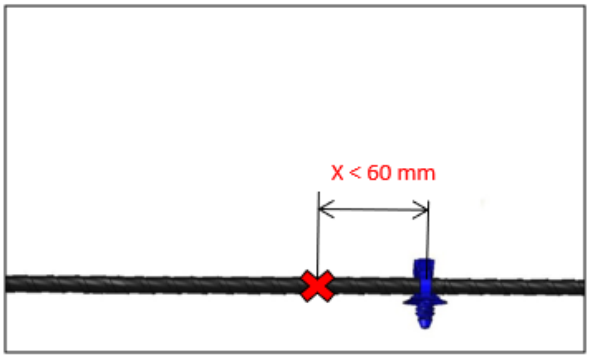
Reference measuring point is the center of the splice and center of component.

For CAN splices: Please review also DFM concept 6.17.

OK



NG



Correct Method:

Minimum distance between splice and component is 60 mm.

Incorrect Method:

Distance between splice and the component is less than 60 mm.

Benefits/Effects:

Splices create a bigger bundle diameter and the harness is more inflexible at this point especially when waterproofed splices are used. So it is much more difficult to add components to the harness or to assemble takeouts which are too close or directly located on a splice. Having an adequate distance between splices and takeouts or components eases the assembly of e.g. fixing features on the branch and it ensures a better flexibility of the harness in the area of takeouts. Additionally, the splices are located in an unstressed area which reduces the risk of damaging the splice.

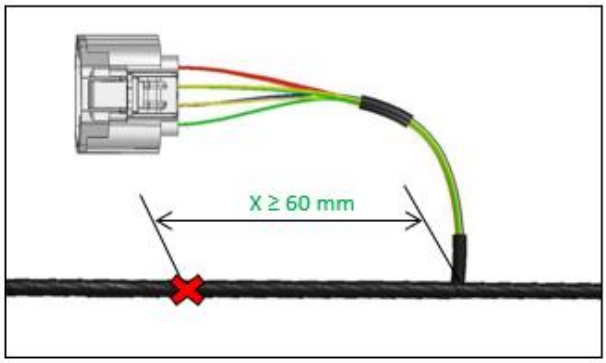
Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

Splicing: 6
Item: 12

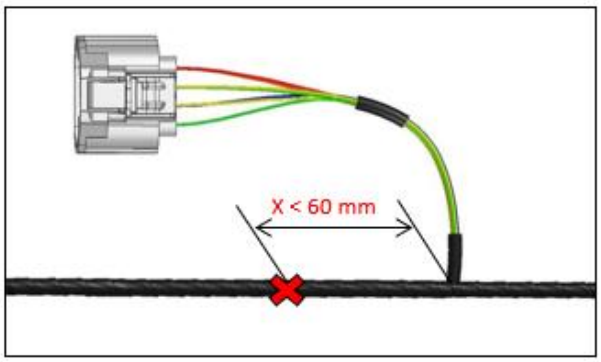
4.6.12 Position of splices in relation to takeouts

DFM Concept:
The position of a splice (waterproof and non-waterproof) in relation to takeout should have a minimum distance of 60 mm.
Reference measuring point is the center of the splice.
For CAN Network splices: Please review also DFM concept 6.17.

OK



NG



Correct Method:

Minimum distance between splice and takeout is 60 mm.

Incorrect Method:

Distance between splice and takeout is less than 60 mm.

Benefits/Effects:

Splices create a bigger bundle diameter and the harness is less flexible at this point especially when waterproofed splices are used. So, it is much more difficult to add components to the harness or to assemble takeouts which are too close or directly located on a splice. Having an adequate distance between splices and takeouts or components eases the assembly of e.g. fixing features on the branch and it ensures a better flexibility of the harness in the area of takeouts. Additionally, the splices are located in an unstressed area which reduces the risk of damaging the splice.

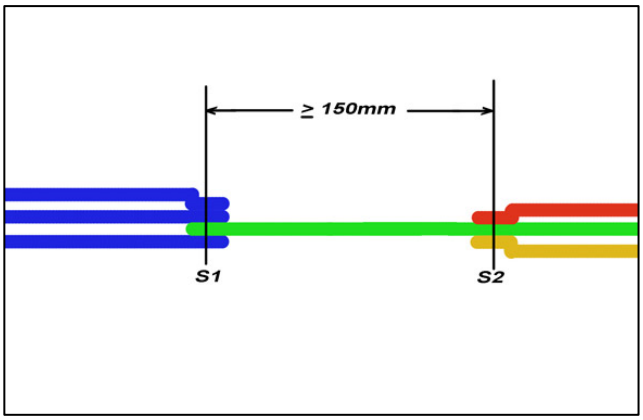
Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

Splicing: 6
Item: 13

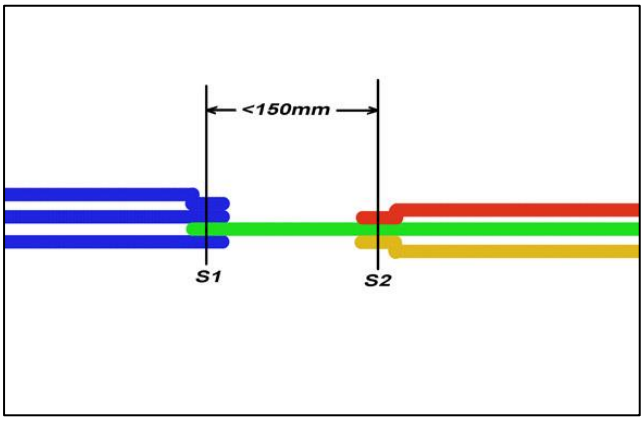
4.6.13 Minimum length of circuits connecting two splices

DFM Concept:
A circuit ≤ 25 mm² connecting two splices should have a minimum length L of 150 mm.
For a circuit > 25 mm² the minimum length L should be 350 mm.
For CAN splices: Please review also DFM concept 6.17.

OK



NG



Correct Method:

Minimum clearance between two splices connected by one circuit is adequate.

Incorrect Method:

Clearance between two splices connected by one circuit is too small and therefore inadequate.

Benefits/Effects:

Having a minimum distance L between two splices guarantees the usage of a heat shrinking machine (Please review also DFM concept 6.19). Therefore, this minimum clearance between two splices eases the assembly process and decreases the overall process time. Additionally, it improves the operators safety during the welding, shrinking or crimping process.

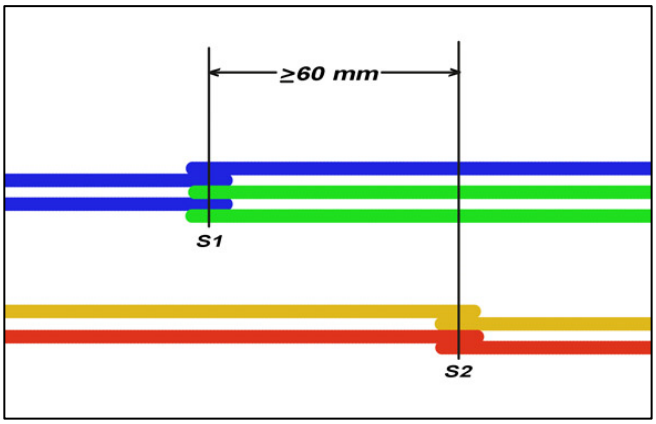
Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

Splicing: 6
Item: 14

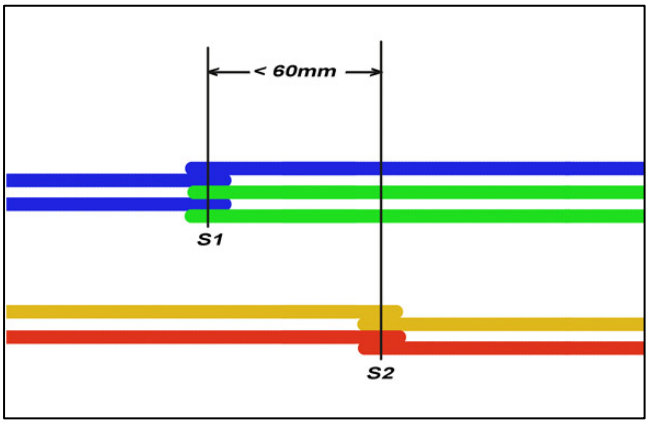
4.6.14 Distance of splices to each other within a W/H bundle (≥ 60 mm)

DFM Concept:
Two splices within a W/H bundle should have a minimum distance of 60 mm to each other.
Reference measuring point is the center of the splice.
For CAN splices: Please review also DFM concept 6.17.

OK



NG



Correct Method:

Splice [S1] and splice [S2] have a minimum distance of 60 mm to each other.

Incorrect Method:

Distance between splice [S1] and splice [S2] is less than 60 mm. Distance between both splices is too short.

Benefits/Effects:

Splices create a bigger bundle diameter and the harness is less flexible at this point. Therefore, the bundle becomes too stiff if splices are located too close to each other. Additionally, the heat shrink tube used for waterproofed splices are 50 mm, 65 mm or 75 mm long. The width of the used protection tape for non-waterproofed splices is 50 mm. It is easier for the manufacturing operators and for the packaging process to handle harnesses with an adequate distance between splices.

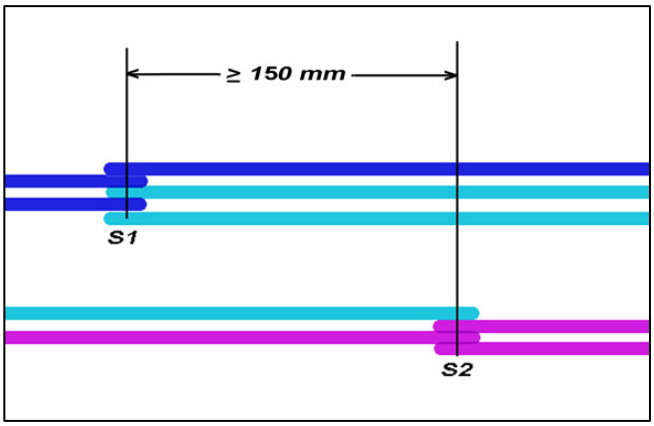
Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

Splicing:	6
Item:	15

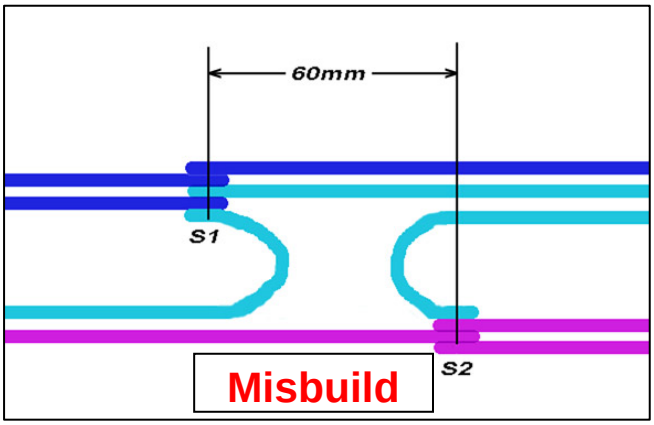
4.6.15 Distance between splicing (≥ 150 mm) (only for on-board splice welding)

DFM Concept:
A minimum distance of 150 mm must be maintained between splices which contain wires of the same color and same wire size range.
Same wire size range means that the difference between the wire is less than 2 wire size steps.

OK



NG



Correct Method:

The distance is greater than or equal to 150 mm.

Incorrect Method:

The distance is less than 150 mm.

Benefits/Effects:

Having these types of splices less than 150 mm apart a risk exists that the operator can swap the wire between the splices. This leads to manufacturing inefficiency and possible quality issues.

NOTE: Splicing is usually done in the "traditional style" (not on-board) unless Manufacturing is required to do it on-board.

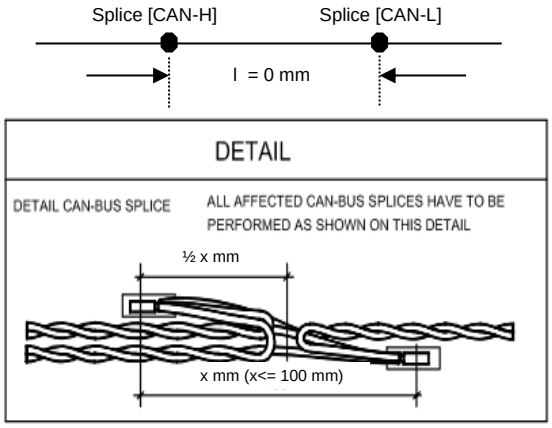
Assembly:
Cost:
Environment & Safety:
Quality:
High Voltage:

Splicing: 6
Item: 16

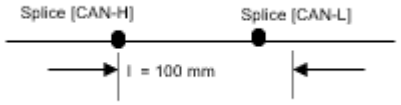
4.6.16 Distance of splices within a CAN network system (x = 0 mm)

DFM Concept:
The wires for a CAN system should have the same circuit cutting length for CAN High and CAN Low.

OK



NG



Correct Method:

Circuits of one twisted pair [CAN-H] and [CAN-L] have always the same length.

Incorrect Method:

Circuits of one twisted pair [CAN-H] and [CAN-L] don't have the same lengths. Due to the different WH drawing physical lengths.

Benefits/Effects:

The benefit is to have a better workflow and process for the separate twist operations. If the splices have to be covered with heat shrink tubes, the shrink operation of two heat shrinks for the two splices can be done with one shrinking operation by store them together in the heater.

NOTE:
For a better understanding how to manufacture CAN splices and to avoid misbuilds, the above shown detail should be added to the W/H drawing.

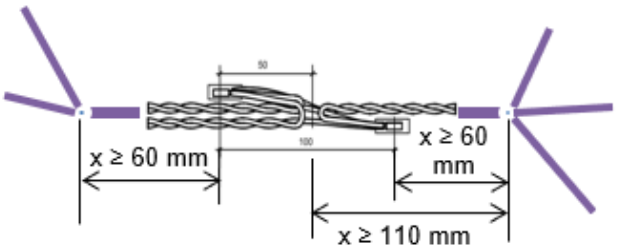
Assembly:
Cost:
Environment & Safety:
Quality:
High Voltage:

Splicing: 6
Item: 17

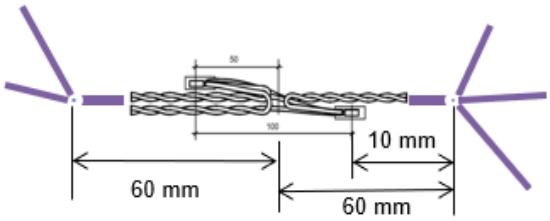
4.6.17 Position of CAN splices within a W/H

DFM Concept:
Due to the special handling of CAN splices (bending of the splices), the distance to takeouts, components or other splices has to be increased.
Please see also DFM concept: 6.11, 6.12, 6.13 and 6.14.

OK



NG



Correct Method:

Between the actually position of the splice and the next takeout is sufficient space.

Incorrect Method:

The specialties of the CAN splice arrangement are not taken into consideration.

Benefits/Effects:

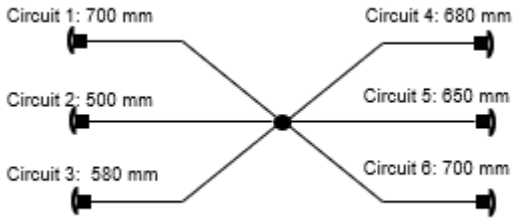
Splices create a bigger bundle diameter and the harness is less flexible at this point, especially when waterproofed splices are used. It is much more difficult to add components to the harness or to assemble takeouts which are too close or directly located on a splice. Having an adequate distance between splices and takeouts or components eases the assembly of e.g. fixing features on the branch and it ensures a better flexibility of the harness in the area of takeouts. Additionally, the splices are located in an unstressed area which reduces the risk of damaging the splice.

Assembly:		**Splicing:**	6
Cost:		**Item:**	18
Environment & Safety:			
Quality:			
High Voltage:			

4.6.18 Minimum circuit length between an ultrasonic splice and crimped terminals (500 mm)

DFM Concept:
The wiring harness design engineer should specify a minimum distance of 500 mm between joints manufactured by using the ultrasonic welding method and a crimped terminal.
Note: The wire size has no influence to the minimum distance of 500 mm.
(According to USCAR38, no collateral damage may result from terminal being welded if the wire length is equal or more than 500mm. If this can´t be done, Inspection must include components on the same wire within 500mm and welded prior to WUT. If OEM Customer and/or Supplier specify something different, you MUST follow what they are requiring).

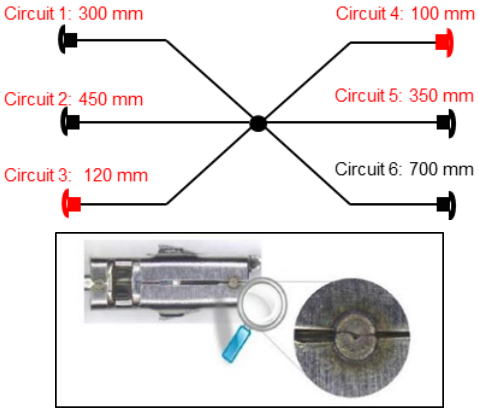
OK



Correct Method:

The wiring harness design engineer specified only circuits with a minimum length of 500 mm between an ultrasonic welded splice and a crimped terminal.

NG



Incorrect Method:

The wiring harness design engineer specified several circuits with a length shorter than 500 mm between an ultrasonic welded splice and a crimped terminal.

Benefits/Effects:

When using the ultrasonic welding method, wires are bounded together by friction. In relation to this welding process wave effects (resonances) appears. Those wave effects can lead to broken terminals when the minimum distance between joint and crimped terminal is not kept. Following this requirement leads to a reduced amount of rework. Additionally, there is the possibility that slight breakages are not visible. This situation can lead to quality issues after the part was assembled to the customers product.

Assembly:		**Splicing:**	6
Cost:		**Item:**	18
Environment & Safety:		**Attachment:**	6.18a
Quality:			
High Voltage:			

4.6.18 Attachment a
Minimum circuit length between an ultrasonic splice and crimped terminals (500 mm)

Development Flow:

STEP ONE:
The wire harness design engineer should check what is the minimum recommendation length from terminal supplier.

STEP TWO:
The wire harness design engineer should specify a minimum distance of 500 mm between a joint manufactured by using the ultrasonic welding method and a crimped terminal.

STEP THREE:
If a minimum wire lengths of 500 mm or a process change is not possible, exactly this terminal, wire type, wire size and wire lengths combination has to be tested according to EA-MF-WH-S-029-A05.
The test has to be requested from the W/H responsible CSBU engineer.
The test can be performed by the manufacturing.
A failed test has to be discussed with the customer and the design has to be changed accordingly.

STEP FOUR:
If it is not possible to have a minimum wire lengths of 500 mm the design engineer has to contact the manufacturing plant and ask to change the process flow in the way that the terminal will be crimped to the affected wire after the ultrasonic welding operation. This means that the terminal will be crimped in a manual process (of course on an automatic press and no hand crimp). Another (not preferred) possibility is to switch the process from ultrasonic welding to resistance welding.

Possible reasons why it is not feasible to have 500 mm wire lengths between splice and terminal:
- It is not enough space to package the necessary wire lengths. Most likely inside hard wired boxes.
- Special customer requirements for the splice location.
- Routing: To avoid splice location in high vibration area or in bending locations.

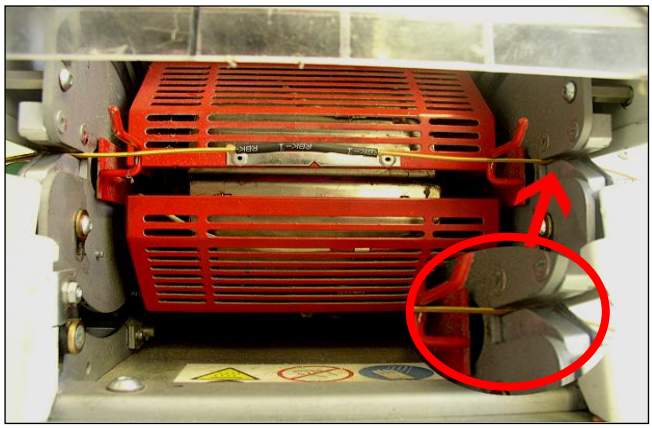
Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

Splicing:	6
Item:	19

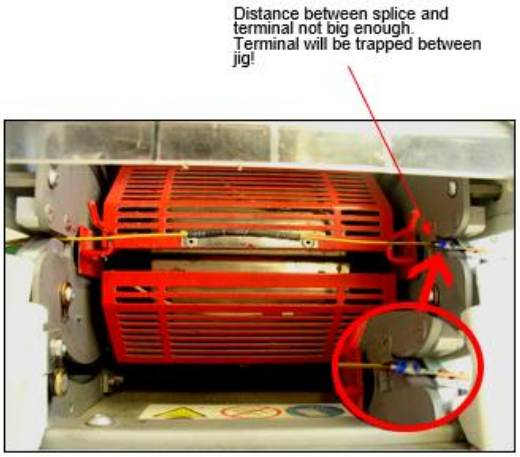
4.6.19 Minimum circuit length to allow using heat shrink machines

DFM Concept:
The wiring harness design engineer should specify a minimum circuit length of 150 mm on both sides of a splice that should be protected by a heat shrink tube. This concept is valid for resistance welding and crimp splice technology. Splices produced with ultrasonic welding technology please see concept 6.18.

OK



NG



Correct Method:

The circuit length between a heat shrink tube and a terminal is > 150 mm. It is possible to use shrink machines.

Incorrect Method:

The circuit length between heat shrink tube and a terminal is less than 150 mm. The shrink process has to be done manually. By using the machine the terminal may be damaged!.

Benefits/Effects:

When having an adequate distance between heat shrink tube and the end of all circuits, it is possible to use an automated shrink process which will be decrease the overall process time.

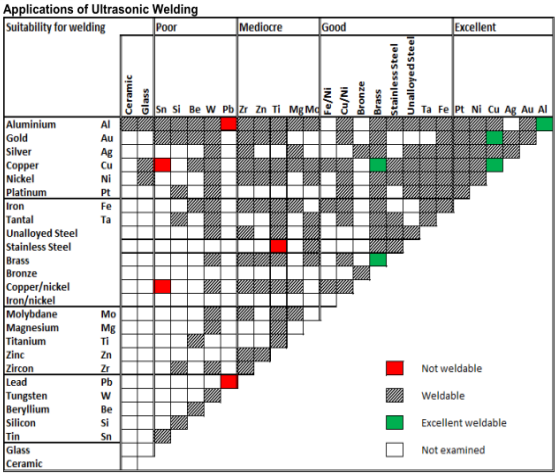
Assembly:
Cost:
Environment & Safety:
Quality:
High Voltage:

Splicing: 6
Item: 20

4.6.20 No Brass materials for Bonder Welding

DFM Concept:
Brass terminals (Cu/Zn) should be included in the design with Ultrasonic welding. Brass terminals with resistance welding are NOT allowed. Also, when the process includes aluminum cable welded to a terminal, use ultrasonic welding.

OK



NG

Using bonder technology for Brass Terminals.

Correct Method:

Select the correct terminal to the used welding Technology.

Incorrect Method:

Bonder welding is not good for brass.

Benefits/Effects:

The brass welded by bonder has lower strength due to:
1) Vaporing point (boiling point) in zinc is below as that in copper.
2) Brass is highly susceptible to stress corrosion cracking. That phenomena caused by weaker grain boundary consisting of zinc especially larger concentration. Cracks may not appear on surface, but internal grain boundary slightly change to be weaker under stress in commercial use. Sometimes welded parts are suddenly broken.

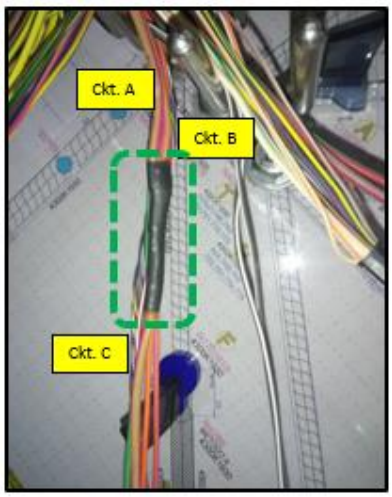
Assembly:
Cost:
Environment & Safety:
Quality:
High Voltage:

Splicing: 6
Item: 21

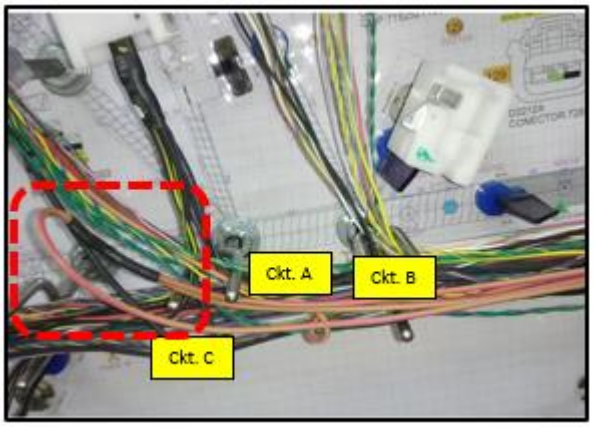
4.6.21 Avoid U-turn splices

DFM Concept:
Due to the balancing of the circuits, some of them must be relocated to the left side of the splice. This results in some circuits making a "U-turn" due to the splice location. The designer must review the splice location to avoid any circuit U-turns.

OK



NG



Correct Method:

Relocate the splice to an area where a U-turn isn´t required.

Incorrect Method:

Circuit makes a U-turn after the splice.

Benefits/Effects:

The U-turn increases material costs and takt time when routing the circuit and coverings over them. Having the U-turn will cause the bundle diameter to increase which may require larger diameter coverings.

In case that the U-turn can't be avoided, the protection of splice must use heat shrink tube. The usage of tape as protection to the splice no matter the splice is located in a dry area is not sufficient robust to have a good CpK for the peel and pull forces. However, when using a heat shrink to protect the splice would give the additional mechanical robustness to have better Cpk for peel and pull forces.

Assembly:		**Splicing:**	6
Cost:		**Item:**	22
Environment & Safety:			
Quality:			
High Voltage:			

4.6.22 Welding Wire to Terminal

DFM Concept:

a) Cross Section: To ensure a good level of defect detection (missing wire) the cross section ratio between the smallest and largest wire must not exceed 3:1 (three to one). Not following this rule increases the risk of missing a small wire that is not detected in the machine during the welding process. If the Customer or Quality Department permits to have a different ratio, then you may follow it, but not to be bigger than 4.5 to 1.
Also, follow the supplier maximum quantity of wires to be used and the minimum and maximum cross section area permitted by the terminal selected.

b) Component selection: If the terminal selected and/or used does not have the necessary minimum cross section area required by the supplier to make the welding, the Designer should select another terminal, otherwise, it would be required to add additional circuits in a loop to increase the cross section area of the weld.

c) Wire Length: It should equal or longer than 500mm with the exception of a folded circuit used to give the sufficient cross section area.

OK





Permissible nominal cross-section of the other conductors (mm^2)

nominal cross-section (mm^2)	0.13*	0.22*	0.35	0.50	0.75	1.00	1.50	2.00	2.50	3.00	4.00	5.00	6.00	7.00	10.00	16.0	25.0
0.13*	x	x	x	x**													
0.22*	x	x	x	x													
0.35	x	x	x	x	x	x											
0.50		x	x	x	x	x	x										
0.75			x	x	x	x	x	x									
1.00			x	x	x	x	x	x	x	x							
1.50				x	x	x	x	x	x	x	x						
2.00					x	x	x	x	x	x	x	x					
2.50						x	x	x	x	x	x	x	x				
3.00						x	x	x	x	x	x	x	x	x			
4.00							x	x	x	x	x	x	x	x	x		
5.00								x	x	x	x	x	x	x	x		
6.00									x	x	x	x	x	x	x	x	
7.00									x	x	x	x	x	x	x	x	x
10.00											x	x	x	x	x	x	x
16.00												x	x	x	x	x	x
25.00															x	x	x

Correct Method:
The wiring harness design engineer specified only circuits with a minimum length of 500 mm between the welding and terminal and meets the ratio 3:1

Incorrect Method:

Benefits/Effects:
According to USCAR38, collateral damage may result from terminal being welded if the wire length is less than 500mm. If this can't be done Inspection must include components on the same wire within 500mm and welded prior to WUT. If OEM Customer and/or Supplier specify something different, you MUST follow what they are requiring.

A no good terminal selection can increase the need to add additional wire and generates more cost in labor and materials.

Assembly:		**Splicing:**	6
Cost:		**Item:**	23
Environment & Safety:			
Quality:			
High Voltage:			

4.6.23 Usage of Anticapillary Wire in a splice

DFM Concept:

The designer must include into the drawing the Yazaki process code P74063305 for each circuit that would be part of a splice in order to give instruction to clean the end of the wire that is going to be welded and remove the silicon that it has.

When an anticapillary wire is used and it is required to be in a splice, it is necessary a cleaning process to brush the silicone in order to have a good performance of the weld and the machine to be used is a resistance welding.

OK



Correct Method:	**Incorrect Method:**
Add the need to remove the silicone at the end of the wire that is going to be in a splice.	

Benefits/Effects:

If the process code is not added to the drawing to clean and remove the silicone of the wire before the welding it is going to impact the cost and labor required to perform the splice causing inefficiencies into production.

4.7 High Voltage

Assembly:
Cost:
Environment & Safety:
Quality:
High Voltage:

High Voltage Item: 7 01

4.7.1 Properly marking wires when multiple wires of the same color are in a single connector

DFM Concept:

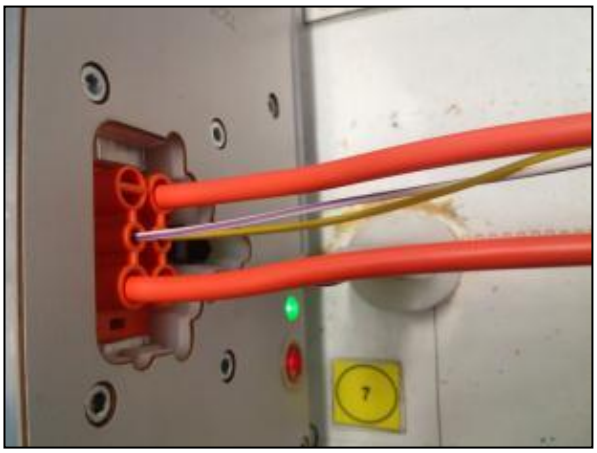
When using the same wire color more than once in a single connector, the different wires must be differently marked, either by ink or by colored tape.

OK





NG



Correct Method:

Each wire can be clearly identified due to the usage of ink or a label (colored tape)

Incorrect Method:

Wires are identical but not specially marked. This can lead to misbuilds (pin-swap).

Benefits/Effects:

Many times the HV wire is not available in different colors or even with a stripe. This means it is sometimes unavoidable to have the same color wire more than once in a connector. If the color of the wire is the same, more than once, in a single connector it causes a high potential for quality issues.

Assembly:
Cost:
Environment & Safety:
Quality:
High Voltage:

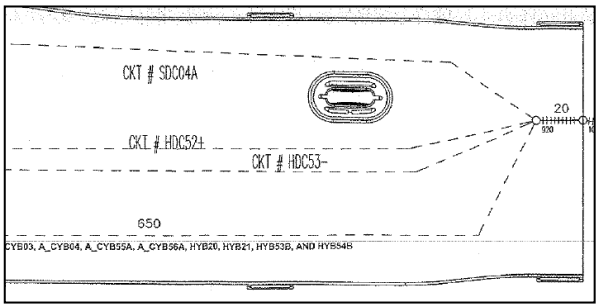
High Voltage: 7
Item: 02

4.7.2 Specify which wires go into each channel of a shield / trough / routing aid

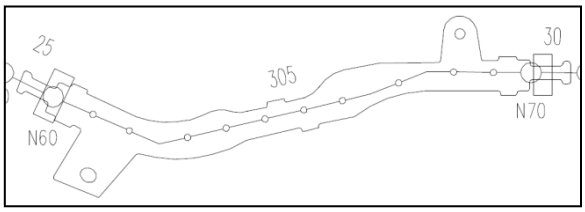
DFM Concept:

If there are shields/troughs/routing aids that have multiple channels for individual wires, the design engineer has to indicate in their W/H drawings which wires go into each of the channels in the shields/troughs/routing aids.

OK



NG



Correct Method:

Shields/troughs/routing aids that have multiple channels for individual wires with indication of routing in the drawing.

Incorrect Method:

Shields/troughs/routing aids that have multiple channels for individual wires without any indication of routing in the drawing.

Benefits/Effects:

The wire length in the large gauge cables, that are in a HV harness, are very sensitive to tolerancing. It is important to specify which wires belong in each trough so that individual wire lengths can be called out on each wire in the shield/trough/routing aid. If the wires are not called out individually in the shields/troughs/routing aids, the routing of the wires could cause length issues in the wires, and there will be inconsistencies in the harness that could result in a quality issue.

Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

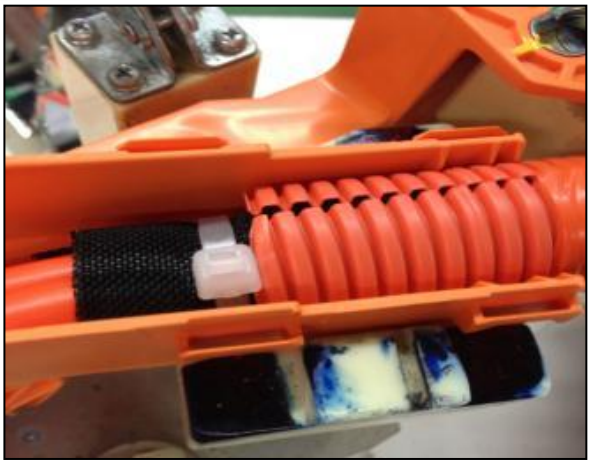
High Voltage:	7
Item:	03

4.7.3 Use a tie strap to secure wires to shields / troughs / routing aids

DFM Concept:

For multiple wires, add a tie strap to fasten the wires in order to prevent them from moving inside the protector. For single wires, tape could be used.

OK



NG



Correct Method:

For multiple wires, add a tie strap to fasten the wires in order to prevent them from moving inside the protector. For single wires, tape could be used.

Incorrect Method:

No tie straps holding the wires in place.

Benefits/Effects:

When large cables are routed in a protector, those cables can move in the protector during shipping & handling. This can cause the harness to fall out of tolerance in other areas of the harness, which is an obvious quality issue. To avoid this, a tie strap needs to be added to securely fastens the wires to the protector when there are multiple wires. When there is only a single wire in the protector, tape can be used to secure it.
This tie strap is needed even if the convolute that covers the wires is secured to the protector.

Assembly:
Cost:
Environment & Safety:
Quality:
High Voltage:

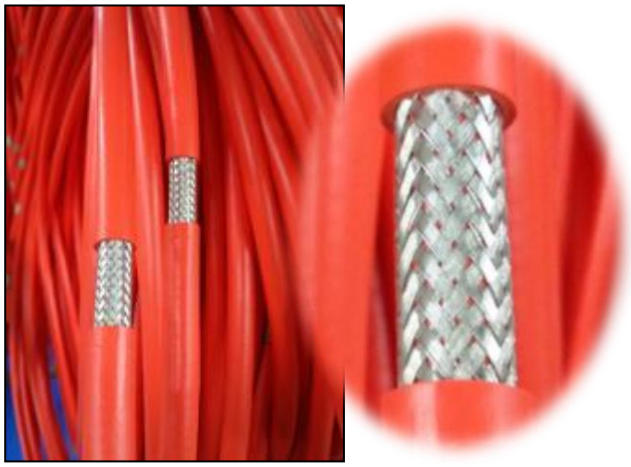
High Voltage: 7
Item: 04

4.7.4 No HV wires with paper

DFM Concept:

Some wire types have paper between the insulation and the wire strands. The automatic wire cutters are not able to remove the paper.

OK



NG





Correct Method:

No paper present under insulation.

Incorrect Method:

Wire insulation has paper under it.

Benefits/Effects:

When the paper is between the insulation and the wire strands, the paper needs to be manually removed from the strands before crimping or welding. This creates scrap in the plant, it increases the operation time for crimping and welding, and it introduces quality issues if the paper is not fully removed. Also, the customer does not pay for us to remove this paper. Therefore, it decreases the efficiency of the plant.

Assembly:	
Cost:	
Environment & Safety:	
Quality:	
High Voltage:	

High Voltage:	7
Item:	05

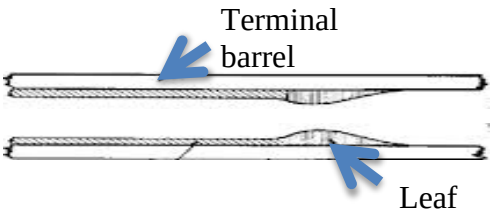
4.7.5 Avoid terminals that have coil springs

DFM Concept:

Avoid using terminals that have 'coil springs' internal to the terminal. Instead, use terminals that have leaf springs or no springs at all.

OK





NG





Correct Method:

Terminals that do not have springs, or that have "leaf springs."

Incorrect Method:

Terminals that have coil springs

Benefits/Effects:

If coil springs are used in the terminals, the springs have a high susceptibility to falling out of the terminal, and are very difficult to detect in the terminal. This means that manufacturing can not guarantee that the spring is in the terminal when the W/H leaves our facilities, and the spring can fall out between our location and the customers.
This has a huge detrimental effect on quality.

5. REFERENCES & RELATED DOCUMENTS:

Number	References
EA-EN-WH-P-010	New DFM Item Verification Process
EA-EN-WH-I-010	DFM Item Check
EA-EN-WH-I-010-F-01	DFM Item Check Form
EA-MF-WH-S-029	WIRE WELDING PROCESS
EA-MF-WH-S-029-A05	Wire Welding: Test Method for USW impact at crimped connection

Number	Related Documents

6. DOCUMENT APPROVAL:

Process Owner:				
Title/ Role	**Name**	**Authorization***	**Revision**	**Date**
Sr. Director Core Eng.	Jens U. Bieler	Via EQMS	4.0	28/FEB/2022
CEE Sr. Manager	Joerg Tillmann	Via EQMS	4.0	28/FEB/2022

Authored by:				
Title / Role	**Name**	**Authorization***	**Revision**	**Date**
CEE Supervisor	Paulo Ascensão	Via EQMS	4.0	28/FEB/2022

Reviewed by:				
Title / Role	**Name**	**Authorization***	**Revision**	**Date**
YE EQ QMS Sr. Specialist	Lina Budre	Via EQMS S	4.0	28/FEB/2022
Cross Functional Team (by functions) EQMS issue # **The signatures from XFR team is not required*				
XFR EQMS 1238				

Approved by:				
Title / Role	**Name**	**Authorization***	**Revision**	**Date**
YEL Head of Engineering	Paul Bradley	EQMS	4.0	28/FEB/2022

Evidence of authorization is either an electronic approval or a physical signature and stored in a secure location.

7. DOCUMENT HISTORY:

For ONLY new or content revisions, the revision date is equal to the Authored by date and all Authorization dates are the same or after. Note: See Section 4 Rules in Yx-QM-xx-Y-001 Process Documentation Management Policy for format or grammar change revision date rules.

Revision Summary	Revision	Date
Initial Release Document number has changed from YEL-RD-G-01 (Revision 8) to EA-EN-WH-G-001 (Revision NEW) Revision of 1.01 / 1.06 / 1.08 / 1.10 / 2.02 / 2.04 / 2.05 / 2.09 / 2.13 / 2.14 / 2.15 / 2.17 / 2.24 / 3.01 / 3.02 / 3.04 / 3.07 / 3.11 / 3.12 / 4.01 / 4.09 / 6.11 New items 1.11 / 1.12 / 3.14 / 3.14a / 3.15 / 6.21 Split of DFM Guideline into several documents (details see chapter 5 References & Related Documents)	New	07/JULY/2019
New items 2.25 and 2.26.	1	24/JUNE/2020
Item 6.18 revised, was missing in the Revision Summary 24. June 2020	1.1	17/AUG/2020
Item 6.18 updated with note and revision of NG image	1.2	08/SEPT/2020
Modified items 1.03, 2.10, 2.14, 2.21, 3.06, 3.14, 4.06, 5.05, 6.01, 6.09, 6.21. Added new DFM items 4.13, 5.13, 5.14, 6.22, 6.23	2.0	05/FEB/2021
Revision of 2.5 resulting of the customer claim GQRS# 1053365	3.0	23/APR/2021
Revision of items 1.01, 1.10,1.11, 1.12, 2.03, 2.15, 2.22, 2.25, 4.01, 4.08, 4.11, 5.03, 6.08, 6.09, 6.10, 6.18, 6.18a, 7.05. Added new DFM items 1.13, 1.14	4.0	28/FEB/2022
Revision of items Added new DFM item 5.15	4.1	04/APR/2022